

Physical Models of Living Systems

Ecology, neuroscience and the physics of the cell — Lecture Notes

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About these notes

Scope

These are lecture notes for *Physical Models of Living Systems* (Master in Physics of Data, University of Padova; Profs. Samir Suweis and Amos Maritan). The course uses the vocabulary of statistical physics to model biological systems at scales from a single microbial population to a whole brain and the inside of a cell: master equations, mean-field limits, phase transitions, random matrices, maximum entropy and criticality.

The notes follow the three modules of the course:

- **Part I: Ecology.** Deterministic and stochastic population dynamics; neutral theory and the upscaling of biodiversity; spatial and temporal macroecological patterns and metapopulations; predator–prey dynamics and stochastic amplification; random matrices and the complexity–stability paradox; mutualistic networks, consumer–resource models and metabolic scaling.
- **Part II: Neuroscience.** The brain and how it is measured; single-neuron models and spike statistics; firing-rate networks and Wilson–Cowan dynamics; Hebbian plasticity and Hopfield networks; Boltzmann machines; whole-brain models and brain criticality.
- **Part III: Physics of the cell.** The cell and gene regulation; thermodynamics at the micro-scale and molecular motors; stochastic gene expression and genetic circuits; inference of interaction networks by maximum entropy; proteins as compact polymers.

How the notes are organised

Each chapter follows the official course notes and the lecture slides of the 2025–26 edition. The books listed below supply depth and examples. Definitions, theorems, worked examples and remarks are set in callouts. Every chapter ends with exercises: the course homework (marked **HW**), the class exercises (notebooks on Oja’s rule, Hopfield networks and datasets), and further problems, all with worked solutions. Numerical results quoted in solutions were checked by simulation or direct computation.

Notation

- $n, N(t)$: population sizes; $P_n(t)$ or $P(x, t)$: probability of a state; $x = n/N$ a density.
- $W^\pm(n), T(n \rightarrow n')$: transition rates; $A(x), B(x)$: drift and diffusion in Fokker–Planck equations.
- r_i, K_i, M_{ij} : growth rates, carrying capacities and interactions in Lotka–Volterra models; J : the Jacobian (community matrix).
- ϕ, r : firing rates; W_{ij} : synaptic weights; $s_i \in \{\pm 1\}$ or $\{0, 1\}$: binary neural states; E, H : energy functions; $\beta = 1/T$.
- m, p : mRNA and protein numbers in gene-expression models.

Main references

- Prof. S. Suweis, *Physical Model of Living System*, course notes 2022–23 (written by G. Cataldi, F. Manzali, L. Rosset), and the 2025–26 lecture slides of S. Suweis and A. Maritan.
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- U. Alon, *An Introduction to Systems Biology*, CRC Press (2019).
- I. Volkov, J. R. Banavar, S. P. Hubbell, A. Maritan, *Neutral theory and relative species abundance in ecology*, *Nature* **424**, 1035 (2003).

Each chapter closes with its own *Sources* section naming the lectures, homework and papers it draws on.

Part I.

Part I: Ecology

1. Introduction: living systems as complex systems

Biology offers systems in which many simple units interact: individuals in an ecosystem, neurons in a brain, genes and proteins in a cell. The behaviour of the whole is not written in any single unit, yet it is often surprisingly *simple* and *reproducible* across systems that share little microscopic detail. This course uses the tools of statistical physics to explain such regularities: master equations, mean-field limits, phase transitions, random matrices and criticality. This chapter sets out the programme, the empirical patterns we want to explain, and the modelling philosophy behind them.

1.1. Three systems, one framework

The course has three modules, each studied with the same sequence of questions:

	Ecological systems	Neural networks	Genetic circuits
one unit	a single species	a single neuron	a single gene
two units	predator and prey	excitatory and inhibitory populations	a regulator and its target
many units	communities, food webs	recurrent networks, the whole brain	regulatory networks

For each we ask how the units evolve, first *deterministically* (ordinary differential equations for mean quantities) and then *stochastically* (master equations, Fokker–Planck and Langevin equations), and which collective, **emergent** properties appear when many units interact.

Definition 1.1 (Complex system and emergence). A **complex system** is made of many units whose interactions, rather than whose individual properties, determine the macroscopic behaviour. A property is **emergent** if it belongs to the system as a whole and cannot be deduced from any single component taken in isolation.

Emergent behaviour would be of little use if it were as intricate as the microscopic details. The reason a physicist can contribute is that, very often, it is not. Interacting units produce simple patterns (power laws, universal distributions, sharp transitions) that recur across systems with completely different microscopic rules. That recurrence suggests common principles that a *minimal* model can capture. Throughout the course we look for the **null model**: the simplest mechanism that reproduces an observed pattern. Departures from it then tell us which biological detail actually matters.

Note

The 2021 Nobel Prize in Physics, shared by G. Parisi “for the discovery of the interplay of disorder and fluctuations in physical systems from atomic to planetary scales”, recognised precisely this programme. Several tools of this course (random matrices, replica calculations, spin-glass-like energy landscapes) come from that tradition.

1.2. Two motivating examples

Example 1.1 (*The collapse of the Atlantic cod*)

The catch record of Atlantic cod over the twentieth century oscillates with the reproductive cycle and shows a huge peak in the 1960s, driven by industrial fishing. After a smaller secondary maximum, the fishery collapsed abruptly in the early 1990s and never recovered, even after fishing stopped. The collapse is not a smooth decline proportional to the fishing effort. It is a **critical transition**, a jump between two qualitatively different states of the ecosystem.

Example 1.2 (*The geography of malaria*)

Deaths from malaria are distributed very unevenly across the globe. Their statistics are strongly heterogeneous, with a few areas carrying most of the burden. Homogeneous fluctuations are well described by Gaussian statistics with a typical scale. Heterogeneous ones are typically described by **power-law** (scale-free) distributions, in which no single scale dominates. Which mechanism produces one shape or the other is a recurring theme of the course.

1.3. Emergent ecological patterns

Ecology offers the cleanest examples of universal patterns. They are measured by counting individuals of each species in plots of increasing size.

Definition 1.2 (Species–area relationship (SAR)). The **SAR** $S(A)$ is the mean number of species found in an area A . Over a wide range of intermediate scales it is well fitted by a power law

$$S(A) = c A^z, \quad z \approx 0.2-0.3,$$

for plants, birds, amphibians and mammals, on every continent: the curves differ in the prefactor c but share the shape.

Definition 1.3 (Endemic–area relationship (EAR)). The **EAR** $E(A)$ is the number of species found *only* inside an area A (endemic to it). It rises slowly at small areas and steeply near the size of the whole region. Since destroying an area A wipes out the species endemic to it, the EAR is the right tool to estimate extinctions from habitat loss (Chapter 4).

Definition 1.4 (Relative species abundance (RSA)). The **RSA**, or species abundance distribution (SAD), $P(n)$, is the fraction of species represented by n individuals. Empirical RSAs have a characteristic shape: many rare species and few very abundant ones. They are classically fitted by Fisher’s log-series or Preston’s log-normal, and neutral theory predicts them from first principles (Chapter 3).

A second celebrated regularity concerns metabolism.

Definition 1.5 (Kleiber’s law). Across organisms whose masses span more than twenty orders of magnitude, from bacteria to whales, the metabolic rate B scales with body mass M as

$$B = B_0 M^a, \quad a \simeq 3/4.$$

Many other traits follow from it as power laws of M : lifespan $\propto M^{1/4}$, heartbeat frequency $\propto M^{-1/4}$, and population density $\propto M^{-3/4}$.

Since a $3/4$ exponent differs from the “surface-to-volume” value $2/3$, its explanation (West, Brown and Enquist’s optimal transport networks, or Banavar, Maritan and Rinaldo’s argument on supply networks) is a classic problem in biological physics. We return to metabolic scaling and its consequences for coexistence in Chapter 7.

1.4. A short history of population dynamics

Population modelling is as old as quantitative science:

- **Fibonacci** (1202) counted rabbits, producing the first growth law with a characteristic ratio.
- **Euler** (1748) wrote exponential growth, $\dot{N} = rN$.
- **Malthus** (1798) argued that unchecked populations grow geometrically while resources grow at most arithmetically.
- **Verhulst** (1838) added self-limitation, $\dot{N} = rN(1 - N/K)$, the logistic equation, which predicts a steady state.

Chapter 2 shows how these laws can be *derived*, rather than postulated, from symmetry and from resource dynamics.

1.5. Critical transitions and alternative stable states

The cod collapse illustrates a mechanism we will meet repeatedly: a system can have two stable states for the same environmental conditions, and slow changes in a parameter can make one of them disappear.

Example 1.3 (*Grazing and sudden collapse (May, 1977)*)

A resource (vegetation, fish stock) grows logistically and is consumed by a predator whose consumption saturates:

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right) - c \frac{N^2}{N_0^2 + N^2}.$$

For small consumption pressure c there is a single, high-abundance equilibrium. Over an intermediate range of c a second, low-abundance stable equilibrium appears, separated from the first by an unstable one. Increasing c beyond a threshold c_2 makes the high branch collide with the unstable branch and vanish (a **saddle-node**, or fold, bifurcation): the population drops to the low branch. To recover the high state one must reduce c well below c_2 , down to the other fold $c_1 < c_2$ (**hysteresis**).

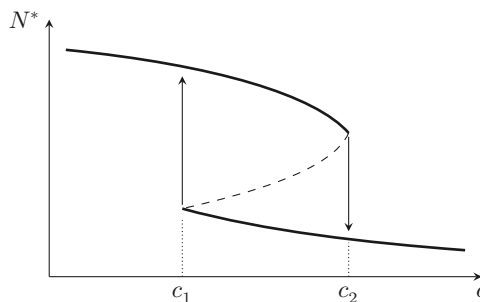


Figure: equilibrium abundance against consumption pressure. Solid: stable branches; dashed: unstable branch. Crossing a fold produces a jump; recovery requires going back to the other fold.

Near a fold the stabilising eigenvalue tends to zero, so perturbations decay more and more slowly. This **critical slowing down** gives measurable *early-warning signals* of an approaching collapse: rising variance and rising autocorrelation of fluctuations. Chapter 5 adds the stochastic side of the story, where demographic noise near a weakly stable equilibrium is amplified into large, coherent oscillations.

1.6. How to read these notes

- **Part I (ecology)** builds the toolbox: master equations and their continuum limits (Chapter 2), neutral theory and scaling (Chapters 3–4), interacting species and random matrices (Chapters 5–7).
- **Part II (neuroscience)** reuses it for neurons and networks: single-neuron models, firing-rate networks, learning and memory, and brain criticality (Chapters 8–13).
- **Part III (physics of the cell)** applies it inside the cell: gene expression, molecular machines, regulatory circuits, inference of interaction networks and proteins (Chapters 14–18).

Each chapter ends with exercises. They include the course homework (marked **HW**), whose solutions are worked out in full.

1.7. Exercises

Exercise 1.1

(Scaling from a SAR) Suppose $S(A) = cA^z$ with $z = 0.25$. (a) By what factor does the number of species grow when the area grows tenfold? (b) If a habitat loses half its area, what fraction of species does the SAR predict to be lost? (c) Why is (b) only a rough estimate of extinctions?

Solution

(a) $10^{0.25} = 1.78$: a tenfold area holds about 78% more species. (b) $S(A/2)/S(A) = 2^{-0.25} = 0.84$, so about 16% of species are “lost”. (c) The SAR counts species *present* in a sample of area $A/2$; species that also live in the destroyed half survive there. The species that disappear are those *endemic* to the destroyed part, which the EAR measures (Chapter 4). Using the SAR overestimates extinctions caused by habitat loss.

Exercise 1.2

(Kleiber scaling) With $B \propto M^{3/4}$, compare the metabolic rate per unit mass of a 3 g shrew and a 3000 kg elephant. How does the energy needed per kilogram compare?

Solution

$B/M \propto M^{-1/4}$. The mass ratio is 10^6 , so the shrew needs $(10^6)^{1/4} \approx 32$ times more energy per kilogram than the elephant. Small animals live “faster”: with lifespan $\propto M^{1/4}$, the total energy spent per kilogram over a lifetime, $\propto M^{-1/4}M^{1/4}$, is roughly independent of size.

Exercise 1.3

(Fisher’s log-series) Suppose the number of species with n individuals is $\phi_n = \theta x^n/n$, with $0 < x < 1$. Compute the number of species S and individuals J . Show that for $J \gg \theta$, $S \simeq \theta \ln(1 + J/\theta)$.

Solution

$S = \sum_{n \geq 1} \theta x^n / n = -\theta \ln(1 - x)$ and $J = \sum_{n \geq 1} n \phi_n = \theta \sum_{n \geq 1} x^n = \theta \frac{x}{1-x}$. From the second, $1 - x = \theta / (J + \theta)$, so $S = \theta \ln(1 + J/\theta)$, exactly. Species richness grows only logarithmically with the number of sampled individuals. θ , the *fundamental biodiversity number*, sets the rate. Neutral theory gives it a meaning (Chapter 3).

Exercise 1.4

(Fold bifurcation) For $\dot{x} = r + x^2$, find the equilibria and their stability as a function of r . Show that the relaxation time near the stable equilibrium diverges as $r \rightarrow 0^-$, and give the exponent.

Solution

For $r < 0$ the equilibria are $x_{\pm} = \pm\sqrt{-r}$. The derivative of the right-hand side is $2x$, so $x_- = -\sqrt{-r}$ is stable (eigenvalue $-2\sqrt{-r}$) and x_+ unstable. For $r > 0$ there are none. Linearising, perturbations decay as $e^{-2\sqrt{-r}t}$, with relaxation time $\tau = 1/(2\sqrt{|r|})$, which diverges with exponent $1/2$. This is critical slowing down. Any noise-driven fluctuations then have a variance $\propto \tau$ that grows as the fold approaches: an early-warning signal.

Exercise 1.5

(Hysteresis in the grazing model) In May's grazing model set $r = N_0 = 1$, so $\dot{N} = N(1 - N/K) - cN^2/(1 + N^2)$. (a) Show that the positive equilibria satisfy $c = c(N) \equiv (1 - N/K)(1 + N^2)/N$. (b) For $K = 1$ and $K = 10$, find the range of c with three positive equilibria.

Solution

(a) Dividing $\dot{N} = 0$ by $N > 0$ gives $1 - N/K = cN/(1 + N^2)$. Each horizontal line $c = \text{const}$ crosses the curve $c(N)$ once per equilibrium. (b) For $K = 1$, $c(N)$ is monotonically decreasing on $(0, 1)$: one equilibrium for every c , and no bistability. For $K = 10$, $c(N)$ has a local minimum $c_1 \approx 1.79$ (at $N \approx 1.14$) and a local maximum $c_2 \approx 2.60$ (at $N \approx 4.78$). For $c_1 < c < c_2$ there are three positive equilibria: the lowest and highest are stable, the middle one unstable. Alternative stable states need a separation of scales between the carrying capacity K and the saturation scale N_0 of consumption.

1.8. Sources

Prof. S. Suweis, *PMLS Course Notes* (2022–23, by G. Cataldi, F. Manzali, L. Rosset), Preface and Chapter 1; Lecture 1 slides (course overview, population dynamics). R. M. May, *Thresholds and breakpoints in ecosystems with a multiplicity of stable states*, *Nature* **269**, 471 (1977). M. Scheffer et al., *Early-warning signals for critical transitions*, *Nature* **461**, 53 (2009). I. Volkov, J. R. Banavar, S. P. Hubbell, A. Maritan, *Neutral theory and relative species abundance in ecology*, *Nature* **424**, 1035 (2003). G. B. West, J. H. Brown, B. J. Enquist, *Science* **276**, 122 (1997); J. R. Banavar, A. Maritan, A. Rinaldo, *Nature* **399**, 130 (1999).

2. Dynamics of a single species

We start from the simplest living system that can be modelled quantitatively: one population of bacteria. The chapter has two halves. The deterministic half derives exponential and logistic growth from a symmetry argument and from an explicit model of resource consumption. The stochastic half replaces the population by a random variable, writes its master equation, and develops the tools used throughout the course: stationary solutions and detailed balance, the Gillespie algorithm, the Fokker–Planck and Langevin limits, and first-passage problems such as the timing of cell division.

2.1. Basic microbiology

A bacterium such as *Escherichia coli* is a rod roughly $2\text{--}4\text{ }\mu\text{m}$ long and $0.8\text{ }\mu\text{m}$ wide. It is enclosed by an inner membrane, a cell wall and (in Gram-negative species) an outer membrane, and it contains a “soup” of DNA, RNA, proteins and small molecules. Nutrients from the surrounding medium cross the membranes and are converted into biomass by **metabolism**. When the cell reaches a critical size it divides into two daughters (binary fission). In rich medium at 37°C , *E. coli* divides every $\tau \approx 20$ minutes.

DNA is a sequence of nucleotides (bases A, C, G, T) organised into **genes**. To a first approximation each gene can be treated as a binary variable, *on* when it is transcribed and its protein produced, *off* otherwise. This picture becomes quantitative in Part III.

2.2. Deterministic growth laws

2.2.1. Growth laws from scale invariance

Let $N(t)$ be the number of individuals living in an area (or volume) A , and suppose the growth rate depends only on these two quantities, $\dot{N} = F(N, A)$. If we consider λ identical, independent copies of the system, the total population must obey the same law:

$$F(\lambda N, \lambda A) = \lambda F(N, A).$$

F is homogeneous of degree one. Choosing $\lambda = 1/A$ gives $F(N, A) = A F(N/A, 1) \equiv A f(\rho)$, so the **density** $\rho = N/A$ obeys an autonomous equation

$$\dot{\rho} = f(\rho).$$

Expanding f for small densities,

$$f(\rho) = f_0 + f_1\rho + \frac{1}{2}f_2\rho^2 + \dots,$$

each coefficient has a meaning:

- $f_0 > 0$ describes **immigration** into an open system; $f_0 = 0$ for a closed population, which must stay extinct once extinct.
- $f_1 = r$ is the per-capita growth rate at low density. If $f_1 > 0$ the empty state is unstable and the population grows exponentially.
- $f_2 < 0$ describes **self-limitation** (crowding, competition for resources).

2. Dynamics of a single species

Truncating at second order with $f_0 = 0$, $f_1 = r > 0$, $f_2 = -2r/K$ gives the logistic law. It is the *generic* behaviour of a closed population near extinction and near saturation, not an ad hoc guess.

Definition 2.1 (Exponential and logistic growth). **Exponential growth:** $\dot{N} = \mu N$, so $N(t) = N_0 e^{\mu t} = N_0 2^{t/T}$, with **doubling time** $T = \ln 2/\mu$. For *E. coli* in optimal conditions $\mu \simeq 2.1 \text{ h}^{-1}$, so $T \simeq 20 \text{ min}$.

Logistic growth (Verhulst, 1838): with **carrying capacity** K ,

$$\dot{N} = \mu N \left(1 - \frac{N}{K}\right), \quad N(t) = \frac{N_0 e^{\mu t}}{1 + \frac{N_0}{K}(e^{\mu t} - 1)}.$$

The solution follows by separation of variables, $\int \frac{dN}{N(1-N/K)} = \mu t$, and partial fractions. A bacterial batch culture fits it well: an exponential phase while $N \ll K$, a saturation (stationary) phase as $N \rightarrow K$. A real culture also has an initial *lag* phase, during which cells adapt to the new medium.

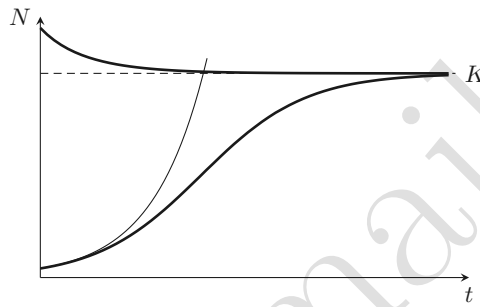


Figure: logistic growth from below and from above the carrying capacity (bold), and the exponential law with the same initial condition (thin).

2.2.2. Linear stability

An equilibrium N^* of $\dot{N} = F(N)$ is stable if small perturbations decay. Writing $N = N^* + \delta$ and expanding, $\dot{\delta} = F'(N^*)\delta$, so N^* is stable if $F'(N^*) < 0$, with relaxation time $1/|F'(N^*)|$. For the logistic law $F'(N) = \mu(1 - 2N/K)$: $N^* = 0$ has $F'(0) = \mu > 0$ (unstable) and $N^* = K$ has $F'(K) = -\mu$ (stable, relaxation time $1/\mu$).

For systems of equations $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$, the role of F' is played by the **Jacobian** $J_{ij} = \partial F_i / \partial x_j$ at the equilibrium. The equilibrium is stable iff all its eigenvalues have negative real part. This criterion will be the starting point of the random-matrix analysis of Chapter 6.

2.3. Consumer–resource models

The carrying capacity hides the resource. To make it explicit, let $R(t)$ be the amount of resource, $g(R)$ the uptake rate per cell, $\mu(R)$ the resource's own supply or growth, γ the **yield** (cells produced per unit of resource), c an uptake efficiency (set to 1 below) and d the death rate:

$$\dot{N} = (\gamma c g(R) - d)N, \quad \dot{R} = \mu(R) - c g(R)N.$$

Common uptake functions are linear, $g(R) = qR$ (mass action), and saturating, the **Monod** law

$$g(R) = \frac{q_M R}{K_M + R},$$

which states that a cell cannot process resource faster than q_M . K_M is the concentration at which uptake is half-maximal.

Definition 2.2 (Quasi-stationary approximation (QSA)). If the resource relaxes much faster than the population, then on the population's time scale R is always at its instantaneous equilibrium, found by setting $\dot{R} = 0$ at fixed N . Substituting it into $\dot{N}$ gives a closed equation for N alone. This is adiabatic elimination of a fast variable, and it is used throughout physics and biochemistry (Michaelis–Menten kinetics is the same idea).

Biotic resource (MacArthur). If the resource is itself a living population with logistic growth, $\mu(R) = r^*R(1 - R/k_c)$, and uptake is linear, then

$$\dot{N} = N(\gamma c R - d), \quad \dot{R} = R \left[r^* \left(1 - \frac{R}{k_c} \right) - cN \right].$$

The QSA with $R \neq 0$ gives $R^* = k_c(1 - cN/r^*)$, and substituting,

$$\dot{N} = N(\tilde{k} - \tilde{a}N), \quad \tilde{k} = \gamma c k_c - d, \quad \tilde{a} = \frac{\gamma c^2 k_c}{r^*},$$

a logistic law with growth rate $\mu = \tilde{k}$ and carrying capacity $K = \tilde{k}/\tilde{a}$. The carrying capacity is now *derived* from the resource's parameters. The population survives only if $\gamma c k_c > d$: the resource must be able to sustain more births than deaths.

Abiotic resource. If the resource is supplied at a constant rate, $\mu(R) = C$ (a chemostat), the QSA gives $g(R)N = C$ and then

$$\dot{N} = \gamma C - dN, \quad N^* = \frac{\gamma C}{d},$$

whatever the uptake function: the population relaxes *linearly* to N^* with time constant $1/d$, not logistically. The uptake function only fixes the resource level at equilibrium, through $\gamma g(R^*) = d$:

$$R_{\text{linear}}^* = \frac{d}{\gamma q}, \quad R_{\text{Monod}}^* = \frac{K_M d}{\gamma q_M - d}.$$

With Monod uptake a population can persist only if $\gamma q_M > d$, i.e. if its maximal growth beats death. The QSA also requires $C/N < q_M$: a small population cannot consume all the supplied resource, which then accumulates.

2.4. Stochastic population dynamics

Populations are made of discrete individuals, and births and deaths are random events. Deterministic equations describe the *mean* behaviour of large populations. Near extinction, or in small populations, **demographic noise** dominates.

i Rates and probabilities

A process occurring at rate r produces, in a time t , a Poisson number of independent events: $\Pr[k \text{ events}] = \frac{(rt)^k}{k!} e^{-rt}$. In a short interval dt the probability of one event is $r dt$ and of two or more is $O(dt^2)$. Rates are not probabilities: they have dimensions of inverse time and can exceed 1.

2.4.1. The birth–death master equation

Let $P_n(t)$ be the probability of having n individuals at time t . With birth and death rates

$$W^+(n) = T(n \rightarrow n+1), \quad W^-(n) = T(n \rightarrow n-1),$$

the probability balance over dt gives the **master equation**

$$\frac{dP_n}{dt} = W^+(n-1)P_{n-1} + W^-(n+1)P_{n+1} - [W^+(n) + W^-(n)]P_n,$$

a gain–loss equation: the first two terms bring probability into state n , the last takes it out. More generally, for a process with transition rates $W_{n'n}$ from n to n' ,

$$\frac{dP_n}{dt} = \sum_{n'} [W_{nn'}P_{n'} - W_{n'n}P_n].$$

A logistic population with immigration has

$$W^+(n) = bn + m, \quad W^-(n) = d(n)n = \frac{b}{K}n^2.$$

Births are multiplicative ($\propto n$) plus an immigration rate m that removes the absorbing state $n = 0$. Deaths increase with crowding. The deterministic limit, $\dot{\bar{n}} = W^+(\bar{n}) - W^-(\bar{n}) = b\bar{n}(1 - \bar{n}/K) + m$, recovers the logistic law.

2.4.2. Probability current and stationary solutions

Write the one-step master equation as a discrete continuity equation, $\dot{P}_n = J_{n-1} - J_n$, with the **probability current** between n and $n+1$

$$J_n = W^+(n)P_n - W^-(n+1)P_{n+1}.$$

At stationarity all J_n are equal. With a reflecting boundary at $n = 0$ ($W^-(0) = 0$), $J_{-1} = 0$, so every current vanishes: $J_n^* = 0$. This is **detailed balance**, $W^+(n)P_n^* = W^-(n+1)P_{n+1}^*$, which can be iterated:

$$P_n^* = P_0^* \prod_{z=1}^n \frac{W^+(z-1)}{W^-(z)}, \quad P_0^* = \left[1 + \sum_{n \geq 1} \prod_{z=1}^n \frac{W^+(z-1)}{W^-(z)} \right]^{-1}.$$

Every one-step process with reflecting boundaries satisfies detailed balance at stationarity. Processes with several variables generally do not, and then stationary states carry non-zero probability currents (a hallmark of non-equilibrium systems, Chapter 15).

Proposition 2.1 (Stationary stochastic logistic). *For $W^+(n) = bn + m$, $W^-(n) = bn^2/K$,*

$$P_n^* = P_0^* \frac{\Gamma(n + m/b)}{\Gamma(m/b)} \frac{K^n}{(n!)^2} \underset{m \ll b}{\simeq} P_0^* \frac{m}{b} \frac{K^n}{n n!} \quad (n \geq 1).$$

Proof. $\prod_{z=1}^n W^+(z-1) = b^n \Gamma(n + m/b)/\Gamma(m/b)$ and $\prod_{z=1}^n W^-(z) = (b/K)^n (n!)^2$. For small m/b , $\Gamma(n + \varepsilon)/\Gamma(\varepsilon) \simeq \varepsilon(n-1)!$. $\square$

The distribution $K^n/(n n!)$ is peaked near $n \simeq K$ with width $\simeq \sqrt{K}$. Unlike the deterministic model, the population can exceed the carrying capacity: for $K = 50$, $b = 1$, $m = 0.1$, the mean is 49.1, the standard deviation 7.1, and $P(n > K) \simeq 0.41$.

2.4.3. Moments and the failure of mean-field closure

Multiplying the master equation by n and summing,

$$\frac{d\langle n \rangle}{dt} = \langle W^+(n) - W^-(n) \rangle = b\langle n \rangle + m - \frac{b}{K}\langle n^2 \rangle.$$

Because the death rate is nonlinear, the mean depends on the second moment, the second moment on the third, and so on. The deterministic equation is recovered only by the **mean-field** approximation $\langle n^2 \rangle \simeq \langle n \rangle^2$, which neglects fluctuations, $\langle n^2 \rangle - \langle n \rangle^2 = O(K)$. Their relative size $O(K^{-1/2})$ vanishes for large systems. This is the first instance of the **system-size expansion** developed below and in Chapter 5.

2.4.4. The Gillespie algorithm

Master equations are rarely solvable, but they can be sampled exactly. Consider reactions $j = 1, \dots, M$ with propensities $a_j(\mathbf{x})$ (rates in state $\mathbf{x}$) and state changes $\mathbf{v}_j$, and let $a_0 = \sum_j a_j$. The probability that nothing happens in $[t, t + \tau]$ satisfies $P_0(\tau + d\tau) = P_0(\tau)(1 - a_0 d\tau)$, so $P_0(\tau) = e^{-a_0 \tau}$. The probability that the next event is reaction j , occurring in $[t + \tau, t + \tau + d\tau]$, is therefore

$$P(\tau, j) d\tau = a_j e^{-a_0 \tau} d\tau = \underbrace{a_0 e^{-a_0 \tau} d\tau}_{\text{when}} \cdot \underbrace{\frac{a_j}{a_0}}_{\text{which}}.$$

The waiting time and the identity of the reaction are independent.

- Definition 2.3** (Gillespie's stochastic simulation algorithm). 1. Initialise $\mathbf{x}(0)$ and $t = 0$.
2. Compute all propensities $a_j(\mathbf{x})$ and $a_0 = \sum_j a_j$.
 3. Draw the waiting time $\tau \sim \text{Exp}(a_0)$, e.g. $\tau = -\ln u_1 / a_0$ with u_1 uniform on $(0, 1)$.
 4. Draw u_2 uniform and choose the smallest j with $\sum_{k \leq j} a_k > u_2 a_0$.
 5. Update $t \rightarrow t + \tau$, $\mathbf{x} \rightarrow \mathbf{x} + \mathbf{v}_j$, and return to step 2.

Each trajectory is an exact sample of the process described by the master equation. Averaging many trajectories reconstructs $P_n(t)$. For the logistic process with $K = 50$, starting from $n = 5$, the average of 300 trajectories at $t = 2, 4, 8$ is 22.0, 40.8, 48.4, against the deterministic 22.5, 42.9, 49.9. The stochastic mean lags slightly behind and saturates slightly below K , as predicted by the moment equation (the $\langle n^2 \rangle$ term exceeds $\langle n \rangle^2$).

Example 2.1 (SIS epidemic)

In a population of N individuals, I infected and $S = N - I$ susceptible, infected individuals recover at rate μ ($I \rightarrow S$) and infect susceptibles on contact ($I + S \xrightarrow{\lambda} 2I$). The rates are

$$W^-(I) = \mu I, \quad W^+(I) = \lambda \frac{I(N - I)}{N}.$$

$I = 0$ is absorbing, so $P(I, t) \rightarrow \delta_{I,0}$ as $t \rightarrow \infty$. Yet for $\lambda > \mu$ the deterministic dynamics has an endemic equilibrium $I^* = N(1 - \mu/\lambda)$. The stochastic system sits there for a time that grows exponentially with N before fluctuations drive it to extinction: a **metastable** (quasi-stationary) state, the relevant one in practice.

2.5. From master equation to Fokker–Planck and Langevin

2.5.1. The continuum limit

When populations are large, let Ω be a system size (the carrying capacity K , the total population N) and $x = n/\Omega$ a density, which becomes continuous as $\Omega \rightarrow \infty$. For rates of the form $W^\pm(n) = \Omega w^\pm(x)$ and $P_n(t) = P(x, t)/\Omega$, the master equation reads

$$\partial_t P(x, t) = \Omega [(\mathbb{E}^- - 1) w^+(x) P(x, t) + (\mathbb{E}^+ - 1) w^-(x) P(x, t)],$$

with the shift operators $\mathbb{E}^\pm f(x) = f(x \pm 1/\Omega)$. Expanding $\mathbb{E}^\pm - 1 = \pm \frac{1}{\Omega} \partial_x + \frac{1}{2\Omega^2} \partial_x^2 + \dots$ (Kramers–Moyal expansion) and truncating at second order gives the **Fokker–Planck equation**

$$\partial_t P = -\partial_x [A(x)P] + \frac{1}{2} \partial_x^2 [B(x)P], \quad A = w^+ - w^-, \quad B = \frac{w^+ + w^-}{\Omega}.$$

The **drift** A is the deterministic law. The **diffusion** B is small, $O(1/\Omega)$, and measures demographic noise. Births and deaths both *add* to it: noise comes from the total event rate, not the net one.

For the stochastic logistic ($\Omega = K$, $m = 0$), $w^+ = bx$ and $w^- = bx^2$:

$$A(x) = bx(1 - x), \quad B(x) = \frac{b}{K} x(1 + x).$$

With immigration, $A = b[x(1 - x) + \frac{m}{bK}]$ and $B = \frac{b}{K^2} [Kx(1 + x) + \frac{m}{b}]$, which no longer vanishes at $x = 0$: immigration removes the absorbing state.

2.5.2. The Langevin equation

The same process can be described by a stochastic differential equation for a single trajectory,

$$\dot{x} = A(x) + \sqrt{B(x)} \xi(t), \quad \langle \xi(t) \rangle = 0, \quad \langle \xi(t) \xi(t') \rangle = \delta(t - t'),$$

whose ensemble of solutions has density $P(x, t)$ obeying the Fokker–Planck equation (in the Itô interpretation). For the logistic case

$$\dot{x} = bx(1 - x) + \sqrt{\frac{bx(1 + x)}{K}} \xi(t).$$

The noise amplitude depends on the state: it is **multiplicative noise**. Numerically, the Euler–Maruyama scheme

$$x_{t+\Delta t} = x_t + A(x_t) \Delta t + \sqrt{B(x_t)} \Delta t \eta_t, \quad \eta_t \sim \mathcal{N}(0, 1),$$

evaluates the noise amplitude at the beginning of the step (Itô). Evaluating it at the midpoint gives the Stratonovich interpretation, which corresponds to a *different* Fokker–Planck equation with an extra “noise-induced drift” $\frac{1}{4} B'(x)$. This is the **Itô–Stratonovich dilemma**: for multiplicative white noise the equation alone is ambiguous, and the interpretation is part of the model. Equations derived from master equations, like ours, are Itô.

2.5.3. Random walk and diffusion

The symmetric random walk, $W^\pm(n) = \lambda$, is the simplest case. With $x = n \Delta x$,

$$\partial_t P(x, t) = \lambda [P(x - \Delta x) + P(x + \Delta x) - 2P(x)] \simeq \lambda \Delta x^2 \partial_x^2 P,$$

the **diffusion equation** $\partial_t P = D \partial_x^2 P$, provided $\lambda \rightarrow \infty$ and $\Delta x \rightarrow 0$ with $D = \lambda \Delta x^2$ fixed. Its Langevin form is $\dot{x} = \sqrt{2D} \xi(t)$, with Gaussian propagator $P(x, t) = (4\pi Dt)^{-1/2} e^{-x^2/4Dt}$.

2.5.4. Stationary solutions of the Fokker–Planck equation

Writing Fokker–Planck as a continuity equation, $\partial_t P = -\partial_x J$, with current $J = AP - \frac{1}{2} \partial_x (BP)$, a stationary state has constant J . With a reflecting boundary $J = 0$ everywhere, so $\partial_x (BP) = 2AP$, which integrates to

$$P^*(x) = \frac{\mathcal{N}}{B(x)} \exp\left(2 \int^x \frac{A(y)}{B(y)} dy\right).$$

Proposition 2.2 (Linear birth and death: a Gamma distribution). *For linear rates $W^+(n) = bn + m$, $W^-(n) = dn$ with $d > b$ (decline compensated by immigration), work directly with n : $A(n) = m - (d-b)n$, $B(n) \simeq (b+d)n$. Then*

$$P^*(n) \propto n^{\frac{2m}{b+d}-1} e^{-\frac{2(d-b)}{b+d} n},$$

a **Gamma distribution**. The exact discrete solution of the master equation is the **negative binomial**

$$P_n^* = \frac{\Gamma(n + m/b)}{\Gamma(m/b) n!} \left(\frac{b}{d}\right)^n \left(1 - \frac{b}{d}\right)^{m/b},$$

whose continuum limit is a Gamma distribution with shape m/b and rate $1 - b/d$. The two agree when $b \simeq d$, the regime where diffusion is accurate.

Proof. With $A/B = \frac{m}{(b+d)n} - \frac{d-b}{b+d}$, $2 \int A/B = \frac{2m}{b+d} \ln n - \frac{2(d-b)}{b+d} n$, and dividing by $B \propto n$ gives the result. The discrete one follows from the product formula with $W^+(z-1)/W^-(z) = (b(z-1) + m)/(dz)$. $\square$

The negative binomial is the cornerstone of neutral theory (Chapter 3). There, each species follows exactly this linear birth–death–immigration process, and the near-critical case $b \rightarrow d$ produces the log-series of species abundances.

2.6. Stochastic doubling and the timing of division

So far each cell divided at a constant rate. Real *E. coli* cells grow exponentially in size, $x(t) = x_0 e^{\alpha t}$, and divide at a size x_f that varies from generation to generation (Osella et al., 2014). Division can be modelled as a random event with a **hazard rate** $h_d(x, t)$: the probability of dividing in $[t, t + dt]$, given no division so far, is $h_d dt$.

Let $P_0(t | x_0)$ be the probability that a cell born with size x_0 has not divided by time t (the *survival function*). Not dividing up to $t + dt$ means not dividing up to t and then not in $[t, t + dt]$:

$$P_0(t + dt) = P_0(t)[1 - h_d(x(t), t) dt] \implies P_0(t | x_0) = \exp\left(-\int_0^t h_d(x(s), s) ds\right).$$

The distribution of division times is then

$$P(t | x_0) = h_d(x(t), t) P_0(t | x_0) = -\frac{d}{dt} P_0(t | x_0).$$

Because size grows monotonically, the same can be written in terms of size, with $h_d^*(x) dx = h_d dt$ and $dx/dt = \alpha x$:

$$P(x_f | x_0) = h_d^*(x_f) \exp\left(-\int_{x_0}^{x_f} h_d^*(s) ds\right).$$

The functional form of h_d distinguishes the classical models of cell-size control:

2. Dynamics of a single species

- a **timer** (h_d depends on time since birth only),
- a **sizer** (h_d^* depends on current size only),
- an **adder** (h_d^* depends on the size *added* since birth, $x - x_0$).

The data favour a “concerted” control, close to the adder, which makes the added size $x_f - x_0$ independent of x_0 and keeps the size distribution stable over generations. A constant hazard gives back exponentially distributed division times: the Poisson birth process of the master equation.

2.7. Exercises

Exercise 2.1

(HW 1.1) Linear stability. (a) Perform the linear stability analysis of the logistic equation. (b) Do the same for the consumer–resource model with abiotic resource, $\dot{N} = (\gamma q R - d)N$, $\dot{R} = C - qRN$.

Solution

- (a) $F(N) = \mu N(1 - N/K)$, $F'(N) = \mu(1 - 2N/K)$. $N^* = 0$: $F' = \mu > 0$, unstable. $N^* = K$: $F' = -\mu < 0$, stable, with relaxation time $1/\mu$.
- (b) The interior equilibrium is $R^* = d/(\gamma q)$, $N^* = \gamma C/d$. The Jacobian there is

$$J = \begin{pmatrix} 0 & \gamma q N^* \\ -q R^* & -q N^* \end{pmatrix},$$

with $\text{tr } J = -q N^* < 0$ and $\det J = \gamma q^2 R^* N^* = qC > 0$. Both eigenvalues have negative real part, so the equilibrium is stable. It is a stable node if $(\text{tr } J)^2 > 4 \det J$, i.e. $q^2 N^{*2} > 4qC$, which reads $q\gamma^2 C/d^2 > 4$. Otherwise it is a stable focus, approached through damped oscillations. With the numbers of HW 1.2, $q\gamma^2 C/d^2 = 0.2 \times 0.64 \times 5/0.16 = 4$: the system sits exactly at the boundary.

Exercise 2.2

(HW 1.2) The QSA in the abiotic case. Apply the QSA to the abiotic consumer–resource model with linear uptake. Check the analytical result with simulations, e.g. with $\gamma = 0.8$, $q = 0.2$, $C = 5$, $d = 0.4$. How does the solution change with Monod uptake ($q_M = 3$, $K_M = 8$)?

Solution

QSA: $\dot{R} = 0$ gives $R = C/(qN)$; then $\dot{N} = (\gamma q C/(qN) - d)N = \gamma C - dN$, so $N(t) = N^* + (N_0 - N^*)e^{-dt}$ with $N^* = \gamma C/d = 10$, and $R^* = C/(qN^*) = d/(\gamma q) = 2.5$. Simulating the full system from, e.g., $N_0 = 1$, $R_0 = 0$, the population relaxes to 10 and the resource to 2.5. The match improves the faster the resource relaxes (large qN). With Monod uptake, QSA gives $g(R)N = C$ and again $\dot{N} = \gamma C - dN$: **the same** $N^* = 10$. Only the resource level changes, $R^* = K_M d/(\gamma q_M - d) = 8 \times 0.4/2 = 1.6$. With Monod the QSA fails when $N < C/q_M = 5/3$: the cells cannot take up all the supply and R accumulates, so early transients from small N_0 differ from the QSA prediction.

Exercise 2.3

(HW 1.3, optional) Stochastic versus deterministic logistic growth. Simulate the birth–death process with $W^+ = bn + m$, $W^- = bn^2/K$ ($K = 50$, $b = 1$, $m = 0.1$) with the Gillespie algorithm and

compare the mean trajectory with the deterministic solution. Compare the long-time histogram with the exact stationary distribution.

Solution

Averaging 300 trajectories from $n_0 = 5$: $\langle n \rangle = 22.0, 40.8, 48.4, 49.1$ at $t = 2, 4, 8, 20$, against the deterministic $22.5, 42.9, 49.9, 50.0$. The stochastic mean is slightly lower because $\frac{d\langle n \rangle}{dt}$ involves $\langle n^2 \rangle > \langle n \rangle^2$, which enhances death. At long times the histogram matches $P_n^* \propto \frac{\Gamma(n+m/b)}{(n!)^2} K^n$: mean 49.1 , standard deviation $7.1 \approx \sqrt{K}$, and 41% of the time $n > K$. Individual trajectories fluctuate by $O(\sqrt{K})$ around K , while the deterministic solution sits exactly at K .

Exercise 2.4

(Detailed balance) Show that for any one-step process on $\{0, 1, 2, \dots\}$ with $W^-(0) = 0$, the stationary currents $J_n = W^+(n)P_n - W^-(n+1)P_{n+1}$ vanish. Why does this fail for the symmetric random walk on a ring of L sites with a bias ($W^+ \neq W^-$)?

Solution

At stationarity $\dot{P}_n = J_{n-1} - J_n = 0$, so J_n is independent of n . At the boundary $J_{-1} = 0$ (nothing flows below 0), hence all $J_n = 0$. On a ring there is no boundary: the stationary state is uniform, $P_n = 1/L$, and carries a constant current $J = (W^+ - W^-)/L \neq 0$. It is a non-equilibrium steady state, which does not satisfy detailed balance.

Exercise 2.5

(Linear birth–death–immigration) For $W^+(n) = bn + m$, $W^-(n) = dn$, $d > b$: (a) find the stationary distribution from the product formula; (b) compute its mean and compare with the deterministic equilibrium; (c) what happens as $b \rightarrow d$?

Solution

(a) $P_n^* = P_0^* \prod_{z=1}^n \frac{b(z-1)+m}{dz} = P_0^* \frac{\Gamma(n+m/b)}{\Gamma(m/b)n!} \left(\frac{b}{d}\right)^n$, a negative binomial with $P_0^* = (1 - b/d)^{m/b}$. (b) Its mean is $\frac{m/b}{1-b/d} = \frac{m}{d-b}$, equal to the deterministic equilibrium of $\dot{n} = m - (d-b)n$: for linear rates the mean obeys the deterministic equation exactly. (c) As $b \rightarrow d$ the mean diverges and the distribution develops a long tail $P_n \propto n^{m/b-1}(b/d)^n$. For $m/b \rightarrow 0$ this is $\propto x^n/n$, Fisher's log-series (Chapter 3).

Exercise 2.6

(Fokker–Planck coefficients) For the logistic process with immigration, derive $A(x)$ and $B(x)$ by the Kramers–Moyal expansion and write the associated Langevin equation. Show that $x = 0$ is no longer absorbing.

Solution

With $x = n/K$: $w^+(x) = W^+/K = bx + m/K$ and $w^-(x) = W^-/K = bx^2$. Then $A = w^+ - w^- = b(x(1-x) + \frac{m}{bK})$ and $B = (w^+ + w^-)/K = \frac{b}{K^2}(Kx(1+x) + \frac{m}{b})$. The Langevin equation is $\dot{x} = A(x) + \sqrt{B(x)}\xi$. At $x = 0$: $A(0) = m/K > 0$ and $B(0) = m/K^2 > 0$, so the drift pushes the

population away from zero. Without immigration $A(0) = B(0) = 0$ and extinction is permanent.

Exercise 2.7

(Division with constant hazard) (a) If $h_d = k$ is constant, find the distribution of division times and its mean. (b) For a sizer with $h_d^*(x) = \kappa(x - x_c)^2$ for $x > x_c$ (0 otherwise), and exponential growth from $x_0 < x_c$, find $P(x_f | x_0)$ and explain why it does not depend on x_0 .

Solution

(a) $P_0(t) = e^{-kt}$ and $P(t) = ke^{-kt}$, exponential with mean $1/k$: memoryless division, as in the birth process. (b) $P_0(x_f) = \exp(-\int_{x_c}^{x_f} \kappa(s - x_c)^2 ds) = e^{-\kappa(x_f - x_c)^3/3}$, so $P(x_f) = \kappa(x_f - x_c)^2 e^{-\kappa(x_f - x_c)^3/3}$ for $x_f > x_c$, a Weibull distribution. Because the hazard is zero below x_c , the birth size only shifts when x_c is reached, not the distribution of division sizes. This is the defining property of a sizer: the final size carries no memory of the initial one. An adder instead gives $x_f - x_0$ independent of x_0 .

2.8. Sources

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3. Many species: neutral theory and upscaling

From one population we move to communities of hundreds or thousands of species, such as the trees of a tropical forest. Writing a model with species-specific parameters for each of them is hopeless. Yet community-level patterns, above all the distribution of species abundances, are remarkably regular. This chapter shows that a minimal model, in which every species obeys the *same* stochastic birth–death dynamics (**neutral theory**), predicts these patterns. Its solution, the negative binomial distribution, has a scaling property that lets us infer the biodiversity of a whole forest from a small sample. Real forests turn out to sit close to a critical point of the model.

3.1. Species abundance distributions

A census of a plot records the abundance n_i of every species $i = 1, \dots, S$. Two equivalent summaries are used:

- the **relative species abundance** (RSA) ϕ_n : the number of species with exactly n individuals;
- the **species abundance distribution** (SAD) $P(n) = \phi_n/S$: the probability that a randomly chosen species has n individuals.

Empirical RSAs are dominated by **rare** species: many species have one or two individuals, and a few are hyper-abundant. Plotted on a linear scale they are hollow curves with very sparse tails. Ecologists therefore use **Preston plots**, which group abundances into octaves $[2^k, 2^{k+1})$ and plot the number of species per octave against $k = \log_2 n$.

⚠ Be careful with binning

An octave collects many more abundance values at large k than at small k . The count in octave k is $\sum_{n=2^k}^{2^{k+1}-1} \phi_n \simeq 2^k \phi_{2^k} \ln 2$, which is $n \phi_n$ rather than ϕ_n . A log-series $\phi_n = \theta x^n/n$, monotonically decreasing in n , therefore looks *flat* per octave (each octave holds about $\theta \ln 2$ species) up to $n \sim 1/(1-x)$, and then drops. A genuine interior maximum in a Preston plot indicates that $n \phi_n$ increases, which the log-series cannot produce but a negative binomial with $r > 1$ can. Always state the binning before interpreting a shape.

3.2. From individual dynamics to the negative binomial

3.2.1. Density-dependent birth and death

Let $P_n^s(t)$ be the probability that species s has n individuals. If each species follows a one-step process with rates b_n^s and d_n^s , the master equation is

$$\frac{dP_n^s}{dt} = b_{n-1}^s P_{n-1}^s + d_{n+1}^s P_{n+1}^s - (b_n^s + d_n^s) P_n^s.$$

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The per-capita rates may depend on abundance. Expanding them for large n ,

$$\frac{b_n^s}{n} = b_s + \frac{r_s}{n} + O(n^{-2}), \quad \frac{d_n^s}{n} = d_s + \frac{d_1}{n} + O(n^{-2}).$$

Such **density dependence** is observed: seedlings close to adults of the same species suffer more from specialised pests (the *Janzen–Connell effect*), so being rare is an advantage. The simplest choice keeping the leading terms is

$$b_n^s = b_s n + r_s, \quad d_n^s = d_s n,$$

exactly the linear birth–death–immigration process of Chapter 2. The constant r_s plays the role of immigration (or speciation) and keeps rare species from going extinct.

Treating each species independently amounts to replacing the joint distribution $P(n_1, \dots, n_S)$ by a product $\prod_s P^s(n_s)$: interactions are absorbed into effective, mean-field rates.

3.2.2. The stationary solution

By the product formula of Chapter 2,

$$P_n^{s*} = P_0^{s*} \prod_{j=1}^n \frac{b_{j-1}}{d_j} = P_0^{s*} \frac{\Gamma(n+r)}{\Gamma(r) n!} \xi^n, \quad \xi = \frac{b}{d}, \quad r = \frac{r_s}{b_s},$$

dropping the label s . This is the **negative binomial** distribution

$$P_n^{\text{NB}}(r, \xi) = \binom{n+r-1}{n} \xi^n (1-\xi)^r, \quad n = 0, 1, 2, \dots,$$

which requires $\xi < 1$ (deaths outweigh births) and $r > 0$. Its generating function is particularly simple:

$$G(z) = \sum_{n \geq 0} P_n^{\text{NB}} z^n = \left(\frac{1-\xi}{1-\xi z} \right)^r,$$

from which the mean $\langle n \rangle = G'(1) = \frac{r\xi}{1-\xi}$ and the variance $\frac{r\xi}{(1-\xi)^2}$ follow. In a census only species that are present are counted, so the RSA is the NB **truncated** to $n \geq 1$:

$$P_n^*(r, \xi) = \frac{1}{1 - (1-\xi)^r} \binom{n+r-1}{n} \xi^n (1-\xi)^r, \quad n \geq 1.$$

Proposition 3.1 (The log-series as a critical limit). *As $r \rightarrow 0$ the truncated NB tends to Fisher's log-series*

$$P_n^* \rightarrow \frac{1}{-\ln(1-\xi)} \frac{\xi^n}{n}.$$

Proof. $\binom{n+r-1}{n} = \frac{r(r+1)\dots(r+n-1)}{n!} \simeq \frac{r(n-1)!}{n!} = \frac{r}{n}$ and $1 - (1-\xi)^r \simeq -r \ln(1-\xi)$ as $r \rightarrow 0$. $\square$

3.3. Neutral theory

3.3.1. The neutrality hypothesis

Fitting species-specific (r_s, ξ_s) to thousands of species would need thousands of parameters. Hubbell's **neutral theory** (2001) assumes instead that species on the same trophic level (e.g. all tree species of a

forest) are *demographically equivalent*: they have the same per-capita birth, death, dispersal and speciation rates. Different species are then different realisations of the *same* stochastic process, and their abundances differ only by chance (**ecological drift**), not by fitness.

Under neutrality the S species are S independent draws from one distribution, so the expected RSA is simply $\phi_n = S P_n^*(r, \xi)$, with two parameters for the whole community. Fits of the NB to the RSA of large tropical forest plots (Barro Colorado Island, Pasoh, Korup, Lambir, Yasuní, Sinharaja) are excellent. Neutrality is surprising, since species obviously differ, and controversial. Its success says that, for the *abundance distribution*, differences between species average out in very diverse communities. It is a null model, and departures from it point to the biology that matters.

3.3.2. The voter model with speciation

Neutral theory can also be formulated in a spatially explicit, zero-sum way. A landscape (say, a lattice) is saturated by J individuals. At each step a random individual dies and is replaced either by the offspring of a neighbour (probability $1 - \nu$) or by an individual of a brand-new species (speciation, probability ν). Without speciation this is the **voter model** of statistical physics: domains of the same “opinion” coarsen and one species eventually takes over. Speciation injects diversity and produces a statistically stationary community.

In the well-mixed (mean-field) version, the abundance of each species performs a birth–death walk with $b \lesssim d$, and new species enter at rate ν . The resulting metacommunity RSA is the log-series

$$\langle \phi_n \rangle = \theta \frac{x^n}{n}, \quad \theta \simeq \frac{2J\nu}{1-\nu},$$

where θ is Hubbell’s **fundamental biodiversity number** and $x = b/d \rightarrow 1$ for large communities. The mean number of species found among J individuals is $\langle S \rangle = \theta \ln(1 + J/\theta)$ (Ewens’ sampling formula).

3.3.3. A local community fed by a metacommunity

Real plots are open: they receive immigrants from a large surrounding **metacommunity**, the “continent” to the plot’s “island”. Consider an island of fixed size J (zero-sum dynamics). At each event an individual dies, and the vacancy is filled with probability $1 - m$ by local reproduction and with probability m by an immigrant drawn from the metacommunity, where species k has relative abundance μ_k/J_M . The rates for species k with n individuals on the island are

$$b_{n,k} = (1 - m) \left(1 - \frac{n}{J}\right) \frac{n}{J-1} + m \left(1 - \frac{n}{J}\right) \frac{\mu_k}{J_M},$$

$$d_{n,k} = (1 - m) \frac{n}{J} \frac{J-n}{J-1} + m \frac{n}{J} \left(1 - \frac{\mu_k}{J_M}\right).$$

In $b_{n,k}$ an individual of *another* species dies (probability $1 - n/J$) and is replaced by a local offspring of k or by an immigrant of k . In $d_{n,k}$ an individual of k dies and is replaced by another species. With

$$\lambda_k = \frac{m}{1-m} (J-1) \frac{\mu_k}{J_M}, \quad \theta_k = J + \frac{m}{1-m} (J-1) \left(1 - \frac{\mu_k}{J_M}\right),$$

the rates become $b_{n,k} = \frac{1-m}{J} \frac{J-n}{J-1} (n + \lambda_k)$ and $d_{n,k} = \frac{1-m}{J-1} \frac{n}{J} (\theta_k - n)$, and the stationary distribution is

$$P_{n,k}^* = P_{0,k} \frac{J!}{n! (J-n)!} \frac{\Gamma(n + \lambda_k)}{\Gamma(\lambda_k)} \frac{\Gamma(\theta_k - n)}{\Gamma(\theta_k)}.$$

On a log scale it resembles a log-normal, but it comes from a mechanism, which is why Volkov et al. (2003) prefer it to the phenomenological log-normal. The island RSA follows by averaging over the metacommunity

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abundances, $\phi_n = S_M \sum_k P_{n,k}^* P(\mu_k)$. Strong dispersal limitation (small m) depletes rare species locally and produces the hump of empirical Preston plots.

3.4. Upscaling: from a sample to the whole forest

3.4.1. The problem

The Amazon holds about 10^{11} trees; no census can count them. Typically, plots covering a fraction $p = a/A \ll 1$ of a forest of area A are sampled. How many species does the whole forest contain? Rare species are mostly missed by small samples, so the RSA is **scale dependent**. At small p it is a hollow curve dominated by singletons. At larger p , species accumulate individuals, mass moves out of the lowest abundance classes, and the curve develops an interior maximum. Linear extrapolation of S fails.

3.4.2. Binomial sampling and self-similarity

Assume individuals are spread over the forest independently of each other (**random placement**, or well-mixed). If a species has n individuals in total, the number k found in a region covering a fraction p of the area is binomial:

$$P_p(k | n) = \binom{n}{k} p^k (1-p)^{n-k}.$$

The abundance distribution at scale p is then

$$P_p(k) = \sum_{n \geq k} \binom{n}{k} p^k (1-p)^{n-k} P_n(r, \xi).$$

Theorem 3.2 (Self-similarity of the negative binomial). *If abundances at scale 1 are $NB(r, \xi)$, then abundances at scale p are $NB(r, \xi_p)$ with the same r and*

$$\xi_p = \frac{p\xi}{1 - \xi(1-p)}.$$

Proof. Binomial thinning acts on generating functions as $G_p(z) = G(1-p+pz)$: each individual is kept with probability p . For the NB,

$$G_p(z) = \left(\frac{1-\xi}{1-\xi+\xi p-\xi p z} \right)^r = \left(\frac{1-\xi_p}{1-\xi_p z} \right)^r, \quad 1-\xi_p = \frac{1-\xi}{1-\xi(1-p)},$$

again an NB, with ξ_p as stated. $\square$

The NB is a fixed form under coarse-graining, like a renormalisation-group fixed point: only ξ flows with scale, r does not. Inverting,

$$\xi = \frac{\xi_p}{p + \xi_p(1-p)}.$$

Zero size probability of a species

At scale p there is a non-zero probability of having a zero size population species, since

$$P_p(k) \equiv NB(r, \xi_p)$$

Which normalization is:

$$\sum_{i=0}^n NB(r, \xi_p) = 1$$

3.4.3. Estimating the number of species

A species is missed by the sample with probability $P_p(0)$. If S^* species are observed at scale p , then $S^* = S/[1 - P_p(0)]$. For the NB truncated at scale 1, $P_p(0) = \frac{(1-\xi_p)^r - (1-\xi)^r}{1-(1-\xi)^r}$, hence

$$S = S^* \frac{1 - (1 - \xi)^r}{1 - (1 - \xi_p)^r}.$$

The recipe: fit a truncated NB to the sample RSA to get (r, ξ_p) ; compute ξ from self-similarity; compute S . The same machinery predicts the RSA and the species–area relationship at any scale $p' \in (p, 1]$.

The method was validated on computer-generated forests, where the truth is known, and on fully censused plots sub-sampled at random. It outperforms classical estimators based on the log-series or on non-parametric corrections (Chao). Applied to tropical forests (Tovo et al., 2017), it predicts, for instance, about 366 ± 15 tree species at Barro Colorado from 301 observed, and an Amazonian richness of order 10^4 species. The large number of hyper-rare species implied (*hyper-rarity*), together with the dominance of a few hundred species (*hyper-dominance*), is the main ecological outcome.

3.4.4. How much data is enough?

A bootstrap answers whether a sample is large enough. Upscale the sample from p^* to 1 (estimate S); generate synthetic samples at a scale p_{pred} by binomial thinning; upscale them again and compare the new estimate S' with S . The smallest p_{pred} for which $S' \simeq S$ within the desired precision is the scale that *encodes sufficient information*. If $p_{\text{pred}}/p^* > 1$ more fieldwork is needed; if it is below 1 the sample is more than sufficient.

3.5. Tropical forests are near a critical point

The relative fluctuations of a species' abundance are measured by the coefficient of variation

$$CV = \frac{\sqrt{\langle (n - \langle n \rangle)^2 \rangle}}{\langle n \rangle} = \frac{1}{\sqrt{r\xi}} \quad (\text{untruncated NB}),$$

and for the truncated NB it diverges in the limit $\xi \rightarrow 1$, $r \rightarrow 0$, where the distribution becomes the log-series with a cut-off $1/(1 - \xi) \rightarrow \infty$. In that limit the system has no characteristic abundance scale: births balance deaths ($b = d$), the hallmark of a critical point. Fits of (r, ξ) to real forests consistently give $r \lesssim 1$ and ξ extremely close to 1 ($\ln(1 - \xi) \sim -10$ to -20): tropical forests sit very near criticality, with large fluctuations of species abundances across forests.

3.6. A phenomenological renormalisation group

Random placement neglects spatial clustering: trees of the same species aggregate because seeds fall near their parents. Clustering is quantified by the **pair correlation function**

$$g(r) = \frac{\langle n_{\mathbf{x}} n_{\mathbf{y}} \rangle}{\langle n \rangle^2}, \quad r = |\mathbf{x} - \mathbf{y}|,$$

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where $n_{\mathbf{x}}$ is the abundance in a small cell at $\mathbf{x}$ and the average runs over positions and, by neutrality, over species. A spatially explicit neutral model with dispersal predicts

$$g(r) = 1 + \frac{1}{T} \left(\frac{\rho}{\lambda} \right)^2 K_0 \left(\frac{r}{\lambda} \right),$$

with K_0 the modified Bessel function of the second kind, λ a correlation length set by dispersal and speciation, and T, ρ model constants. Taking the continuum (Gamma) version of the NB as the RSA in a circle of radius R ,

$$P(n | \alpha, \beta) = \frac{1}{\beta \Gamma(\alpha)} \left(\frac{n}{\beta} \right)^{\alpha-1} e^{-n/\beta},$$

the parameters at each scale are fixed by matching the first two moments,

$$\alpha(R) = \frac{\langle N(R) \rangle^2}{\sigma^2(R)}, \quad \beta(R) = \frac{\sigma^2(R)}{\langle N(R) \rangle},$$

with $\langle N(R) \rangle = \pi R^2 \langle n \rangle$ and $\sigma^2(R)$ obtained by integrating g over pairs of points in the circle. The flow of (α, β) with R is an empirical renormalisation group: it predicts the RSA, and from it the SAR, at every scale, now including clustering (Azade et al., 2015).

3.7. Biodiversity from presence–absence data

Abundances are expensive to measure; presence/absence on a grid of cells is cheaper. Two summaries replace the RSA:

- the **relative species occupancy** (RSO): the number of species occupying c cells;
- the **species accumulation curve** (SAC): the mean number of species $\langle S_a \rangle$ found in randomly placed sub-areas of size $a < a^*$, with a^* the surveyed area.

Under the NB model the probability that a species is absent from a cell of relative area p is $P_p(0)$, computed above. The expected SAC is therefore

$$\langle S_a \rangle = S[1 - P_{a/A}(0)],$$

a closed form in (r, ξ, S) . Fitting it to the empirical SAC inside the surveyed area, and extrapolating to $a = A$, estimates total biodiversity from presence–absence data alone (Tovo et al., 2019). The fitted parameters also predict the RSO, which provides an independent check.

3.8. Exercises

Exercise 3.1

(Negative binomial from the master equation) For $b_n = bn + r_s$, $d_n = dn$, show that the stationary distribution is NB(r, ξ) with $r = r_s/b$, $\xi = b/d$. Compute its mean and variance from the generating function.

Solution

$P_n^* = P_0^* \prod_{j=1}^n \frac{b(j-1)+r_s}{dj} = P_0^* \frac{\xi^n}{n!} \prod_{j=1}^n (j-1+r) = P_0^* \frac{\Gamma(n+r)}{\Gamma(r)n!} \xi^n$. Normalisation gives $P_0^* = (1-\xi)^r$ by the binomial series $\sum_n \binom{n+r-1}{n} \xi^n = (1-\xi)^{-r}$. With $G(z) = ((1-\xi)/(1-\xi z))^r$: $G'(1) = \frac{r\xi}{1-\xi}$ and $G''(1) = \frac{r(r+1)\xi^2}{(1-\xi)^2}$, so the variance is $G'' + G' - G'^2 = \frac{r\xi}{(1-\xi)^2}$. The variance-to-mean ratio $1/(1-\xi) > 1$

makes the distribution over-dispersed compared to a Poisson.

Exercise 3.2

(Preston octaves of a log-series) For $\phi_n = \theta x^n/n$ with $x = 0.99$ and $\theta = 50$, compute the number of species in octaves $\{1\}, \{2, 3\}, \{4, \dots, 7\}, \dots$. Explain the shape.

Solution

For $x \rightarrow 1$ each octave $k \geq 1$ contains $\theta \sum_{n=2^k}^{2^{k+1}-1} 1/n \rightarrow \theta \ln 2 \simeq 34.7$ species. The first octave has $\theta x = 49.5$, the second $\theta(x^2/2 + x^3/3) \simeq 40.7$. With $x = 0.99$ the cut-off $1/(1-x) = 100$ makes octaves beyond $n \sim 100$ ($k \geq 7$) decay exponentially. The Preston plot is a plateau followed by a fall, not a hump: a monotone ϕ_n looks flat when binned in octaves.

Exercise 3.3

(Self-similarity) Verify by direct summation, for $r = 1$, that binomial sampling maps $\text{NB}(1, \xi)$ (a geometric distribution) into $\text{NB}(1, \xi_p)$. Then show that ξ_p decreases monotonically as p decreases, and interpret.

Solution

$P_n = (1-\xi)\xi^n$. $P_p(k) = \sum_{n \geq k} \binom{n}{k} p^k (1-p)^{n-k} (1-\xi)\xi^n = (1-\xi)(p\xi)^k \sum_{m \geq 0} \binom{m+k}{k} [\xi(1-p)]^m = \frac{(1-\xi)(p\xi)^k}{[1-\xi(1-p)]^{k+1}}$, which is $(1-\xi_p)\xi_p^k$ with $\xi_p = \frac{p\xi}{1-\xi(1-p)}$. Since $\frac{d\xi_p}{dp} = \frac{\xi(1-\xi)}{[1-\xi(1-p)]^2} > 0$, smaller samples have smaller ξ_p : the distribution is pushed toward low abundances, giving the hollow curves of small plots.

Exercise 3.4

(Upscaling a forest) The file `RSA_sampled_1percent.xlsx` contains the abundances of the $S^* = 4283$ species observed in a 1% random sample of a forest (total 39 779 individuals). Fit a truncated NB by maximum likelihood, upscale it to $p = 1$, and estimate the total number of species. Compare with the estimate from Fisher's log-series, $S(J) = \theta \ln(1 + J/\theta)$, extrapolated to 100 times more individuals.

Solution

Maximising $\sum_i \ln P_{n_i}^*(r, \xi_p)$ gives $r = 0.82$ and $\xi_p = 0.906$. Inverting self-similarity with $p = 0.01$: $\xi = \xi_p/(p + \xi_p(1-p)) = 0.99897$. Then $S = S^* \frac{1-(1-\xi)^r}{1-(1-\xi_p)^r} \simeq 4976$. The forest is a computer-generated one with 4974 species (Lecture 2), so the NB estimate is essentially exact. The log-series fit to the sample gives $\theta = 1218$ and, extrapolated to $100J$ individuals, $S \simeq 9856$, a twofold overestimate. Fixing $r = 0$ forces too many rare species.

Exercise 3.5

(Criticality) For the untruncated $\text{NB}(r, \xi)$ show that $\text{CV}^2 = 1/(r\xi)$. For the truncated NB with $r \rightarrow 0$ (log-series with parameter ξ), compute $\langle n \rangle$ and $\langle n^2 \rangle$ and show that the CV diverges as $\xi \rightarrow 1$.

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Solution

Mean $\frac{r\xi}{1-\xi}$, variance $\frac{r\xi}{(1-\xi)^2}$, so $CV^2 = \frac{1}{r\xi}$. For the log-series $P_n = \frac{\xi^n}{nL}$ with $L = -\ln(1-\xi)$: $\langle n \rangle = \frac{\xi}{(1-\xi)L}$ and $\langle n^2 \rangle = \frac{\xi}{(1-\xi)^2 L}$. Then $CV^2 = \frac{\langle n^2 \rangle}{\langle n \rangle^2} - 1 = \frac{L}{\xi} - 1 \simeq -\ln(1-\xi) \rightarrow \infty$. Fluctuations grow without bound as births and deaths balance: the critical point of the neutral model.

Exercise 3.6

(Neutral community simulation) Simulate the mean-field neutral model: a community of $J = 2000$ individuals; at each step one random individual dies and is replaced with probability $1 - \nu$ by the offspring of a random individual, and with probability ν by a new species. Run to stationarity for $\nu = 0.005$ and compare the RSA with the log-series $\theta x^n/n$, $\theta = 2J\nu/(1-\nu)$.

Solution

After a transient of order J/ν deaths, the number of species fluctuates around Ewens' $\langle S \rangle = \sum_{i=0}^{J-1} \frac{\theta}{\theta+i} \simeq \theta \ln(1 + J/\theta)$. With $\theta \simeq 20.1$ this is about 93 species. The time-averaged RSA follows $\theta x^n/n$, with x fixed by $\sum_n n\phi_n = J$. Singletons are the most common abundance class, and a few species hold hundreds of individuals. Many rare species and few dominant ones arise from pure drift, without fitness differences.

3.9. Sources

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4. Spatial and temporal patterns of biodiversity

Chapter 3 explained the distribution of abundances in a community. Ecologists measure several other macroecological patterns: how many species are found in areas of increasing size, how many are endemic to an area, how long species persist before disappearing locally. This chapter derives them from the same null model of independent, randomly placed species. It then shows how space changes the picture, gives a “grand-canonical” formulation in which the number of species itself fluctuates, and closes with metapopulations: species living on a network of habitat patches, whose persistence becomes a spectral problem.

4.1. Species–area and endemic–area relationships

4.1.1. Three regimes of the SAR

Over many orders of magnitude of area, the SAR has three phases. At small scales S grows quickly, roughly linearly, because every new individual tends to be a new species. At intermediate scales it grows sublinearly, close to a power law $S \propto A^z$ with $z \approx 0.2$ – 0.3 . At continental scales it steepens again, because new areas contain new, independent biotas. We model the intermediate regime.

4.1.2. The random-placement null model

Following Coleman (1981), divide a region of area A containing S species into cells and consider a sub-area a , a fraction $\alpha = a/A$ of the total. Individuals are placed independently and uniformly (**random placement**). A species with n_i individuals is absent from a with probability $q_i = (1 - \alpha)^{n_i}$ and present with probability $p_i = 1 - q_i$. The number of species found in a is a sum of independent Bernoulli variables, with generating function

$$G(z) = \prod_{i=1}^S (q_i + p_i z).$$

Its mean and variance, given the abundances $\mathbf{n} = (n_1, \dots, n_S)$, are

$$\langle s(a | \mathbf{n}) \rangle = \sum_i p_i = S - \sum_{i=1}^S (1 - \alpha)^{n_i}, \quad \sigma^2(a | \mathbf{n}) = \sum_i p_i q_i = \sum_i [(1 - \alpha)^{n_i} - (1 - \alpha)^{2n_i}].$$

If the abundances themselves are independent draws from the SAD $P(n)$, averaging over $\mathbf{n}$ gives

$$\text{SAR}(a) = S \left[1 - \sum_{n \geq 1} (1 - \alpha)^n P(n) \right] = S [1 - \mathcal{G}(1 - \alpha)],$$

where $\mathcal{G}(z) = \sum_n P(n) z^n$ is the generating function of the SAD. The SAR is the SAD’s generating function evaluated at $1 - \alpha$.

4.1.3. The endemic–area relationship

A species is **endemic** to a if *all* its individuals fall inside a , which happens with probability α^{n_i} . Hence

$$\text{EAR}(a | \mathbf{n}) = \sum_i \alpha^{n_i}, \quad \text{EAR}(a) = S \sum_{n \geq 1} \alpha^n P(n) = S \mathcal{G}(\alpha).$$

Proposition 4.1 (SAR and EAR are complementary). *Under random placement, for any SAD,*

$$\text{EAR}(a) = S - \text{SAR}(A - a).$$

Proof. $S - \text{SAR}(A - a) = S \mathcal{G}(1 - (1 - \alpha)) = S \mathcal{G}(\alpha)$. A species is endemic to a exactly when it is absent from the complement $A - a$. $\square$

4.1.4. The log-series case

For the log-series $P(n) = \frac{x^n}{n |\ln(1-x)|}$ (with $x = 1 - \nu$ close to 1), $\mathcal{G}(z) = \frac{\ln(1-xz)}{\ln(1-x)}$, so

$$\frac{\text{SAR}(\alpha)}{S} = 1 - \frac{\ln(1 - x(1 - \alpha))}{\ln(1 - x)} = 1 + \frac{\ln(\nu(1 - \alpha) + \alpha)}{|\ln \nu|}, \quad \frac{\text{EAR}(\alpha)}{S} = \frac{\ln(1 - x\alpha)}{\ln(1 - x)}.$$

At small α the SAR grows linearly ($\simeq \alpha x / [(1 - x) |\ln \nu|]$); at intermediate α it grows logarithmically, which over a couple of decades is hard to distinguish from a power law. The EAR stays tiny until α approaches 1 and then shoots up.

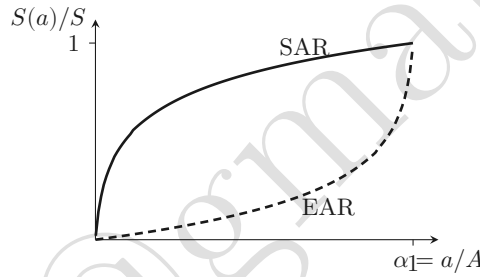


Figure: SAR (solid) and EAR (dashed) under random placement with a log-series SAD, $x = 0.99$.

4.1.5. Habitat loss and extinctions

Destroying a fraction of a region eliminates the species endemic to the destroyed part. The correct estimate of extinctions is therefore $\text{EAR}(a_{\text{lost}})$, which by the complementarity above equals $S - \text{SAR}(A - a_{\text{lost}})$. The common practice of fitting a power-law SAR, $S \propto A^z$, and predicting a loss $1 - (1 - f)^z$ for a lost fraction f uses an *approximate* SAR and typically overestimates extinctions. He and Hubbell (2011) showed that the EAR rises much more slowly than this backward extrapolation of the SAR suggests.

4.2. Persistence times

4.2.1. The species persistence-time distribution

Monitoring a region over time, a species appears (by speciation, immigration or recolonisation), persists for a while, and disappears. The **species persistence time** (SPT) τ is the length of such an interval. Breeding-bird surveys in North America, grassland plots in Kansas, fish and forest censuses all give SPT distributions with a power-law body and a cut-off. Long-term data provide many intervals $\tau_1, \tau_2, \dots$, treated as realisations of the same process, from which the distribution can be estimated.

4.2.2. Mean-field solution by generating functions

Take a species with linear rates $b_n = bn$, $d_n = dn$ (no immigration, $b \lesssim d$), starting from m individuals. The generating function $G(z, t) = \sum_n P_n(t) z^n$ turns the master equation into a first-order PDE. Multiplying by z^n and summing,

$$\frac{\partial G}{\partial t} = (d + bz^2 - (b + d)z) \frac{\partial G}{\partial z} = (1 - z)(d - bz) \frac{\partial G}{\partial z},$$

with $G(z, 0) = z^m$. It is solved by the **method of characteristics**. Along curves with $\frac{dz}{dt} = -(1 - z)(d - bz)$, G is constant. Integrating by partial fractions with $\delta = d - b > 0$,

$$\frac{(z - 1)(bz_0 - d)}{(z_0 - 1)(bz - d)} = e^{-\delta t},$$

which gives the initial point z_0 of the characteristic through (z, t) :

$$z_0 = \frac{d - g}{b - g}, \quad g(z, t) = e^{-\delta t} \frac{bz - d}{z - 1}.$$

Hence $G(z, t) = \left[\frac{d - g(z, t)}{b - g(z, t)} \right]^m$. The probability of being extinct at time t is $P_0(t) = G(0, t)$:

$$P_0(t) = \left[\frac{d(1 - e^{-\delta t})}{d - b e^{-\delta t}} \right]^m.$$

For $m = 1$ the **persistence probability** $\mathcal{P}(t) = 1 - P_0(t)$ is

$$\mathcal{P}(t) = \frac{\delta e^{-\delta t}}{d - b e^{-\delta t}} \simeq \frac{1}{dt} \mathcal{F}(\delta t), \quad \mathcal{F}(x) = \frac{x}{e^x - 1},$$

in the scaling regime $\delta \rightarrow 0$, $t \rightarrow \infty$ with δt fixed. $\mathcal{F}$ is a **finite-size scaling function**: $\mathcal{F}(x) \rightarrow 1$ for $x \rightarrow 0$ and decays exponentially for $x \gg 1$. The SPT density is

$$p(\tau) = -\frac{d\mathcal{P}}{d\tau} \propto \tau^{-2} f(\delta\tau),$$

a power law with exponent 2 and an exponential cut-off at $\tau_c = 1/\delta = 1/(d - b)$. At criticality ($b = d$) the cut-off disappears. The exponent differs from that of the first return of an unbiased random walk (3/2) because the step rates are proportional to n : large populations fluctuate faster.

If instead $b > d$, a finite fraction of species never goes extinct; if $b < d$ with a large gap, lifetimes are exponential with mean $1/(d - b)$.

4.2.3. Space changes the exponent

The mean-field model ignores space. In the spatial voter model with speciation (Chapter 3), individuals sit on a lattice or network, die, and are replaced by a neighbour's offspring or, with probability ν , by a new species. Simulations give power-law SPT distributions whose exponent depends on the geometry. In terms of the cumulative persistence $\mathcal{P}(t) \sim t^{-\beta}$:

- $\beta \simeq 1/2$ on a one-dimensional chain (lifetimes $\propto t^{-3/2}$, as for a random walk),
- intermediate values on river-like networks and in two dimensions (where logarithmic corrections appear),
- $\beta \simeq 1$ in three dimensions and in the well-mixed (global) case, recovering the mean-field t^{-2} density.

Spatial constraints make species persist *longer*: a species confined to a region of a chain can only be eroded from its edges.

4.2.4. Sampling effects

Observation windows are finite. A species already present at the start, or still present at the end, has a truncated measured lifetime. Censoring piles up probability at the longest observable time and produces a spurious bump at the end of the empirical distribution. Suweis et al. (2012) model the *effective* lifetime L_e in terms of the true one L , the observation window $[x_0, x_f]$ and the appearance time, and derive the distribution of L_e . This model fits breeding-bird, grassland, forest and fish data, including the final bump, with the same underlying power-law SPT.

4.3. The grand-canonical formulation of neutral theory

So far the number of individuals was fixed. Let it fluctuate, and describe the community by the occupation numbers $\phi = (\phi_1, \phi_2, \dots)$: ϕ_k is the number of species with abundance k . Species are created (speciation or immigration) at rate b_0 with one individual. A species with k individuals gains one at rate b_k and loses one at rate d_k , moving from class k to $k \pm 1$. Writing $\hat{e}_k$ for the unit vector of class k ,

$$\begin{aligned} \dot{P}(\phi) = & b_0 P(\phi - \hat{e}_1) + d_1(\phi_1 + 1)P(\phi + \hat{e}_1) - d_1\phi_1 P(\phi) + \sum_{k \geq 2} [b_{k-1}(\phi_{k-1} + 1)P(\phi + \hat{e}_{k-1} - \hat{e}_k) - d_k\phi_k P(\phi)] \\ & + \sum_{k \geq 1} [d_{k+1}(\phi_{k+1} + 1)P(\phi + \hat{e}_{k+1} - \hat{e}_k) - b_k\phi_k P(\phi)] - b_0 P(\phi). \end{aligned}$$

Detailed balance for each move, together with the ansatz of **independent occupation numbers**, $P_{\text{st}}(\phi) = \prod_k P_k(\phi_k)/Z$, gives

$$P_k(\phi_k) \propto \frac{1}{\phi_k!} \left(\frac{b_0}{d_1} \prod_{i=2}^k \frac{b_{i-1}}{d_i} \right)^{\phi_k}.$$

For linear rates $b_i = bi$, $d_i = di$, with $x = b/d$ and $\gamma = b_0/d$,

$$P_k(\phi_k) = \frac{1}{\phi_k!} \left(\gamma \frac{x^k}{k} \right)^{\phi_k} e^{-\gamma x^k/k} ;$$

each ϕ_k is an independent **Poisson** variable with mean $\langle \phi_k \rangle = \gamma x^k/k$, which is again the log-series. The normalisation is $Z = \exp(\gamma \sum_k x^k/k) = (1 - x)^{-\gamma}$.

The total number of species $S = \sum_k \phi_k$ is a sum of independent Poisson variables, hence Poisson:

$$P(S) = \frac{\langle S \rangle^S}{S!} e^{-\langle S \rangle}, \quad \langle S \rangle = -\gamma \ln(1 - x).$$

4.3.1. Relations between patterns

The grand-canonical picture suggests a coarse-grained view of a community. Species appear at random times, at a rate λ (speciation or immigration), and each persists for a time τ drawn from the SPT distribution, independently of the others: an “ $M/G/\infty$ queue”. Then:

- the number of species present is Poisson, with mean given by **Little’s law**: $\langle S \rangle = \lambda \langle \tau \rangle$;
- the SPT and the species richness are linked. With $p(\tau) \propto \tau^{-2}$ cut off at $\tau_c = 1/\delta$, $\langle \tau \rangle \propto \ln \tau_c$, so $\langle S \rangle \propto \lambda \ln(1/\delta)$. Compare $-\gamma \ln(1 - x)$ above, with $1 - x = \delta/d$;
- making appearances also random in space, each species occupying a region that grows with its age, turns the SPT into a range-size distribution and hence into a SAR (Suweis, Rinaldo, Maritan, 2012). Temporal and spatial patterns are two faces of the same birth–death–speciation process.

4.4. Metapopulations

Many species live on **patchy** habitats: ponds, meadows, islands, forest fragments. Levins (1969) introduced the **metapopulation**, a “population of populations” in which each local population has a substantial risk of extinction and the species persists only regionally, through recolonisation. Hanski applied and extended the idea (e.g. to the Glanville fritillary butterfly in the Åland islands).

4.4.1. The Hanski model

Let $P_x(t) \in [0, 1]$ be the probability that patch x is occupied. Occupied patches go extinct at rate v_x ; an empty patch is colonised from occupied patches y at rate K_{xy} :

$$\dot{P}_x = -v_x P_x + (1 - P_x) \sum_y K_{xy} P_y, \quad K_{xy} = c e^{-|x-y|/\xi} \quad (y \neq x),$$

with ξ the dispersal distance and c a colonisation rate (which can include patch sizes). The dynamics keeps $0 \leq P_x \leq 1$.

4.4.2. Persistence as an eigenvalue problem

The empty state $P = 0$ is always an equilibrium. Linearising, $\dot{P}_x \simeq \sum_y (K_{xy} - v_x \delta_{xy}) P_y$. With equal extinction rates $v_x = v$, the empty state is unstable, and the species persists, if and only if

$$\lambda_M > v,$$

where λ_M is the largest eigenvalue of the colonisation matrix K , the **metapopulation capacity** of the landscape. Since K has positive entries (and is irreducible if every patch can be reached), the **Perron–Frobenius theorem** guarantees that λ_M is real, simple and dominant, with an eigenvector of positive entries. The landscape’s ability to support the species is summarised in a single number. With heterogeneous v_x , the criterion becomes that the leading eigenvalue of $E^{-1}K$, with $E = \text{diag}(v_x)$, exceeds 1. Data on the fraction of occupied patches against $\log \lambda_M$ across patch networks show the predicted threshold.

4.4.3. Translation-invariant landscapes

On a regular lattice with $v_x = v$ and $K_{xy} = \hat{K}_{x-y}$, Fourier modes $\hat{P}(q) = \sum_x e^{iqx} P_x$ decouple at linear order:

$$\dot{\hat{P}}(q) = (-v + \hat{K}(q)) \hat{P}(q), \quad \hat{K}(q) = \sum_x e^{iqx} \hat{K}_x.$$

Since $|\hat{K}(q)| \leq \hat{K}(0) = \sum_x \hat{K}_x$, the most unstable mode is uniform. The species survives iff $\hat{K}(0) > v$, and the homogeneous non-trivial equilibrium is

$$P^* = 1 - \frac{v}{\hat{K}(0)}.$$

This is a mean-field transcritical bifurcation, the same structure as the SIS epidemic threshold (Chapter 2) and the contact process.

4.4.4. Heterogeneous extinction rates and localisation

Let $\hat{K}(0) = 1$ and let the v_x be drawn independently from a distribution $Q(v)$. In mean field each patch feels the average occupancy $m = \frac{1}{N} \sum_y P_y$:

$$\dot{P}_x = -v_x P_x + (1 - P_x)m \implies P_x^* = \frac{m^*}{v_x + m^*},$$

with self-consistency $m^* = \int dv Q(v) \frac{m^*}{v + m^*}$. Besides $m^* = 0$, a positive solution exists iff $\int dv Q(v)/v > 1$. As the transition is approached, $m^* \rightarrow 0$ and patches with small v_x dominate the occupancy. How localised the occupancy becomes is measured by the participation ratio

$$I = \frac{\langle P^2 \rangle}{\langle P \rangle^2}, \quad \langle P^\ell \rangle = \int dv Q(v) \left(\frac{m^*}{v + m^*} \right)^\ell.$$

If Q is bounded away from zero (e.g. uniform on $[\bar{v} - \Delta, \bar{v} + \Delta]$), I stays finite as $m^* \rightarrow 0$: the species is spread over the landscape. If $Q(v) \propto v^{\delta-1}$ near $v = 0$ (e.g. a Gamma distribution with shape $\delta < 1$), then $\langle P \rangle \propto m^{*\delta}$ and $\langle P^2 \rangle \propto m^{*\delta}$, so $I \propto m^{*-\delta} \rightarrow \infty$. Near extinction the species survives only in a vanishing fraction of exceptionally good patches: **localisation**. Conservation must then target these refugia.

4.5. Exercises

Exercise 4.1

(HW 2a) SAR and EAR. Using the log-series SAD $P(n) = c(1 - \nu)^n/n$, plot $\text{SAR}(\alpha)/S$ and $\text{EAR}(\alpha)/S$ against the relative area α . Compare the SAR with a power law $k\alpha^z$, with $z \in (0.2, 0.3)$ and k tuned to match the random-placement solution as closely as possible.

Solution

With $x = 1 - \nu$: $\text{SAR}/S = 1 - \ln(1 - x(1 - \alpha))/\ln(1 - x)$ and $\text{EAR}/S = \ln(1 - x\alpha)/\ln(1 - x)$. The SAR rises linearly at very small α , bends over, and approaches 1 logarithmically. The EAR is concave-up and negligible until $\alpha \gtrsim 0.5$. A straight-line fit of $\ln \text{SAR}$ against $\ln \alpha$ over $\alpha \in [10^{-3}, 1]$ gives $z \simeq 0.31$ for $x = 0.999$ and larger z for smaller x (0.55 for $x = 0.99$, 0.75 for $x = 0.95$). Exponents in the empirical range 0.2–0.3 correspond to x extremely close to 1, i.e. to the near-critical forests of Chapter 3. A single power law matches the curve only over an intermediate window; on log–log axes the random-placement SAR is visibly curved.

Exercise 4.2

(HW 2b) Habitat loss. If one third of the area is removed, how many species go extinct? Estimate it with the EAR and with the SAR, and compare with the power-law estimate.

Solution

Extinct species are those endemic to the removed third: $\text{EAR}(1/3)/S = \ln(1 - x/3)/\ln(1 - x)$. Using the SAR, the survivors are the species present in the remaining two thirds, $\text{SAR}(2/3)$, so the loss is $1 - \text{SAR}(2/3)/S$, which is the *same* number by complementarity. For $x = 0.9, 0.95, 0.99, 0.999$ the loss is 15.5%, 12.7%, 8.7%, 5.9%. A power-law SAR with $z = 0.25$ gives $1 - (2/3)^{0.25} = 9.6\%$, and with z fitted over the whole range (e.g. $z = 0.55$ for $x = 0.99$) it gives 20%, more than twice the exact 8.7%. Approximating the SAR by a power law and extrapolating overestimates extinctions; the EAR is the right tool.

Exercise 4.3

(Variance of the number of species) Under random placement with fixed abundances, show that the variance of the number of species in a is maximal for intermediate α , and compute it for S species all with the same abundance n .

Solution

$\sigma^2 = \sum_i p_i q_i$ with $q_i = (1-\alpha)^{n_i}$. Each term vanishes at $\alpha = 0$ and $\alpha = 1$ and is maximal when $q_i = 1/2$. For equal abundances $\sigma^2 = S q(1-q)$ with $q = (1-\alpha)^n$, maximal ($= S/4$) at $\alpha^* = 1 - 2^{-1/n} \simeq \ln 2/n$. The species count fluctuates most at the scale where a typical species is equally likely to be found or missed.

Exercise 4.4

(Extinction probability) For the linear birth–death process with $b < d$ and m initial individuals, show from $P_0(t)$ that eventual extinction is certain, and compute the mean time to extinction for $m = 1$. What happens for $b > d$?

Solution

As $t \rightarrow \infty$, $P_0(t) \rightarrow [d/d]^m = 1$ for $b < d$. For $m = 1$ the mean lifetime is $\int_0^\infty \mathcal{P}(t) dt = \int_0^\infty \frac{\delta e^{-\delta t}}{d - b e^{-\delta t}} dt = \frac{1}{b} \ln \frac{d}{d-b} = \frac{1}{b} \ln \frac{1}{1-b/d}$, which diverges logarithmically at criticality $b \rightarrow d$: the t^{-2} tail with cut-off $1/\delta$. For $b > d$, $e^{-\delta t}$ grows: $P_0(\infty) = (d/b)^m < 1$, so a species survives forever with probability $1 - (d/b)^m$.

Exercise 4.5

(Grand-canonical community) In the grand-canonical model with $x = 0.99$ and $\gamma = 30$, compute $\langle S \rangle$, $\text{Var}(S)$, the expected number of singletons, and the mean total abundance.

Solution

$\langle S \rangle = -\gamma \ln(1-x) = 30 \ln 100 = 138.2$ and, being Poisson, $\text{Var}(S) = 138.2$ (standard deviation 11.8). Singletons: $\langle \phi_1 \rangle = \gamma x = 29.7$. Total abundance: $\sum_k k \langle \phi_k \rangle = \gamma \sum_k x^k = \gamma \frac{x}{1-x} = 2970$ individuals.

Exercise 4.6

(Metapopulation capacity) Three patches on a line at positions 0, 1, 3, with $K_{xy} = e^{-|x-y|}$ for $x \neq y$ and $K_{xx} = 0$. Compute λ_M and the critical extinction rate. Which patch matters most?

Solution

$K = \begin{pmatrix} 0 & e^{-1} & e^{-3} \\ e^{-1} & 0 & e^{-2} \\ e^{-3} & e^{-2} & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0.368 & 0.050 \\ 0.368 & 0 & 0.135 \\ 0.050 & 0.135 & 0 \end{pmatrix}$. Its largest eigenvalue is $\lambda_M \simeq 0.41$, so the species persists iff $v < 0.41$. The Perron eigenvector $\simeq (0.66, 0.69, 0.31)$ weights the middle patch most: it is close to both others. The isolated patch at $x = 3$ contributes little, and removing it lowers λ_M only slightly, to $e^{-1} = 0.37$.

4.6. Sources

Prof. S. Suweis, *PMLS Course Notes*, Sections 3.4–3.5; Lecture 3 slides (SAR, EAR, persistence times, spatial and sampling effects) and Lecture 4 slides (A. Maritan: Hanski metapopulation model); Homework 2. B. D. Coleman, *On random placement and species–area relations*, Math. Biosci. **54**, 191 (1981). F. He and S. P. Hubbell, *Species–area relationships always overestimate extinction rates from habitat loss*, Nature **473**, 368 (2011). S. Pigolotti, A. Flammini, M. Marsili, A. Maritan, PNAS **102**, 15747 (2005). E. Bertuzzo et al., PNAS **108**, 4346 (2011). S. Suweis et al., J. Theor. Biol. **303**, 15 (2012); S. Suweis, A. Rinaldo, A. Maritan, J. Stat. Mech. P07017 (2012). J. Grilli, S. Azale, J. R. Banavar, A. Maritan, J. Theor. Biol. **313**, 87 (2012). I. Hanski and O. Ovaskainen, *The metapopulation capacity of a fragmented landscape*, Nature **404**, 755 (2000).

5. Two interacting species: predators, prey and noise

Introducing a single new species into an ecosystem (wolves into Yellowstone, a predatory fish into a lagoon) can reorganise the whole community through a *trophic cascade*. To understand how, we start with two species. This chapter treats the predator–prey problem twice. Deterministically, the Lotka–Volterra equations and their variants explain damped or sustained oscillations. Stochastically, a microscopic individual-based model shows that demographic noise, far from being a small correction, is **resonantly amplified** into large, persistent cycles. The machinery developed here (reaction schemes, multivariate master equations, the van Kampen system-size expansion, power spectra) is reused for neurons (Chapter 10) and for genes (Chapter 16).

5.1. The Lotka–Volterra equations

5.1.1. Historical model and its conserved quantity

Lotka (1920) and Volterra (1926) wrote, independently,

$$\dot{x} = ax - bxy, \quad \dot{y} = -cy + dxy,$$

for prey x and predators y . Volterra was prompted by his son-in-law, the biologist Umberto D’Ancona, who had noticed that the fraction of predatory fish in the markets of Trieste, Fiume and Venice rose during the First World War, when fishing stopped. Prey grow exponentially at rate a when alone; predators die at rate c without prey; encounters (mass action, rate $\propto xy$) convert prey into predators.

Besides the trivial equilibrium, there is a coexistence point $(x^*, y^*) = (c/d, a/b)$. The model has a **conserved quantity**

$$H(x, y) = dx - c \ln x + by - a \ln y, \quad \dot{H} = 0,$$

so trajectories are closed orbits around (x^*, y^*) , whose amplitude is fixed by the initial condition: **neutral cycles**. The Jacobian at (x^*, y^*) has purely imaginary eigenvalues $\pm i\sqrt{ac}$, giving small oscillations of period $2\pi/\sqrt{ac}$. This is a structurally unstable situation: any perturbation of the equations (self-limitation, saturation, noise) turns the centres into spirals.

Volterra’s model also explains D’Ancona’s observation. Fishing adds mortality f to both species, $a \rightarrow a - f$ and $c \rightarrow c + f$, so the time-averaged abundances (equal to the equilibrium values on a closed orbit) become $\langle x \rangle = (c + f)/d$, $\langle y \rangle = (a - f)/b$. Less fishing means *fewer* prey and *more* predators: **Volterra’s principle**.

5.1.2. Self-limited prey

Adding logistic self-limitation to the prey gives

$$\dot{x} = \lambda x \left(1 - \frac{x}{k}\right) - pxy, \quad \dot{y} = -\eta y + pxy.$$

The equilibria are $z_1^* = (0, 0)$, $z_3^* = (k, 0)$ and the coexistence point

$$z_2^* = \left(\frac{\eta}{p}, \frac{\lambda}{p} \left(1 - \frac{\eta}{pk}\right)\right),$$

5. Two interacting species: predators, prey and noise

which is feasible (positive) iff $pk > \eta$: the prey at carrying capacity must sustain the predators. The Jacobian

$$J = \begin{pmatrix} \lambda(1 - 2x/k) - py & -px \\ py & px - \eta \end{pmatrix} \text{ gives:}$$

- at z_1^* , eigenvalues $\lambda > 0$ and $-\eta$: a saddle (unstable);
- at z_3^* , eigenvalues $-\lambda$ and $pk - \eta > 0$: a saddle, unstable when coexistence is feasible;
- at z_2^* ,

$$J_2 = \begin{pmatrix} -\frac{\lambda\eta}{kp} & -\eta \\ \lambda(1 - \frac{\eta}{pk}) & 0 \end{pmatrix}, \quad \vartheta_{1,2} = \frac{1}{2kp} \left[-\lambda\eta \pm \sqrt{\lambda\eta(\lambda\eta - 4kp(kp - \eta))} \right].$$

Since $\text{tr } J_2 < 0$ and $\det J_2 = \lambda\eta(1 - \eta/pk) > 0$, z_2^* is **always stable**. For large k the discriminant is negative and the eigenvalues are complex: populations approach the equilibrium through **damped oscillations** with frequency $\simeq \sqrt{\lambda\eta(1 - \eta/pk)}$ and damping rate $\lambda\eta/(2kp)$, which is small when k is large.

5.1.3. Real cycles

Real predator–prey systems often oscillate for decades: the snowshoe hare and lynx pelts of the Hudson’s Bay Company, rodents, plankton in chemostats. The deterministic model above predicts that the oscillations die out. There are two classical ways out.

1. **Limit cycles.** If predators saturate (Holling type II functional response, $pxy \rightarrow \frac{pxy}{1+hx}$), the coexistence equilibrium can lose stability through a **Hopf bifurcation** when the carrying capacity is increased. A stable limit cycle is born, with amplitude independent of initial conditions. Enriching the environment destabilises it: Rosenzweig’s *paradox of enrichment*.
2. **Stochastic amplification.** Even when the deterministic equilibrium is stable, the discreteness of individuals produces noise that excites the weakly damped oscillatory mode. The result is large, irregular but persistent cycles with a well-defined dominant frequency. This mechanism requires no fine tuning, and is the subject of the rest of the chapter.

5.2. From reactions to master equations

Individual-based models are best written as **reaction schemes**, as in chemistry. Consider N_s types of “molecules” (individuals of different species, or empty sites) and M reactions R_j . Each reaction has:

- a **state-change vector** $\mathbf{v}_j$: the change in the vector of abundances $\mathbf{x}$ when R_j fires (e.g. $B \rightarrow A + A$ has $\mathbf{v} = (+2, -1)$);
- a **propensity** $a_j(\mathbf{x})$: $a_j \Delta t$ is the probability that R_j fires in $[t, t + \Delta t]$.

For a unimolecular reaction $A \rightarrow \dots$ with rate constant C_j , $a_j = C_j x_A$. For a bimolecular one, $A + B \rightarrow \dots$, $a_j = C_j x_A x_B / \Omega$, where Ω is the system size (volume, number of sites). Two molecules must meet, and in a larger volume they are less likely to. In general, a reaction involving k reactants has $a_j \propto x_1 \dots x_k / \Omega^{k-1}$, so propensities are **extensive**: $a_j = \Omega \alpha_j(\mathbf{x}/\Omega)$. The master equation balances inflow from $\mathbf{x} - \mathbf{v}_j$ and outflow from $\mathbf{x}$:

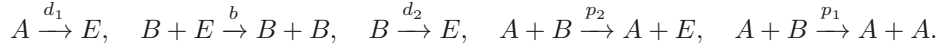
$$\frac{\partial P(\mathbf{x}, t)}{\partial t} = \sum_{j=1}^M [a_j(\mathbf{x} - \mathbf{v}_j) P(\mathbf{x} - \mathbf{v}_j, t) - a_j(\mathbf{x}) P(\mathbf{x}, t)],$$

and can be sampled exactly with the Gillespie algorithm of Chapter 2.

5.3. A stochastic predator–prey model

5.3.1. Microscopic rules

Following McKane and Newman (2005), consider N sites, each empty (E), occupied by a predator (A) or by a prey (B). Let n , m and $e = N - n - m$ be their numbers. Individuals interact as if well mixed (spatial effects neglected), through five reactions:



Predators die. Prey reproduce into empty sites and die. Predation either just removes the prey (p_2) or also converts it into a new predator (p_1). Choosing pairs at random, with a factor 2 from the two orderings, the propensities are

$$\begin{aligned} T(n-1, m \mid n, m) &= d_1 n, & T(n, m+1 \mid n, m) &= 2b \frac{m e}{N}, \\ T(n, m-1 \mid n, m) &= d_2 m + 2p_2 \frac{nm}{N}, & T(n+1, m-1 \mid n, m) &= 2p_1 \frac{nm}{N}. \end{aligned}$$

5.3.2. Drift and diffusion

With densities $x = n/N$ (predators) and $y = m/N$ (prey), each propensity is N times a function of (x, y) . Expanding the master equation to second order in $1/N$ (with shift operators $\mathbb{E}_A^\pm$, $\mathbb{E}_B^\pm$ as in Chapter 2) gives a two-dimensional Fokker–Planck equation

$$\partial_t P = - \sum_i \partial_i [A_i P] + \frac{1}{2N} \sum_{ij} \partial_i \partial_j [B_{ij} P].$$

The drift is the vector of average changes:

$$A_1 = 2p_1 xy - d_1 x, \quad A_2 = 2by(1 - x - y) - d_2 y - 2(p_1 + p_2)xy.$$

The diffusion matrix collects, for each reaction, $\mathbf{v}_j \mathbf{v}_j^\top$ weighted by its rate:

$$B = \begin{pmatrix} d_1 x + 2p_1 xy & -2p_1 xy \\ -2p_1 xy & 2by(1 - x - y) + d_2 y + 2(p_1 + p_2)xy \end{pmatrix}.$$

The off-diagonal element is negative because the conversion reaction changes n and m in opposite directions, $\mathbf{v} = (+1, -1)$. Noise in the two populations is correlated.

As $N \rightarrow \infty$ the noise vanishes and the drift gives the **mean-field equations**

$$\dot{x} = 2p_1 xy - d_1 x, \quad \dot{y} = ry \left(1 - \frac{y}{k}\right) - 2(p_1 + p_2 + b)xy, \quad r = 2b - d_2, \quad k = \frac{2b - d_2}{2b},$$

a predator–prey system with logistic prey. Its interior equilibrium

$$y^* = \frac{d_1}{2p_1}, \quad x^* = \frac{2b(1 - y^*) - d_2}{2(b + p_1 + p_2)}$$

is a stable focus, so the mean-field dynamics predicts damped oscillations and no cycles.

5.3.3. Langevin equations

The equivalent Langevin system is

$$\dot{x} = A_1 + \frac{1}{\sqrt{N}} \sum_j g_{1j} \xi_j, \quad \dot{y} = A_2 + \frac{1}{\sqrt{N}} \sum_j g_{2j} \xi_j, \quad \langle \xi_i(t) \xi_j(t') \rangle = \delta_{ij} \delta(t - t'),$$

where g is any matrix with $gg^\top = B$. For instance $g = S^\top \Lambda^{1/2} S$ from the diagonalisation $B = S^\top \Lambda S$, or the Cholesky factor of B . Only B is physical; different square roots give the same statistics.

5.4. The van Kampen expansion and stochastic amplification

5.4.1. Linear noise approximation

Fluctuations around the deterministic equilibrium are $O(N^{-1/2})$. Following van Kampen, write

$$x(t) = x^* + \frac{\zeta_1(t)}{\sqrt{N}}, \quad y(t) = y^* + \frac{\zeta_2(t)}{\sqrt{N}}.$$

Substituting and keeping the leading order, the drift contributes the Jacobian $J = \partial \mathbf{A} / \partial \mathbf{x}$ at (x^*, y^*) , and the noise amplitude is frozen at its equilibrium value:

$$\dot{\zeta} = J\zeta + \eta(t), \quad \langle \eta_i(t) \eta_j(t') \rangle = B_{ij}(x^*, y^*) \delta(t - t').$$

This is a multivariate Ornstein–Uhlenbeck process. The fluctuations are Gaussian, of size independent of N in the rescaled variables, and their statistics follow from linear algebra. This is the **linear noise approximation**. It is exact to leading order in $1/\sqrt{N}$ around a stable fixed point.

5.4.2. The power spectrum

Fourier transforming, $(i\omega - J)\tilde{\zeta}(\omega) = \tilde{\eta}(\omega)$ with $\langle \tilde{\eta}_i(\omega) \tilde{\eta}_j^*(\omega') \rangle = B_{ij} 2\pi \delta(\omega - \omega')$. The **power spectrum** $P_i(\omega) = \langle |\tilde{\zeta}_i(\omega)|^2 \rangle$ (per unit frequency) is the diagonal of

$$\mathcal{P}(\omega) = (i\omega - J)^{-1} B (-i\omega - J^\top)^{-1}.$$

For two variables, with $\Omega_0^2 = \det J$ and $\Gamma = -\text{tr } J$, the denominator is $|\det(i\omega - J)|^2 = (\omega^2 - \Omega_0^2)^2 + \Gamma^2 \omega^2$, and solving the 2×2 system,

$$P_1(\omega) = \frac{\alpha_1 + B_{11} \omega^2}{(\omega^2 - \Omega_0^2)^2 + \Gamma^2 \omega^2}, \quad \alpha_1 = B_{11} J_{22}^2 - 2B_{12} J_{12} J_{22} + B_{22} J_{12}^2,$$

and similarly for P_2 with indices exchanged. For our model $J_{11} = 0$ at equilibrium (the predator equation is $\dot{x} = x(2p_1 y - d_1)$), so $\Omega_0^2 = J_{12} |J_{21}|$ and $\Gamma = |J_{22}|$.

This is the response of a **damped harmonic oscillator** with natural frequency Ω_0 and damping Γ , driven by white noise. When $\Gamma \ll \Omega_0$ (weak damping, as for a large carrying capacity) the spectrum has a sharp peak near Ω_0 , of height $\propto 1/(\Gamma^2 \Omega_0^2)$. Demographic noise has power at all frequencies; the system filters it and amplifies the frequency of its own damped mode. The result is **resonant amplification** without external forcing and without fine tuning.

Example 5.1 (*Numbers*)

Take $b = 0.1$, $d_1 = 0.1$, $d_2 = 0$, $p_1 = 0.25$, $p_2 = 0.05$. Then $x^* = y^* = 0.2$ and

$$J = \begin{pmatrix} 0 & 0.10 \\ -0.16 & -0.04 \end{pmatrix}, \quad B = \begin{pmatrix} 0.040 & -0.020 \\ -0.020 & 0.048 \end{pmatrix}.$$

The eigenvalues are $-0.02 \pm 0.125i$: a weakly damped focus with $\Omega_0 = 0.126$, $\Gamma = 0.04$. The predator spectrum peaks at $\omega \simeq 0.125$, about 27 times higher than at $\omega = 0$. A direct simulation of the linear Langevin equations gives a peak at $\omega \simeq 0.12$. Gillespie simulations of the full model with N in the thousands show noisy cycles with this period, $2\pi/0.125 \approx 50$ time units, and amplitude $\propto 1/\sqrt{N}$ in density, i.e. $\propto \sqrt{N}$ in numbers of individuals.

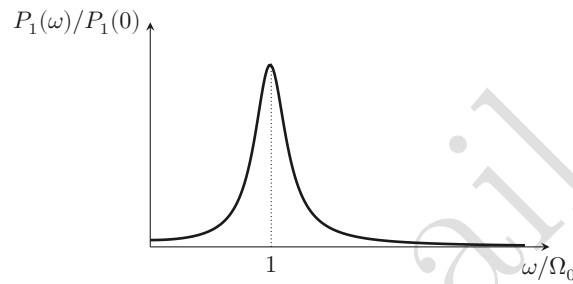


Figure: predator power spectrum of the stochastic model (linear noise approximation), normalised to its zero-frequency value.

Note

Stochastic amplification is generic: whenever a stable fixed point of a large but finite system is a focus with weak damping, intrinsic noise produces quasi-cycles. The same mechanism explains oscillations in epidemics (measles, whooping cough), in biochemical networks (Chapter 16) and in neural populations (Chapter 10), where “noise-driven gamma oscillations” appear in Wilson–Cowan models.

5.5. Two-species neutral competition

In the absence of interactions, two species competing for the same N sites with identical rates perform a neutral **voter model** (or Moran process). At each event an individual dies and is replaced by the offspring of a random individual. The number n of species A is a birth–death chain with $W^+(n) = W^-(n) = n(N-n)/N$: no drift, only noise. In the continuum limit $x = n/N$,

$$\partial_t P = \frac{1}{N} \partial_x^2 [x(1-x)P],$$

so one species always fixates. The probability that A wins is its initial fraction x_0 (the process is a martingale), and the mean fixation time is $-N[x_0 \ln x_0 + (1-x_0) \ln(1-x_0)]$ generations. Neutral coexistence of two species is transient; long-term diversity requires speciation or immigration (Chapter 3) or stabilising interactions (Chapters 6–7).

5.6. Exercises

Exercise 5.1

(Conserved quantity) Show that $H = dx - c \ln x + by - a \ln y$ is conserved by the classical Lotka–Volterra equations, and that the time averages over a closed orbit equal the equilibrium values.

Solution

$\dot{H} = (d-c/x)\dot{x} + (b-a/y)\dot{y} = (d-c/x)x(a-by) + (b-a/y)y(dx-c) = (dx-c)(a-by) + (by-a)(dx-c) = 0$. For the averages, $\frac{d}{dt} \ln x = a - by$; integrating over a period T gives $0 = aT - b \int y dt$, so $\langle y \rangle = a/b = y^*$. Similarly $\langle x \rangle = c/d$.

Exercise 5.2

(Self-limited prey) For $\lambda = 1$, $\eta = 0.5$, $p = 1$, $k = 10$, find the coexistence equilibrium, its eigenvalues, and the period and decay time of the damped oscillations.

Solution

$x^* = \eta/p = 0.5$, $y^* = \frac{\lambda}{p}(1 - \frac{\eta}{pk}) = 0.95$. $J_2 = \begin{pmatrix} -0.05 & -0.5 \\ 0.95 & 0 \end{pmatrix}$, with $\text{tr} = -0.05$ and $\det = 0.475$.

Eigenvalues $-0.025 \pm i\sqrt{0.475 - 0.000625} = -0.025 \pm 0.689i$. Period $2\pi/0.689 = 9.1$; amplitude decay time $1/0.025 = 40$. About four oscillations occur per decay time: on short records such dynamics look cyclic.

Exercise 5.3

(Drift and diffusion from reactions) For a general reaction scheme with propensities $a_j = N\alpha_j(\mathbf{x})$ and state changes $\mathbf{v}_j$, show that the Kramers–Moyal expansion gives $\mathbf{A} = \sum_j \mathbf{v}_j \alpha_j$ and $B = \sum_j \mathbf{v}_j \mathbf{v}_j^\top \alpha_j$. Verify the predator–prey B .

Solution

The master equation is $\partial_t P = N \sum_j [\alpha_j(\mathbf{x} - \mathbf{v}_j/N) P(\mathbf{x} - \mathbf{v}_j/N) - \alpha_j P]$. Expanding $f(\mathbf{x} - \mathbf{v}_j/N) = f - \frac{1}{N} \mathbf{v}_j \cdot \nabla f + \frac{1}{2N^2} (\mathbf{v}_j \cdot \nabla)^2 f$ gives $\partial_t P = -\nabla \cdot (\mathbf{A}P) + \frac{1}{2N} \sum_{kl} \partial_k \partial_l (B_{kl}P)$ with the stated forms. For predator–prey: $\mathbf{v} = (-1, 0)$ rate d_1x ; $(0, +1)$ rate $2by(1-x-y)$; $(0, -1)$ rate $d_2y + 2p_2xy$; $(+1, -1)$ rate $2p_1xy$. Then $B_{11} = d_1x + 2p_1xy$, $B_{22} = 2by(1-x-y) + d_2y + 2p_2xy + 2p_1xy$, and $B_{12} = -2p_1xy$, as in the text.

Exercise 5.4

(Power spectrum) Derive $P_1(\omega)$ for $\dot{\zeta} = J\zeta + \eta$ with $\langle \eta \eta^\top \rangle = B\delta(t-t')$ and a general 2×2 J . Where is the maximum when $\Gamma \ll \Omega_0$?

Solution

$\tilde{\zeta}_1 = \frac{(i\omega - J_{22})\tilde{\eta}_1 + J_{12}\tilde{\eta}_2}{D(\omega)}$, $D = \det(i\omega - J) = \Omega_0^2 - \omega^2 + i\Gamma\omega$. Averaging $|\tilde{\zeta}_1|^2$: $(\omega^2 + J_{22}^2)B_{11} + J_{12}^2 B_{22} + 2\text{Re}[(i\omega - J_{22})J_{12}]B_{12} = B_{11}\omega^2 + B_{11}J_{22}^2 - 2J_{12}J_{22}B_{12} + J_{12}^2 B_{22}$, divided by $|D|^2 = (\omega^2 - \Omega_0^2)^2 + \Gamma^2\omega^2$. For $\Gamma \ll \Omega_0$ the denominator is smallest at $\omega^2 = \Omega_0^2 - \Gamma^2/2 \simeq \Omega_0^2$, where it equals $\simeq \Gamma^2\Omega_0^2$. A sharp peak at the natural frequency, with width $\simeq \Gamma$.

Exercise 5.5

(Simulation) Simulate the McKane–Newman model with the parameters of the example and $N = 3200$ using the Gillespie algorithm. Estimate the power spectrum of the predator density from a long trajectory and compare it with the linear-noise prediction. How does the amplitude of the cycles in numbers of individuals scale with N ?

Solution

After a transient the densities fluctuate around $(0.2, 0.2)$ with irregular oscillations of period $\simeq 50$, predators lagging prey by about a quarter period. The periodogram, averaged over windows, peaks at $\omega \simeq 0.125$ and matches $P_1(\omega)/N$ within statistical error. Fluctuations in density scale as $N^{-1/2}$, so in numbers of individuals they grow as $\sqrt{N}$: the cycles never disappear in a finite system. Doubling N leaves the rescaled spectrum unchanged.

Exercise 5.6

(Neutral fixation) For the two-species voter model, use the backward Kolmogorov equation $\frac{x(1-x)}{N} \phi''(x) = 0$ to show that the fixation probability of A from fraction x_0 is x_0 . Then solve $\frac{x(1-x)}{N} T''(x) = -1$ with $T(0) = T(1) = 0$ for the mean fixation time.

Solution

$\phi'' = 0$ with $\phi(0) = 0$, $\phi(1) = 1$ gives $\phi(x) = x$. For T : $T'' = -\frac{N}{x(1-x)} = -N(\frac{1}{x} + \frac{1}{1-x})$; integrating twice with the boundary conditions, $T(x) = -N[x \ln x + (1-x) \ln(1-x)]$. Starting from equal shares, $T = N \ln 2$ generations: drift eliminates one of two equivalent species in a time proportional to the population size.

5.7. Sources

Prof. S. Suweis, *PMLS Course Notes*, Sections 4.1–4.2; Lecture 5 slides (two-species models: Volterra equations, stability, stochastic microscopic model, Fokker–Planck, Langevin, van Kampen expansion, power spectrum). R. M. May, *Stability and Complexity in Model Ecosystems*, Princeton (1973), Chapter 2. A. J. McKane and T. J. Newman, *Predator–prey cycles from resonant amplification of demographic stochasticity*, PRL **94**, 218102 (2005). N. G. van Kampen, *Stochastic Processes in Physics and Chemistry*, Chapter 10. M. L. Rosenzweig, Science **171**, 385 (1971).

6. Many interacting species: random matrices and stability

Real communities contain hundreds of species linked by thousands of interactions whose strengths are unknown and vary in time. Specifying them one by one is impossible. Physics suggests another route. Treat the interaction network as a random sample from an *ensemble* with a few statistical parameters (number of species, connectance, interaction strength, correlations), and ask which properties hold for almost every member of the ensemble. This chapter develops the generalised Lotka–Volterra model, reduces its stability to the spectrum of a random matrix, derives the classical spectral laws (semicircle, circular, elliptic), and obtains May’s complexity–stability criterion and its extensions to structured ecological networks.

6.1. Generalised Lotka–Volterra dynamics

For S species with abundances N_i ,

$$\frac{dN_i}{dt} = N_i \left[r_i - \sum_{j=1}^S M_{ij} N_j \right], \quad i = 1, \dots, S.$$

The diagonal $M_{ii} = 1/K_i$ encodes self-regulation (carrying capacities K_i when $r_i = 1$). The off-diagonal M_{ij} encode interactions: species j harms i if $M_{ij} > 0$ and benefits it if $M_{ij} < 0$. The *sign pairs* (M_{ij}, M_{ji}) classify interactions: $(+, -)$ predation, $(-, -)$ mutualism (both benefit), $(+, +)$ competition, $(0, \pm)$ commensalism or amensalism.

Two questions arise. **Coexistence**: starting from S species, how many survive? **Stability**: is the resulting equilibrium stable, locally or globally, and can a new species invade?

A coexistence equilibrium solves $\mathbf{r} = M\mathbf{N}^*$. It is **feasible** if all $N_i^* > 0$. Linearising, $\delta \dot{N}_i = \sum_j J_{ij} \delta N_j$ with **community matrix**

$$J_{ij} = -N_i^* M_{ij}.$$

The equilibrium is locally stable iff all eigenvalues of J have negative real parts.

Proposition 6.1 (Antisymmetric interactions give oscillations). *If M is antisymmetric ($M_{ij} = -M_{ji}$, pure predator–prey with no self-regulation) and $N_i^* > 0$, the eigenvalues of $J = -DM$, $D = \text{diag}(N_i^*)$, are purely imaginary.*

Proof. J is similar to $D^{-1/2} J D^{1/2} = -D^{1/2} M D^{1/2}$, which is real and antisymmetric, hence has imaginary eigenvalues. $\square$

Neutral oscillations, as in the two-species Lotka–Volterra model, are structurally unstable. Self-regulation turns them into damped oscillations.

Theorem 6.2 (Global stability for symmetric positive-definite interactions). *If M is symmetric and*

positive definite, the function

$$Q(\mathbf{N}) = \sum_i \left[N_i - N_i^* - N_i^* \ln \frac{N_i}{N_i^*} \right]$$

is a Lyapunov function and the feasible equilibrium is globally stable. More generally, for symmetric positive-definite M the dynamics maximises

$$\mathcal{L}(\mathbf{N}) = \sum_i r_i N_i - \frac{1}{2} \sum_{ij} N_i M_{ij} N_j,$$

which increases along trajectories; the stable state is the unique maximiser of $\mathcal{L}$ on $\mathbf{N} \geq 0$, where some species may be extinct.

Proof. $\dot{Q} = \sum_i (N_i - N_i^*) (r_i - (MN)_i) = - \sum_{ij} (N_i - N_i^*) M_{ij} (N_j - N_j^*) \leq 0$, using $r_i = (MN^*)_i$. Equality holds only at $\mathbf{N}^*$. $\square$

The final state is a **non-invadable** equilibrium $(N_1^*, \dots, N_k^*, 0, \dots, 0)$: the $S - k$ extinct species cannot re-invade, because their growth rate at the equilibrium is negative. Predicting k from the statistics of M is a hard problem. For random M and $\langle r \rangle = 0$, the probability that k of n species coexist is computable (Serván et al., 2018): roughly half survive.

Since the stability of $J = -DM$ is what matters, and many choices of $(\mathbf{r}, M)$ share the same $\mathbf{N}^*$, it is natural to model J (or M) directly as a **random matrix** from an ensemble.

6.2. Random matrices: spectral laws

Let X be an $S \times S$ matrix with independent entries of mean zero and variance σ^2 . Its eigenvalues scale as $\sigma\sqrt{S}$. Rescaled by this factor, their distribution becomes deterministic as $S \rightarrow \infty$ (self-averaging) and **universal**: it does not depend on the entry distribution beyond its first two moments.

Theorem 6.3 (Wigner's semicircle law). *If X is symmetric with i.i.d. upper-triangular entries of mean 0 and variance σ^2 , the empirical density of the eigenvalues of $X/\sqrt{S}$ converges to*

$$\rho(\lambda) = \frac{1}{2\pi\sigma^2} \sqrt{4\sigma^2 - \lambda^2}, \quad |\lambda| \leq 2\sigma.$$

Theorem 6.4 (Circular law (Ginibre, Girko, Tao–Vu)). *If all entries of X are i.i.d. with mean 0 and variance σ^2 , the eigenvalues of $X/\sqrt{S}$ are asymptotically **uniformly distributed in the disk** of radius σ centred at the origin of the complex plane.*

Theorem 6.5 (Elliptic law (Sommers et al., 1988)). *If the pairs (X_{ij}, X_{ji}) are i.i.d. with means 0, variances σ^2 and correlation $\rho = \mathbb{E}[X_{ij}X_{ji}]/\sigma^2$, the eigenvalues of $X/\sqrt{S}$ are uniformly distributed in the ellipse with horizontal semi-axis $\sigma(1 + \rho)$ and vertical semi-axis $\sigma(1 - \rho)$.*

$\rho = 0$ gives the circle. $\rho = 1$ (symmetric) gives a segment, $[-2\sigma, 2\sigma]$, on which the density is the semicircle (the projection of a uniform ellipse that degenerates). $\rho = -1$ (antisymmetric) gives a segment on the imaginary axis.

6.2.1. Derivation of the semicircle law

The method, standard in the physics of disordered systems, is worth knowing.

Resolvent. The spectral density is $\rho(\lambda) = \frac{1}{S} \sum_{\alpha} \delta(\lambda - \lambda_{\alpha})$. Using $\frac{1}{x-i\varepsilon} = \mathcal{P}\frac{1}{x} + i\pi\delta(x)$ as $\varepsilon \rightarrow 0^+$,

$$\rho(\lambda) = \frac{1}{\pi} \operatorname{Im} G(\lambda - i\varepsilon), \quad G(z) = \left\langle \frac{1}{S} \operatorname{Tr} \frac{1}{z - X} \right\rangle = \frac{1}{S} \frac{\partial}{\partial z} \langle \ln \det(z - X) \rangle.$$

For $S \rightarrow \infty$ the average over the ensemble is equivalent to a single large matrix.

Gaussian integral. For a matrix with positive-definite real part,

$$\det(z - X)^{-1/2} = \int \prod_j \frac{dx_j}{\sqrt{2\pi}} e^{-\frac{1}{2} \mathbf{x}^\top (z - X) \mathbf{x}} \equiv \mathcal{Z},$$

so that $\ln \det(z - X) = -2 \ln \mathcal{Z}$ and $G(z) = -\frac{2}{S} \partial_z \langle \ln \mathcal{Z} \rangle$.

Replica trick. Averaging a logarithm is hard. Use $\ln \mathcal{Z} = \lim_{n \rightarrow 0} (\mathcal{Z}^n - 1)/n$: compute $\langle \mathcal{Z}^n \rangle$ for integer n (n independent copies, the *replicas*), then continue to $n \rightarrow 0$. With Gaussian entries of variance σ^2/S (so that the spectrum stays $O(1)$), the average over X_{ij} is a Gaussian integral:

$$\left\langle e^{\frac{1}{2} \sum_{\alpha} \mathbf{x}^{\alpha} \cdot X \mathbf{x}^{\alpha}} \right\rangle = \exp \left[\frac{\sigma^2}{4S} \sum_{ij} \left(\sum_{\alpha} x_i^{\alpha} x_j^{\alpha} \right)^2 \right] = \exp \left[\frac{\sigma^2 S}{4} \sum_{\alpha\beta} Q_{\alpha\beta}^2 \right],$$

where $Q_{\alpha\beta} = \frac{1}{S} \sum_i x_i^{\alpha} x_i^{\beta}$ is the **overlap** between replicas. The disorder has disappeared, at the cost of coupling the replicas through Q .

Saddle point. Enforcing the definition of Q with a Lagrange multiplier $\hat{Q}$, the integral over $\mathbf{x}$ factorises over sites and

$$\langle \mathcal{Z}^n \rangle = \int dQ d\hat{Q} e^{S \mathcal{S}(Q, \hat{Q})}, \quad \mathcal{S} = -\frac{z}{2} \operatorname{Tr} Q + \frac{\sigma^2}{4} \operatorname{Tr} Q^2 + \frac{1}{2} \operatorname{Tr}(\hat{Q} Q) - \frac{1}{2} \ln \det \hat{Q} + \text{const.}$$

For $S \rightarrow \infty$ the integral is dominated by the stationary point. Eliminating $\hat{Q} = Q^{-1}$ gives the matrix equation

$$\sigma^2 Q^2 - zQ + \mathbb{1} = 0.$$

The replicas are equivalent, so the physical solution is **replica symmetric**, $Q_{\alpha\beta} = q \delta_{\alpha\beta}$ (replicas are independent copies of the same Gaussian integral, so off-diagonal overlaps vanish). Then $\sigma^2 q^2 - zq + 1 = 0$, and since $G(z) = \langle \frac{1}{S} \sum_i x_i^2 \rangle$ in the Gaussian representation, $G = q$:

$$G(z) = \frac{z - \sqrt{z^2 - 4\sigma^2}}{2\sigma^2},$$

choosing the root with $G \sim 1/z$ at large $|z|$. Taking the imaginary part on the real axis,

$$\rho(\lambda) = \frac{1}{\pi} \operatorname{Im} G(\lambda - i0^+) = \frac{\sqrt{4\sigma^2 - \lambda^2}}{2\pi\sigma^2} \mathbb{1}[|\lambda| < 2\sigma]. \quad \square$$

The same quadratic equation for G follows from a “cavity” argument: adding one row and column to the matrix and using the independence of the new entries. For a deterministic rank-one matrix $X = \frac{M_0}{S} \mathbf{1}\mathbf{1}^\top$, the Gaussian representation gives the exact spectrum $\{M_0, 0, \dots, 0\}$, a useful check of the method. The circular and elliptic laws need non-Hermitian generalisations (Hermitisation, or the quaternionic resolvent), but rest on the same ideas.

6.3. May's complexity–stability criterion

6.3.1. The random community

Following May (1972), let the community matrix be

$$J = -d\mathbf{1} + X,$$

where each off-diagonal X_{ij} is non-zero with probability C (the **connectance**) and then drawn from a distribution with mean 0 and variance σ^2 , and $-d < 0$ is the self-regulation. The entries of X have variance $C\sigma^2$, and shifting by $-d\mathbf{1}$ translates all eigenvalues by $-d$. (The trace $\text{Tr } J = -dS$ equals the sum of the eigenvalues, so the disk is centred at $-d$.) By the circular law the eigenvalues fill a disk of radius $\sigma\sqrt{SC}$ centred at $-d$. The equilibrium is stable iff the disk lies in the left half-plane:

Theorem 6.6 (May's stability criterion). *For large S , a random community is almost surely stable if*

$$\sigma\sqrt{SC} < d$$

and almost surely unstable if $\sigma\sqrt{SC} > d$.

The **complexity** $\sigma\sqrt{SC}$ (diversity S , connectance C , interaction strength σ) is bounded by self-regulation. Large, highly connected communities with strong interactions should not exist. Yet they do: this is the **complexity–stability paradox** that has shaped theoretical ecology for fifty years. The transition is sharp: the probability of stability drops from 1 to 0 within a window that shrinks as S grows, like a phase transition in the control parameter $\sigma\sqrt{SC}/d$.

6.3.2. Non-zero mean interactions

If the off-diagonal entries have mean $\mathbb{E}[X_{ij}] = \mu C \neq 0$ (e.g. predominantly competitive or mutualistic), write $X = \mu C(\mathbf{1}\mathbf{1}^\top - \mathbf{1}) + Y$ with Y zero-mean. The rank-one part has the eigenvalue $(S-1)\mu C$ on the uniform vector, and $-\mu C$ on the others. Hence:

- the **bulk** is a disk of radius $\sqrt{SV}$, $V = C[\sigma^2 + (1-C)\mu^2]$, centred at $-d - \mu C$;
- one **outlier** eigenvalue sits near $-d + (S-1)\mu C$, the common row sum.

Stability requires

$$\max\{\sqrt{SV} - \mu C, (S-1)\mu C\} < d.$$

For $\mu > 0$ in the convention of J (mutual benefit) the outlier grows linearly with S and dominates: predominantly mutualistic random communities are strongly destabilised. For $\mu < 0$ (competition) the outlier moves left and stability is decided by the bulk.

6.4. Ecological structure

Real interactions are not independent within pairs. Predation is $(+, -)$, so $J_{ij}J_{ji} < 0$. Mutualism and competition have $J_{ij}J_{ji} > 0$. Following Allesina and Tang (2012), construct the matrices pairwise. With probability C a pair (i, j) interacts, and the two strengths are drawn from half-normal distributions $|\mathcal{N}(0, \sigma^2)|$ with signs set by the interaction type:

- **predator–prey**: $(J_{ij}, J_{ji}) = (|x|, -|y|)$ or $(-|x|, |y|)$ with equal probability;
- **competition/mutualism mixture**: $(|x|, |y|)$ or $(-|x|, -|y|)$ with equal probability;

- **mutualism:** $(+|x|, +|y|)$; **competition:** $(-|x|, -|y|)$.

For the first two the mean is zero, the variance is $C\sigma^2$, and the pair correlation is $\rho = \pm \frac{\mathbb{E}|x|^2}{\sigma^2} = \pm \frac{2}{\pi}$ (minus for predator–prey). By the elliptic law the eigenvalues fill an ellipse centred at $-d$ with horizontal semi-axis $\sigma\sqrt{SC}(1 + \rho)$. The criteria become:

structure	correlation ρ	stability criterion
random	0	$\sigma\sqrt{SC} < d$
predator–prey	$-2/\pi$	$\sigma\sqrt{SC}(1 - 2/\pi) < d$
competition + mutualism	$+2/\pi$	$\sigma\sqrt{SC}(1 + 2/\pi) < d$
mutualism	mean > 0	$(S - 1)C\sigma\sqrt{2/\pi} < d$ (outlier)

Predator–prey interactions are stabilising: the ellipse is squeezed horizontally and stretched vertically, so for the same complexity the rightmost eigenvalue is $1 - 2/\pi \simeq 0.36$ times that of the random case. A random network is effectively a mixture (half predator–prey, a quarter each of competition and mutualism). Purely mutualistic networks are the least stable. The paradox persists, since every criterion still bounds $\sqrt{SC}$, but structure changes the numbers by large factors. Further stabilisation comes from correlations between strengths and topology (weak links in long loops), from food-web architecture, and from the fact that feasibility and stability are tied together in GLV.

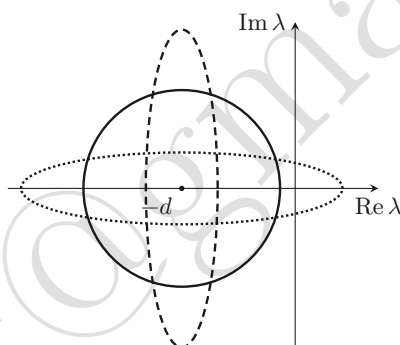


Figure: eigenvalue supports for the same complexity. Circle: random; dashed vertical ellipse: predator–prey; dotted horizontal ellipse: competition–mutualism mixture. Only the last crosses into the unstable half-plane here.

6.5. Food webs

Natural predator–prey networks are hierarchical. Species are arranged in **trophic levels**: **basal** species (B) consume no other species (plants, algae, detritus), **top** predators (T) have no predators, and **intermediate** species (I) are both. Let $a_{ij} = 1$ if j eats i . Classical descriptors are:

- the **generality** of j , its number of prey (in-degree $k_j^{\text{in}} = \sum_i a_{ij}$), and the **vulnerability** of i , its number of predators, both normalised by L/S ;
- the **trophic similarity** of i and j : the number of predators and prey they share, divided by S ;
- **omnivory** (feeding on several levels) and **cannibalism** (self-loops).

Definition 6.1 (Trophic level and coherence). The **trophic level** s_i is 1 for basal species and otherwise

one plus the average level of the prey,

$$s_j = 1 + \frac{1}{k_j^{\text{in}}} \sum_i a_{ij} s_i,$$

a linear system with a unique solution if every species is connected to a basal one. For each link $i \rightarrow j$ the **trophic distance** is $x_{ij} = s_j - s_i$, whose mean over links is exactly 1. The **trophic incoherence** is the standard deviation

$$q = \sqrt{\langle x^2 \rangle - 1}.$$

A perfectly layered web has $q = 0$; omnivory makes q larger. Johnson et al. (2014) showed that coherent food webs (q small) have few feedback loops and a small leading eigenvalue: **trophic coherence is stabilising**. Random networks with the same S , C , number of basal species and q reproduce the stability of real food webs much better than May's ensemble.

Two generative models reproduce the topology of empirical webs from a single “niche axis”:

- **Cascade model** (Cohen–Newman): each species gets a niche value $\eta_i \sim \mathcal{U}[0, 1]$; species j eats i with probability $2CS/(S-1)$ only if $\eta_i < \eta_j$, so loops are impossible.
- **Niche model** (Williams–Martinez, 2000): species j with niche value n_j eats *all* species in an interval of width $r_j = n_j y$ around a centre $c_j \in [r_j/2, n_j]$, with $y \sim \text{Beta}(1, \beta)$ and $\beta = 1/(2C) - 1$ so that the connectance is C . Diets are contiguous, which reproduces the observed intervality of food webs, and omnivory and cannibalism arise naturally.

6.6. Exercises

Exercise 6.1

(HW 3.1) Eigenvalue clouds. Choose an ecological structure (e.g. predator–prey or mutualism). Following Allesina and Tang's recipe, generate $S \times S$ matrices with connectance C for the completely random case and for the chosen structure, for $S = 20, 50, 100, 200$, and plot their eigenvalues in the complex plane.

Solution

For the random case each off-diagonal entry is non-zero with probability C and drawn from $\mathcal{N}(0, \sigma^2)$. For the structured cases pairs are drawn as described in the text. With $\sigma = 1$, $C = 0.2$, $d = 0$:

- the random eigenvalues fill a disk of radius $\sqrt{SC} = 2.0, 3.2, 4.5, 6.3$;
- the predator–prey eigenvalues fill a vertical ellipse with real semi-axis $\sqrt{SC}(1-2/\pi)$ and imaginary semi-axis $\sqrt{SC}(1+2/\pi)$;
- the mutualistic eigenvalues form a horizontal ellipse plus an isolated real outlier at $\simeq (S-1)C\sigma\sqrt{2/\pi}$, about 3, 7.8, 15.8, 31.8.

The clouds sharpen as S grows, as expected from self-averaging.

Exercise 6.2

(HW 3.2) Largest eigenvalue and transition. Plot $\max \text{Re } \lambda$ against S for the two ensembles. Then, adding self-regulation $-d$, plot the probability that $\max \text{Re } \lambda > 0$ against the proper control parameter (for the random case $R = \sigma\sqrt{SC}$).

Solution

Averaging 20 matrices ($C = 0.2$, $\sigma = 1$, $d = 0$):

S	random	predator-prey	prediction (PP)	mutualism	prediction (outlier)
20	1.57	0.48	0.73	3.63	3.03
50	2.91	1.07	1.15	8.54	7.82
100	4.26	1.48	1.63	16.6	15.8
200	6.17	2.28	2.30	32.5	31.8

The random case approaches $\sqrt{SC}$, the predator-prey case $\sqrt{SC}(1 - 2/\pi)$, with finite-size corrections, and the mutualistic outlier grows linearly in S . With $d > 0$, the probability of instability as a function of R/d is a sigmoid that sharpens with S and crosses $1/2$ at $R/d = 1$ (random) or at $R(1 - 2/\pi)/d = 1$ (predator-prey). Plotted against the correct control parameter for each ensemble, the curves collapse: the transition is universal.

Exercise 6.3

(HW 3.3, optional) The GLV Jacobian. For a GLV with $r_i = 1$, a random feasible equilibrium $N_i^* \sim \mathcal{U}[1, 10]$ and random interactions, the community matrix is $J = -\text{diag}(\mathbf{N}^*)M$. Are its eigenvalues still distributed according to the circular (or elliptic) law? Does the complexity–stability paradox survive?

Solution

Multiplying the rows of a random matrix by heterogeneous positive factors destroys the uniform density. The eigenvalues still fill a roughly circular region, but the density is no longer flat, and the radius is set by an effective variance: without self-regulation it is $\sigma\sqrt{SC\langle N^{*2} \rangle}$. For $S = 400$, $C = 0.2$, $\sigma = 1$, $N^* \sim \mathcal{U}[1, 10]$ ($\langle N^{*2} \rangle = 37$) the rightmost eigenvalue is 54, against $\sqrt{SC\langle N^{*2} \rangle} = 54$. Adding self-regulation d gives a heterogeneous shift $-dN_i^*$; stability is reached at $d \simeq 9$, i.e. when $d\langle N^* \rangle$ slightly exceeds the radius. The circular law no longer holds exactly, but the complexity–stability trade-off survives: the threshold on d still grows as $\sqrt{SC}\sigma$. The exact density follows from the theory of products of diagonal and random matrices (Gibbs, Grilli, Rogers, Allesina, 2018).

Exercise 6.4

(Antisymmetric spectrum) Prove that the eigenvalues of $J = DA$ are purely imaginary when A is real antisymmetric and D is diagonal with positive entries. What happens if a negative diagonal $-d\mathbf{1}$ is added to A ?

Solution

$D^{-1/2}(DA)D^{1/2} = D^{1/2}AD^{1/2}$ is real antisymmetric, so its eigenvalues are imaginary, and similar matrices share eigenvalues. With $A - d\mathbf{1}$, $J = DA - dD$. If $D = c\mathbf{1}$ the spectrum shifts to $-cd + iw$ and every mode is damped. For general D , $-dD$ is negative definite and the symmetric part of the similar matrix $D^{1/2}(A - d\mathbf{1})D^{1/2}$ is $-dD < 0$, so every eigenvalue has negative real part.

Exercise 6.5

(Outlier eigenvalue) For a matrix with diagonal $-d$ and off-diagonal entries all equal to μ (no randomness), find the spectrum. Compare with the random case with mean μ .

Solution

$J = -(d + \mu)\mathbf{1}\mathbf{1}^\top + \mu\mathbf{1}\mathbf{1}^\top$. On **1**: $\lambda = -d - \mu + S\mu = -d + (S - 1)\mu$. On the $(S - 1)$ -dimensional complement: $\lambda = -d - \mu$. In the random case the complement becomes a disk of radius $\sqrt{SV}$ around $-d - \mu C$, and the outlier is at $-d + (S - 1)\mu C$ up to $O(1)$ fluctuations. The deterministic mean decides the outlier; the randomness decides the bulk.

Exercise 6.6

(Trophic levels) Species B is basal. Species 2 eats B , species 1 eats B and 2, species 3 eats 1 and 2. Compute the trophic levels and the incoherence q .

Solution

$s_B = 1$, $s_2 = 1 + s_B = 2$, $s_1 = 1 + \frac{1}{2}(s_B + s_2) = 2.5$, $s_3 = 1 + \frac{1}{2}(s_1 + s_2) = 3.25$. Links and distances: $B \rightarrow 2$: 1; $B \rightarrow 1$: 1.5; $2 \rightarrow 1$: 0.5; $1 \rightarrow 3$: 0.75; $2 \rightarrow 3$: 1.25. The mean is $5/5 = 1$, as it must be. $\langle x^2 \rangle = (1 + 2.25 + 0.25 + 0.5625 + 1.5625)/5 = 1.125$, so $q = \sqrt{0.125} = 0.35$.

Exercise 6.7

(A real food web) The file `Food-web-Dataset-PNAS-05.xls` contains the weighted interaction matrix of a marine food web (prey in rows, predators in columns). Build the binary adjacency matrix, compute S , L , C , the trophic levels and the incoherence q . How does the stability of this structure compare with a random matrix with the same S and C ?

Solution

The web has $S = 249$ nodes and $L = 3313$ links, so $C = L/S^2 \simeq 0.053$. Five nodes have no prey (detritus and autotroph compartments) and are basal. Solving the linear system for s , levels reach $s_{\max} \simeq 5.8$, and the trophic distances have mean 1 and standard deviation $q \simeq 0.75$: substantial omnivory. Assign random half-normal strengths ($\sigma = 1$, $d = 0$) with the observed sign structure: the predator gains, $J_{ji} = +|x|$, and the prey loses, $J_{ij} = -|y|$. The rightmost eigenvalue averages 2.7. A May matrix with the same S and C gives $3.6 \simeq \sqrt{SC}$, and a predator-prey matrix with random topology would give $\sqrt{SC}(1 - 2/\pi) \simeq 1.3$. The predator-prey signs stabilise the real web relative to May's ensemble. Its heterogeneous topology (a few highly connected consumers) partly undoes the gain, which is why null models must preserve degree structure and trophic coherence to reproduce real food webs.

6.7. Sources

Prof. S. Suweis, *PMLS Course Notes*, Sections 4.3–4.7; Lectures 6–8 slides (GLV, Lyapunov functions, circular and elliptic laws, the complexity–stability paradox, structured interactions, food chains); Homework 3 and its detailed recipe (Allesina and Tang, SI). R. M. May, *Will a large complex system be stable?*, *Nature* **238**, 413 (1972). S. Allesina and S. Tang, *Stability criteria for complex ecosystems*, *Nature* **483**, 205 (2012). H. J. Sommers, A. Crisanti, H. Sompolinsky, Y. Stein, *PRL* **60**, 1895 (1988). C. A. Serván et al., *Nat. Ecol. Evol.* **2**, 1237 (2018). S. Johnson et al., *Trophic coherence determines food-web stability*, *PNAS* **111**, 17923 (2014). R. J. Williams and N. D. Martinez, *Nature* **404**, 180 (2000). T. Gibbs et al., *PRE* **98**, 022410 (2018).

Exercise dataset: Food-web-Dataset-PNAS-05.xls.

7. Mutualism, competition for resources and metabolic scaling

The random-matrix picture of Chapter 6 treats interactions as given. This chapter asks where they come from and what structure they acquire. Three themes recur. **Optimisation**: species that adapt their interactions or metabolism to maximise their own growth produce collective structures (nested mutualistic networks, coexistence of many competitors). **Resources**: modelling them explicitly turns competition into a geometric problem and resolves the paradox that many more species coexist than there are resources. **Scaling**: metabolism, growth and the organisation of whole forests follow power laws that can be derived from optimal use of space and energy.

7.1. Mutualistic networks and nestedness

7.1.1. Bipartite structure

Pollination, seed dispersal and many symbioses are **mutualistic**: both partners benefit. A plant–pollinator community with P pollinators and F flowers is a bipartite network, with interaction matrix (connectance $c = L/(PF)$)

$$M = \begin{pmatrix} \Omega_{PP} & \Gamma \\ \Gamma^\top & \Omega_{FF} \end{pmatrix},$$

where Γ encodes mutualism between the two guilds and the diagonal blocks encode competition *within* each guild (pollinators compete for flowers, flowers for pollinators).

Ordering rows and columns by degree, empirical Γ matrices (databases such as web-of-life.es) are strikingly **nested**: specialists interact with subsets of the partners of generalists, so the occupied entries fill a roughly triangular region in the upper-left corner. Nestedness is quantified, e.g., by the overlaps $o_{ij}^P = \sum_k \Gamma_{ik} \Gamma_{jk}$ (flowers shared by pollinators i and j) and by the NODF index (nestedness based on overlap and decreasing fill), between 0 and 1.

7.1.2. Is nestedness random?

Comparison with null models answers it:

- **Null model 1** (same S and c , edges placed at random): empirical networks are far more nested than random ones, well beyond three standard deviations.
- **Null model 2** (same degree sequences, edges reshuffled; configuration model): most of the nestedness is explained by the degree heterogeneity, but real networks remain systematically more nested, especially the most nested ones.

Degree heterogeneity alone is not enough: some mechanism must organise *which* partners each species chooses.

7.1.3. Adaptive foraging

Suweis et al. (2013) proposed that species rewire their interactions to increase their own abundance. Start from GLV dynamics with mutualism and competition,

$$\dot{x}_i = x_i \left(\alpha_i - \sum_j M_{ij} x_j \right),$$

with $\gamma_{ij} \sim -|\mathcal{N}(0, \sigma_\Gamma)|$ in the mutualistic block (negative M means benefit) and $\omega_{ij} \sim |\mathcal{N}(0, \sigma_\Omega)|$ in the competitive blocks, chosen so that the equilibrium $\mathbf{x}^* = M^{-1}\alpha$ is feasible and stable. At each step a species k tries to move one of its links from partner m to partner n , keeping its degree. The swap $M \rightarrow M' = M + \delta M$ is accepted if it increases k 's equilibrium abundance, $x_k^* > x_k^*$.

Simulations show that random networks evolve toward highly nested ones, and the **total** abundance increases, although each species optimises only its own. With competition turned off, the structure is different (symmetric block). Competition breaks the symmetry between the guilds and produces nestedness.

Why does the total increase? To first order in δM , $\delta \mathbf{x}^* = -M^{-1} \delta M \mathbf{x}^*$. In a mean-field approximation where competition is homogeneous, $M_0 = \mathbb{1} + \Omega$ with $\Omega_{ij} = c\omega(1 - \delta_{ij})$, the swap changes only the row and column of k at the positions of m and n , with $\delta M_{km} = -\delta M_{kn}$ (degree conserved). The changes of x_m and x_n then cancel, and

$$\delta x_{\text{tot}}^* \simeq \delta x_k^* > 0.$$

Selfish optimisation, accepted only when it benefits the swapping species, also benefits the community on average.

Why nestedness? Expanding $M^{-1} = M_0^{-1} - M_0^{-1} V M_0^{-1} + M_0^{-1} V M_0^{-1} V M_0^{-1} - \dots$ with the mutualistic part V , and using the explicit inverse of the homogeneous competition matrix (Exercise),

$$x_{\text{tot}}^* \propto \sum_{ij} (M^{-1})_{ij} \simeq k_0 + c_1 o, \quad o = \frac{1}{\gamma^2} \sum_{ij} \sum_k (\Gamma_{ki} \Gamma_{kj} + \Gamma_{ik} \Gamma_{jk}),$$

the second-order term is proportional to the total **overlap** o , a measure of nestedness. Maximising total abundance therefore maximises nestedness.

Stability. Optimised networks are *less* stable than random ones. With weak interactions, perturbation theory gives the eigenvalues of $J = -\text{diag}(\mathbf{x}^*)M$ as $\lambda_i \simeq -x_i^* + O(\text{interactions})$. Stability is controlled by the rarest species, and optimisation, by favouring abundant generalists, pushes some species toward extinction. The eigenvector associated with λ_{max} becomes **localised** on a few species: perturbations affect a small part of the community, which mitigates the loss of stability.

7.2. Competition for resources

7.2.1. The competitive exclusion principle

Two bacterial strains grown on the same single resource, with periodic dilution, cannot coexist: the one that grows faster on that resource increases its share at each cycle and eventually excludes the other (Gause, 1934). With two resources, each preferred by one strain, both can persist. The **competitive exclusion principle** (CEP) generalises this: at most as many species as limiting resources can coexist at equilibrium. Yet plankton communities with a handful of limiting nutrients host hundreds of species (Hutchinson's *paradox of the plankton*), and microbial communities show similar richness.

7.2.2. MacArthur's consumer–resource model

For m species (n_σ) and p abiotic resources (c_i),

$$\dot{n}_\sigma = n_\sigma \left(\sum_{i=1}^p v_i \alpha_{\sigma i} r_i(c_i) - \delta_\sigma \right), \quad \dot{c}_i = s_i - \sum_{\sigma=1}^m n_\sigma \alpha_{\sigma i} r_i(c_i),$$

with supply rates s_i , conversion efficiencies v_i , death rates δ_σ , Monod uptake $r_i(c) = c/(c + k_i)$, and **metabolic strategies** $\alpha_{\sigma i} \geq 0$: the rate at which species σ takes up resource i . A coexistence equilibrium requires, for every surviving σ , $\sum_i v_i \alpha_{\sigma i} r_i(c_i^*) = \delta_\sigma$: m linear equations in the p unknowns $r_i(c_i^*)$. For generic parameters there is no solution when $m > p$. This is the CEP in mathematical form.

With biotic resources (logistic growth instead of a constant supply), the QSA eliminates the resources and yields GLV equations among the consumers. Competition coefficients are then **overlaps of metabolic strategies**, $M_{\sigma\tau} \propto \sum_i \alpha_{\sigma i} \alpha_{\tau i}$ (Exercise).

7.2.3. Metabolic trade-offs

Posfai, Taillefumier and Wingreen (2017) noted that producing transporters and enzymes costs energy, and each organism has a finite budget. Impose

$$\sum_i w_i \alpha_{\sigma i} = E \quad \text{for every species } \sigma,$$

with w_i the cost of using resource i , and let all death rates equal δ . At a coexistence equilibrium $\sum_i v_i \alpha_{\sigma i} r_i^* = \delta$ for all σ . With the budget, $r_i^* = \frac{\delta}{E} \frac{w_i}{v_i}$ solves all m equations at once, so the degeneracy that produced the CEP is lifted. The resource equations then give $\sum_\sigma n_\sigma^* \alpha_{\sigma i} = \frac{E}{\delta} \frac{s_i v_i}{w_i}$, and summing the population equations, $\dot{N} = \tilde{s} - \delta N$ with $N = \sum_\sigma n_\sigma$ and $\tilde{s} = \sum_i s_i v_i$: the total population is fixed, $N^* = \tilde{s}/\delta$. In normalised variables

$$x_\sigma = \frac{n_\sigma}{N}, \quad \tilde{\alpha}_{\sigma i} = \frac{w_i \alpha_{\sigma i}}{E}, \quad \hat{s}_i = \frac{s_i v_i}{\sum_j s_j v_j},$$

each with unit sum, the equilibrium condition is

$$\sum_\sigma x_\sigma^* \tilde{\alpha}_\sigma = \hat{\mathbf{s}}.$$

Proposition 7.1 (Coexistence as a convex-hull condition). *The vectors $\tilde{\alpha}_\sigma$ and $\hat{\mathbf{s}}$ lie in the $(p-1)$ -dimensional simplex. All m species can coexist (with positive x_σ^*) iff the normalised supply $\hat{\mathbf{s}}$ lies inside the **convex hull** of the strategies $\{\tilde{\alpha}_\sigma\}$. If $\hat{\mathbf{s}}$ is outside, species go extinct until at most p remain: the CEP is recovered.*

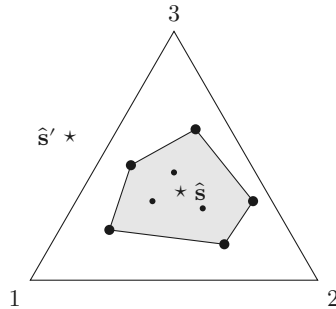


Figure: three resources. Dots are normalised metabolic strategies; the shaded region is their convex hull. With supply $\hat{\mathbf{s}}$ inside, all species coexist; with $\hat{\mathbf{s}}'$ outside, some go extinct.

The trade-off explains the paradox, but it is **fine tuned**. If budgets differ slightly between species, coexistence of more than p species is broken again.

7.2.4. Adaptive metabolic strategies

Microbes do change their metabolism: *E. coli* growing on glucose and lactose first consumes glucose, pauses, and then switches on the *lac* operon to consume lactose (Monod's **diauxic shift**, 1941). Pacciani-Mori et al. (2020) let strategies evolve to increase each species' growth rate $g_\sigma = \sum_i v_i \alpha_{\sigma i} r_i(c_i)$, subject to the *soft* budget constraint $\sum_i \alpha_{\sigma i} \leq E_\sigma$:

$$\dot{\alpha}_{\sigma i} = \alpha_{\sigma i} d \left[v_i r_i(c_i) - \frac{\Theta(\sum_k \alpha_{\sigma k} - E_\sigma)}{\sum_k \alpha_{\sigma k}} \sum_j v_j \alpha_{\sigma j} r_j(c_j) \right].$$

This is gradient ascent of g_σ , projected onto the constraint surface when the budget is saturated. The projection removes the component of the gradient along the constraint's normal. In general, maximising $Q(\mathbf{x})$ under $\rho(\mathbf{x}) = 0$ uses $\dot{\mathbf{x}} = \nabla Q - \frac{\nabla Q \cdot \nabla \rho}{|\nabla \rho|^2} \nabla \rho$, which keeps ρ constant and increases Q by the Cauchy–Schwarz inequality. The factor $\alpha_{\sigma i}$ keeps strategies non-negative, and d is the **adaptation velocity**.

Results:

1. With one species and two resources, the model reproduces diauxic shifts quantitatively (*E. coli*, yeast on galactose and ethanol).
2. With many species, strategies evolve until the supply vector lies inside their convex hull: the CEP is violated **without fine tuning**, even when budgets differ.
3. When the supply fluctuates in time, adaptive communities track it and retain more species than fixed-strategy ones: adaptation increases resilience.
4. If adaptation is too slow relative to ecological dynamics, extinctions occur before strategies adjust and the CEP is recovered: the adaptation velocity is a crucial parameter.

7.2.5. Proteome allocation

Where does the budget come from? Bacterial proteomes divide into sectors (Scott and Hwa, 2010): a fixed housekeeping sector Q , a ribosomal sector R that synthesises proteins, and a metabolic sector P that imports and processes nutrients, with $\Phi_P + \Phi_R + \Phi_Q = 1$. Empirically, at steady exponential growth with rate g ,

$$\Phi_R = \Phi_R^{(0)} + \frac{g}{k_t}, \quad \Phi_P = \frac{g}{k_n},$$

the **bacterial growth laws**. Here k_t is the translational capacity and k_n the nutritional capacity of the medium, which grows with resource concentration through a Monod factor. Growing faster needs more ribosomes and more metabolic machinery from a fixed budget. A consumer–resource model with constrained proteome allocation (each species distributing $\Phi_\sigma^P = \sum_i \Phi_{\sigma i}$ across resources, with Φ_σ^Q incompressible) gives a mechanistic origin for metabolic trade-offs and adaptive strategies, linking cellular physiology to community coexistence.

7.3. Metabolic scaling and the structure of forests

7.3.1. Kleiber's law from optimal transport

Across organisms the basal metabolic rate scales as $B \propto M^{3/4}$ (Kleiber, 1947), not as $M^{2/3}$, the surface-to-volume law expected if metabolism were limited by heat dissipation through the skin. Banavar, Maritan and Rinaldo (1999) gave a geometric argument. A circulatory network delivers nutrients from a source to $N \propto B$

sites filling a D -dimensional region of linear size L , so $B \propto L^D$. The total amount of fluid in transit (blood) is at least proportional to the sum of the distances from the source to all sites, $M_{\text{blood}} \propto L^D \cdot L = L^{D+1}$. Minimising it, $M \propto B^{(D+1)/D}$, i.e.

$$B \propto M^{D/(D+1)} = M^{3/4} \quad (D = 3).$$

Quarter powers follow from efficient supply in three dimensions. They do not depend on fractal details of the network.

7.3.2. Ontogenetic growth

Allocating incoming energy between maintenance and the creation of new tissue gives (West, Brown, Enquist, 2001)

$$\frac{dm}{dt} = a m^{3/4} \left[1 - \left(\frac{m}{M} \right)^{1/4} \right],$$

with M the adult mass. With $r = (m/M)^{1/4}$ and $\tau = at/(4M^{1/4})$, all species collapse onto the **universal growth curve** $r(\tau) = 1 - [1 - r(0)]e^{-\tau}$: chickens, cows and fish follow the same dimensionless curve, with a characteristic time $\propto M^{1/4}$.

7.3.3. Trees and forests

For a tree, metabolism is the water transpired by the leaves, and allometry follows from crown geometry (Simini et al., 2010). If the crown radius scales with height as $r_{\text{crown}} \sim h^H$ (with $H = 1$ for isometric crowns near the tropics), the crown volume, and hence leaf area and metabolism, scale as $B \sim h^{1+2H}$, while the water mass scales as $M \sim h^{2+2H}$. Hence

$$B \sim M^{\frac{1+2H}{2+2H}} \leq M^{3/4},$$

with Kleiber's law recovered for $H = 1$. Since water flux through the trunk $\propto r^2$, the trunk radius scales as $r \sim h^{(1+2H)/2}$.

At the community level, assume that tree crowns fill the available space as efficiently as possible. If $NP(h) dh$ trees have heights in $[h, h + dh]$, requiring that the crown volume per unit area be the same at every height scale gives the ideal distribution

$$P(h) \propto h^{-(1+2H)}, \quad P(r) \propto r^{-7/3} \quad (H = 1).$$

Real forests are limited by total resources, and the ideal power law acquires a finite-size cut-off. Minimising the Kullback–Leibler distance from the ideal distribution under the constraint of fixed total metabolic rate gives

$$P(r) \propto r^{-7/3} \exp[-(r/r_c)^2],$$

which fits trunk-diameter distributions of forests worldwide. Deviations of the measured exponent from the ideal one quantify **disturbance**: disturbed forests lack large trees and have shallower size distributions. The same finite-size-scaling logic (a power law times a scaling function of x/x_c) underlies the distribution of distances to the nearest larger tree and simple packing problems.

7.4. Exercises

Exercise 7.1

(From CRM to GLV) Consider consumers n_σ and biotic resources $\dot{c}_i = r_i c_i (1 - c_i/k) - \sum_\sigma \alpha_{\sigma i} n_\sigma c_i$, with consumer growth $\dot{n}_\sigma = n_\sigma (\sum_i v_i \alpha_{\sigma i} c_i - \delta_\sigma)$. Show that the QSA yields GLV equations among consumers and compute the interaction matrix.

Solution

QSA with $c_i > 0$: $c_i^* = k(1 - \frac{1}{r_i} \sum_\tau \alpha_{\tau i} n_\tau)$. Substituting,

$$\dot{n}_\sigma = n_\sigma \left(\underbrace{k \sum_i v_i \alpha_{\sigma i}}_{a_\sigma} - \delta_\sigma - \sum_\tau \underbrace{k \sum_i \frac{v_i \alpha_{\sigma i} \alpha_{\tau i}}{r_i}}_{M_{\sigma\tau}} n_\tau \right).$$

Competition coefficients are weighted overlaps of metabolic strategies: species that eat the same things compete. M is a Gram-like matrix, symmetric if v_i is uniform and positive semi-definite, so the Lyapunov theorem of Chapter 6 applies and the dynamics converges to a non-invadable equilibrium.

Exercise 7.2

(Mean-field competition inverse) Show that $M_0 = (1 - c\omega)\mathbf{1} + c\omega\mathbf{1}\mathbf{1}^\top$ (size s) has inverse A with $A_{ii} = a$, $A_{ij} = -b$ and $a = \frac{1+c\omega(s-2)}{(1-c\omega)[1+c\omega(s-1)]}$, $b = \frac{c\omega}{(1-c\omega)[1+c\omega(s-1)]}$.

Solution

By the Sherman–Morrison formula, $(u\mathbf{1} + v\mathbf{1}\mathbf{1}^\top)^{-1} = \frac{1}{u}\mathbf{1} - \frac{v}{u(u+sv)}\mathbf{1}\mathbf{1}^\top$ with $u = 1 - c\omega$, $v = c\omega$. Off-diagonal: $-\frac{c\omega}{(1-c\omega)(1+(s-1)c\omega)} = -b$. Diagonal: $\frac{1}{1-c\omega} - b = \frac{1+(s-1)c\omega-c\omega}{(1-c\omega)(1+(s-1)c\omega)} = a$. The row sums $a - (s-1)b = \frac{1}{1+(s-1)c\omega}$ give the mean-field equilibrium abundance $x^* = \alpha/(1 + (s-1)c\omega)$.

Exercise 7.3

(Nestedness) Compute the pairwise overlaps and the total overlap o for the two 3×3 pollinator–flower matrices

$$\Gamma_1 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \Gamma_2 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix},$$

which have the same number of links. Which is more nested?

Solution

Pollinator overlaps $o_{ij}^P = \sum_k \Gamma_{ik} \Gamma_{jk}$ for $i \neq j$. Γ_1 : $o_{12} = 2$, $o_{13} = 1$, $o_{23} = 1$, sum 4; by symmetry the flower overlaps also sum to 4. Γ_2 : each pair of pollinators shares exactly one flower, so the sum is 3, and likewise 3 for flowers. Γ_1 (triangular) has the larger overlap: specialists interact with subsets of the generalists' partners. Γ_2 (circulant) has the same degrees, but its overlaps are spread evenly, and it is not nested.

Exercise 7.4

(Convex hull) Two resources ($p = 2$) and three species with normalised strategies $\tilde{\alpha}_1 = (0.8, 0.2)$, $\tilde{\alpha}_2 = (0.5, 0.5)$, $\tilde{\alpha}_3 = (0.3, 0.7)$. For which supplies $\hat{s} = (\hat{s}, 1 - \hat{s})$ can all three coexist? Is the equilibrium unique?

Solution

On the simplex (a segment) the convex hull of the strategies is $\hat{s} \in [0.3, 0.8]$. Inside it a solution with $x_\sigma \geq 0$ exists, but $\sum_\sigma x_\sigma \tilde{\alpha}_{\sigma 1} = \hat{s}$ and $\sum_\sigma x_\sigma = 1$ are only two equations for three unknowns, so there is a one-parameter family of equilibria: neutral coexistence along a line. With $\hat{s}$ strictly between the extremes, all three can coexist with positive abundances (e.g. for $\hat{s} = 0.5$, $x = (t, 1 - 2t, t)$ for $0 < t < 1/2$, since $0.8t + 0.5(1 - 2t) + 0.3t = 0.5$). For $\hat{s} < 0.3$ or $\hat{s} > 0.8$ the supply is outside the hull and only species at the boundary of the hull can survive.

Exercise 7.5

(Quarter powers in D dimensions) Using the optimal-network argument, find the metabolic exponent in $D = 2$ and explain why the heat-dissipation argument gives $2/3$ instead.

Solution

$M \propto B^{(D+1)/D}$ gives $B \propto M^{D/(D+1)}$: $2/3$ in $D = 2$, $3/4$ in $D = 3$. Heat dissipation through a surface gives $B \propto L^2 \propto M^{2/3}$ for a body of volume L^3 . The network argument instead counts the transported material, which grows faster than the volume served (L^{D+1} versus L^D) and makes metabolism sublinear with the exponent $D/(D+1)$.

Exercise 7.6

(Universal growth curve) Solve $\dot{m} = am^{3/4}[1 - (m/M)^{1/4}]$ with $r = (m/M)^{1/4}$, and show that the time to reach 90% of the adult mass scales as $M^{1/4}$.

Solution

$\dot{r} = \frac{1}{4}(m/M)^{-3/4} \frac{\dot{m}}{M} = \frac{a}{4M^{1/4}}(1 - r)$, so $1 - r = (1 - r_0)e^{-at/(4M^{1/4})}$. Reaching $m = 0.9M$ means $r = 0.9^{1/4} = 0.974$, so $t_{90} = \frac{4M^{1/4}}{a} \ln \frac{1-r_0}{0.026}$, proportional to $M^{1/4}$ for a fixed initial ratio. A mouse (30 g) and a cow (300 kg) differ by 10^4 in mass and by 10 in growth time.

Exercise 7.7

(Tree allometry) With $r_{\text{crown}} \sim h^H$, derive the exponents of $B(h)$, $M(h)$ and $B(M)$, and evaluate them for $H = 1$ and $H = 1/2$.

Solution

Crown volume $\sim h r_{\text{crown}}^2 = h^{1+2H}$, so $B \sim h^{1+2H}$. Water mass: the conducting xylem connects every leaf to the ground, so $M \sim (\text{leaf area}) \times h$; with leaf density constant, leaf area $\propto$ crown volume $\sim h^{1+2H}$, hence $M \sim h^{2+2H}$. Then $B \sim M^{(1+2H)/(2+2H)}$. $H = 1$: $B \sim h^3$, $M \sim h^4$, $B \sim M^{3/4}$ (Kleiber). $H = 1/2$ (narrow crowns, high latitudes): $B \sim h^2$, $M \sim h^3$, $B \sim M^{2/3}$. The exponent never exceeds $3/4$ for $H \leq 1$.

7.5. Sources

Prof. S. Suweis, *PMLS Course Notes*, Sections 4.8–4.11; Lecture 8 slides (mutualistic networks, nestedness, null models, adaptive foraging) and Lecture 9 slides (competitive exclusion, consumer–resource models, metabolic trade-offs, adaptive strategies, diauxic shift, proteome allocation); A. Maritan’s lecture on scaling in ecology, metabolic theory and community patterns (Lecture 22). S. Suweis, F. Simini, J. R. Banavar, A. Maritan, *Nature* **500**, 449 (2013). A. Posfai, T. Taillefumier, N. S. Wingreen, *PRL* **118**, 028103 (2017). L. Pacciani-Mori, A. Giometto, S. Suweis, A. Maritan, *PLoS Comput. Biol.* **16**, e1007896 (2020). M. Scott et al., *Science* **330**, 1099 (2010). J. R. Banavar, A. Maritan, A. Rinaldo, *Nature* **399**, 130 (1999). G. B. West, J. H. Brown, B. J. Enquist, *Nature* **413**, 628 (2001). F. Simini, T. Anfodillo, M. Carrer, J. R. Banavar, A. Maritan, *PNAS* **107**, 7658 (2010).

Part II.

Part II: Neuroscience

8. The brain as a complex system: measuring and describing neural activity

The second module applies the toolbox of Part I to the nervous system. The analogy is direct: individuals become neurons, abundance becomes activity, predator–prey pairs become excitatory–inhibitory pairs, and generalised Lotka–Volterra communities become recurrent neural networks. This chapter introduces the brain as a complex system and the techniques that measure it. It then describes neural activity statistically: firing rates, tuning curves, spike-train statistics and the Poisson null models against which collective phenomena such as **neuronal avalanches** must be tested.

8.1. The brain as a complex system

The human brain contains about 10^{11} neurons, each receiving about 10^4 synapses, for a total of about 10^{14} – 10^{15} connections. It is organised on several scales: molecular (ion channels), cellular (neurons), circuit (microcircuits and cortical columns), mesoscopic (brain areas), and the whole brain. Its functions emerge from the interactions between these levels. No single neuron “understands” a face or remembers a phone number.

The modelling programme parallels ecology:

ecology	neuroscience
one species, $\dot{x} = rx(1 - x)$	one neuron, activity $x(t)$
predator–prey	excitatory–inhibitory neurons
S species, GLV, $\dot{x}_i = r_i x_i (1 - \sum_j M_{ij} x_j)$	N neurons, recurrent networks, $\dot{x}_i = -\tau x_i + \sum_j M_{ij} \tanh(x_j)$

Two **emergent patterns** of collective activity frame the module.

- **Neuronal avalanches** (Beggs and Plenz, 2003). In cortical slices recorded with multi-electrode arrays, activity propagates in bursts separated by silence. Bin time in windows Δt ; an avalanche is a run of consecutive non-empty bins, and its size s is the total number of events. The size distribution follows a power law, $P(s) \sim s^{-3/2}$, with a cut-off set by the number of electrodes, and durations follow $P(T) \sim T^{-2}$. These are the exponents of a critical branching process, which suggests that the cortex operates near a critical point (Chapter 13).
- **Brain waves.** Population activity oscillates in characteristic bands: delta (0.5–4 Hz, deep sleep), theta (4–8 Hz), alpha (8–13 Hz, relaxed wakefulness), beta (13–32 Hz, active thinking) and gamma (32–100 Hz, attention and binding). Oscillations require the synchronisation of large populations and arise from excitatory–inhibitory loops (Chapter 10).

How complex should a model be? The answer depends on the question and on the quality of the data. Models range along a scale of resolution: linear models, mean-field rate models, interacting particles (binary neurons), neural-mass models, networks of neural masses, neural fields, and detailed simulations (e.g. the Blue Brain project). A recurring question is whether **structure determines function**: whether the anatomical network predicts the correlations of spontaneous (“resting-state”) activity.

8.2. Measuring brain activity

Techniques differ in what they measure and at which spatial and temporal resolution.

Non-invasive, whole brain:

- **Electroencephalography (EEG)**: electrodes on the scalp record the electric potential generated by synchronous activity of large populations. It has millisecond resolution but poor spatial resolution, and reconstructing sources from surface signals is an ill-posed inverse problem. Magnetoencephalography (MEG) is the magnetic analogue.
- **Diffusion tensor imaging (DTI)**: MRI measurements of the anisotropic diffusion of water reveal the white-matter fibre tracts. Tractography reconstructs the **structural connectome**: a weighted network between 10^2 – 10^3 brain regions (parcels).
- **Functional MRI (fMRI)**: the blood-oxygen-level-dependent (BOLD) signal measures changes in blood flow and oxygenation that follow neural activity, with a resolution of a few millimetres and a few seconds. Pearson correlations between regional BOLD time series define **functional connectivity** networks, with large individual variability.

Invasive, few neurons:

- **Electrode probes** record the extracellular potential near neurons (spikes and local field potentials) or the membrane potential from inside a cell. Modern silicon probes (Neuropixels) record from hundreds of neurons along a shank simultaneously.
- **Neuronal cultures** grown on multi-electrode arrays: homogeneous or clustered networks whose spontaneous activity can be followed for weeks.
- **Calcium imaging**: genetically encoded fluorescent indicators (GCaMP) light up when calcium enters the cell during spiking. Optical (two-photon) imaging then follows thousands of neurons in behaving animals.
- **Optogenetics**: neurons engineered to express light-sensitive ion channels can be switched on and off with light, giving causal control of activity.

8.3. From stimulus to response

8.3.1. Firing rates

A neuron's output is a sequence of action potentials ("spikes") at times $t_1, \dots, t_n$. Since their shape is stereotyped, a spike train is represented by the **neural response function**

$$\rho(t) = \sum_{i=1}^n \delta(t - t_i), \quad \sum_i h(t - t_i) = \int d\tau h(\tau) \rho(t - \tau).$$

Spike trains vary from trial to trial even with identical stimuli. Useful rates are:

- the **spike-count rate** $r = n/T = \frac{1}{T} \int_0^T \rho(\tau) d\tau$;
- the **time-dependent firing rate** $r(t) = \frac{1}{\Delta t} \int_t^{t+\Delta t} \langle \rho(\tau) \rangle d\tau$, with $\langle \cdot \rangle$ an average over trials. For small Δt , $r(t)\Delta t$ is the probability of a spike in $[t, t + \Delta t]$, and for any smooth h , $\int h(\tau) \langle \rho(t - \tau) \rangle d\tau = \int h(\tau) r(t - \tau) d\tau$.

From a single trial, $r(t)$ is estimated by filtering the spike train with a window w : $r(t) \simeq \int w(\tau) \rho(t - \tau) d\tau$. Options are a rectangular window of width Δt , a Gaussian $w(\tau) \propto e^{-\tau^2/2\sigma_w^2}$, or the causal kernel $w(\tau) = [\alpha^2 \tau e^{-\alpha\tau}]_+$, which uses only past spikes. The choice of window sets a trade-off between temporal resolution and noise.

8.3.2. Tuning curves

The **tuning curve** $f(s) = \langle r \rangle$ gives the mean firing rate as a function of a stimulus parameter s . Examples are a Gaussian tuning to the orientation of a bar in primary visual cortex, $f(s) = r_{\max} e^{-(s-s_{\max})^2/2\sigma^2}$; cosine tuning to arm-movement direction in motor cortex, $f(s) = r_0 + (r_{\max} - r_0) \cos(s - s_{\max})$; and sigmoidal tuning to binocular disparity. Populations of differently tuned neurons encode stimuli collectively.

8.3.3. Psychophysical laws

Sensory systems respond over enormous ranges: the retina detects single photons and works in daylight with 10^6 times more. This is possible through **adaptation**: neurons signal *changes* relative to a background.

- **Weber's law**: the just-noticeable difference Δs is proportional to the stimulus, $\Delta s/s = k$.
- **Fechner's law**: integrating Weber's law, perceived intensity is logarithmic in the stimulus, $P = k \ln(s/s_0)$.

Both follow from coding efficiently a stimulus whose fluctuations scale with its mean (a project topic; see Sims, 2018).

8.3.4. Spike-triggered average

Because of adaptation one studies fluctuations around a zero-mean stimulus. Assuming stationarity and ergodicity (trial averages equal time averages), the **spike-triggered average** stimulus is

$$C(\tau) = \left\langle \frac{1}{n} \sum_{i=1}^n s(t_i - \tau) \right\rangle \simeq \frac{1}{\langle n \rangle} \int_0^T r(t) s(t - \tau) dt = \frac{Q_{rs}(-\tau)}{\langle r \rangle},$$

where $Q_{rs}(\tau) = \frac{1}{T} \int_0^T r(t) s(t + \tau) dt$ is the rate-stimulus correlation. $C(\tau)$ shows what, on average, the stimulus did just before a spike: it is the first step in building a linear model of the neuron's response (*reverse correlation*).

8.4. Spike-train statistics

8.4.1. The homogeneous Poisson process

The simplest null model assumes spikes are independent events at a constant rate r . Divide $[0, T]$ into M bins of width $\Delta t = T/M$, small enough to hold at most one spike. The probability of n spikes is

$$P_T(n) = \lim_{\Delta t \rightarrow 0} \binom{M}{n} (r\Delta t)^n (1 - r\Delta t)^{M-n} = \frac{(rT)^n}{n!} e^{-rT},$$

using $(1 - r\Delta t)^M \rightarrow e^{-rT}$ and $\frac{M!}{(M-n)!} \simeq M^n$. Mean and variance are both rT . The probability density of a specific ordered sequence of spike times is $p(t_1, \dots, t_n) = r^n e^{-rT}$, and the **interspike interval** (ISI) distribution is exponential, $p(\tau) = r e^{-r\tau}$, with mean $1/r$ and coefficient of variation $C_V = \sigma_\tau / \langle \tau \rangle = 1$.

8.4.2. The inhomogeneous Poisson process

If the rate varies in time, the probability of no spike between t_i and t_{i+1} is $\prod_m (1 - r(t_i + m\Delta t)\Delta t) \rightarrow \exp(-\int_{t_i}^{t_{i+1}} r(t)dt)$. Multiplying over intervals,

$$p(t_1, \dots, t_n) = \prod_{i=1}^n r(t_i) \exp\left(-\int_0^T r(t) dt\right).$$

Definition 8.1 (Simulating spike trains by thinning). To sample an inhomogeneous Poisson process with $r(t) \leq r_{\max}$: generate a homogeneous process at rate $r_{\max}$ (exponential gaps of mean $1/r_{\max}$), then keep each candidate spike at t_i with probability $r(t_i)/r_{\max}$ (draw $u \sim \mathcal{U}(0, 1)$, keep if $u < r(t_i)/r_{\max}$). Equivalently, on a grid of width dt , fire with probability $r(t) dt$ in each bin.

8.4.3. Comparison with data

Real spike trains are close to, but not exactly, Poisson. The variance of the spike count across trials grows roughly linearly with the mean, with a **Fano factor** $F = \sigma_n^2 / \langle n \rangle$ near 1 (fits $\sigma_n^2 = A \langle n \rangle^B$ give $A, B \approx 1$, varying with the counting window). ISI histograms, however, are not exponential at short intervals: after a spike a neuron cannot fire again for a few milliseconds (**refractoriness**). A Poisson process with a dead time (or a gamma-distributed ISI) fits much better.

8.5. Avalanches from Poisson null models

Power-law avalanches are often taken as evidence of criticality. A good null model shows how easily independent neurons, driven by fluctuating inputs, produce avalanche statistics.

Consider N neurons, time bins of width dt , and spiking probability $\lambda_j dt$ in bin j for every neuron. The probability that bin j is not empty is $1 - (1 - \lambda_j dt)^N$, so the probability that an avalanche lasts longer than n bins, given the rates, is

$$P_{>}(n | \lambda) = \prod_{j=1}^n [1 - (1 - \lambda_j dt)^N].$$

Rates redrawn at every bin. If λ_j are independent draws from $Q(\lambda)$, $P_{>}(n) = q^n$ with $q = 1 - \int Q(\lambda)(1 - \lambda dt)^N d\lambda$: durations are **geometric** (exponential), with no power law. For an exponential $Q(\lambda) = \gamma e^{-\gamma\lambda}$ and $(1 - \lambda dt)^N \simeq e^{-N\lambda dt}$, the silence probability is $\frac{\gamma}{\gamma + N dt}$.

Rate fixed during each avalanche. If λ is drawn once and kept for the whole avalanche (a slowly varying common input), then with $(1 - \lambda dt)^N \simeq e^{-\delta\lambda}$, $\delta = N dt$,

$$P_{>}(n) = \gamma \int_0^\infty d\lambda e^{-\gamma\lambda} (1 - e^{-\delta\lambda})^n = \frac{\gamma}{\delta} B\left(\frac{\gamma}{\delta}, n+1\right) \simeq \Gamma\left(1 + \frac{\gamma}{\delta}\right) n^{-\gamma/\delta}.$$

For large n the integrand is a sharp step at $\lambda^* = \ln n / \delta$ (saddle point), so $P_{>}(n) \simeq e^{-\gamma\lambda^*} = n^{-\gamma/\delta}$: a **power law**, with an exponent set by the parameters, not by any critical point. Slow, shared fluctuations of the input suffice to produce scale-free avalanches (Touboul and Destexhe, 2017). Criticality claims require more than a power law: exponent relations, data collapse and the dependence on the bin width (Mariani et al., 2022).

8.6. Exercises

Exercise 8.1

(Spike-count statistics) Show that for a homogeneous Poisson process the Fano factor is 1 for any window, the ISI is exponential, and $C_V = 1$. How do refractoriness and bursting change C_V ?

Solution

$\langle n \rangle = \sigma_n^2 = rT$ gives $F = 1$. The probability of no spike in $[t_i, t_i + \tau]$ and one in $[t_i + \tau, t_i + \tau + d\tau]$ is $e^{-r\tau} r d\tau$, so $p(\tau) = re^{-r\tau}$, with $\langle \tau \rangle = 1/r$, $\sigma_\tau = 1/r$, $C_V = 1$. A dead time τ_{ref} shifts the distribution: $p(\tau) = re^{-r(\tau - \tau_{\text{ref}})}$ for $\tau > \tau_{\text{ref}}$, so $C_V = \frac{1/r}{\tau_{\text{ref}} + 1/r} < 1$ (more regular). Bursting produces many short and some long intervals, so $C_V > 1$.

Exercise 8.2

(Thinning) Simulate an inhomogeneous Poisson process with $r(t) = 20[1 + \sin(2\pi t)]$ Hz on $[0, 10]$ s by thinning from $r_{\text{max}} = 40$ Hz. Check that the expected count is 200 and that the peri-stimulus time histogram reproduces $r(t)$.

Solution

Generate exponential gaps of mean $1/40$ s, giving about 400 candidates, and accept each with probability $r(t_i)/40$. The expected number of accepted spikes is $\int_0^{10} r(t) dt = 200$, since the sine integrates to zero over whole periods. Histogramming spike times modulo 1 s over many trials reproduces the sinusoidal profile, with Poisson errors $\propto \sqrt{\text{count}}$. The method is exact: the accepted points form a Poisson process with intensity $r_{\text{max}} \cdot r(t)/r_{\text{max}} = r(t)$.

Exercise 8.3

(HW 4) Poisson null models of avalanches. Generate spike trains for $N = 100$ neurons with $dt = 0.01$, firing probability $\lambda_t dt$ per bin, for at least 10^4 bins, in three cases: (1) independent rates for each neuron, drawn every bin from $p(\lambda) = re^{-r\lambda}$ with $r = 0.1$; (2) a common rate for all neurons drawn every bin from the same exponential; (3) a common rate drawn every bin from a gamma distribution with mean 0.2 and variance 10. Compare the avalanches. Optional: compute the avalanche durations analytically.

Solution

The exponential has mean $1/r = 10$ Hz, so $\langle \lambda dt \rangle = 0.1$. The gamma has shape $k = 0.2^2/10 = 0.004$ and scale $\theta = 50$. With 10^5 bins:

case	fraction of silent bins	number of avalanches	mean duration	mean size
(1) independent	$\simeq 10^{-4}$	a handful	$\sim 10^4$ bins	$\sim 10^5$
(2) shared exponential	0.090	8 179	11.1 bins	123
(3) shared gamma	0.984	1 573	1.02 bins	12.8

- (1) With $N = 100$ independent neurons the probability of a silent bin is $(1 - 0.1)^{100} \simeq 3 \times 10^{-5}$: activity never stops, so avalanches are not even defined.
- (2) Silence probability $\int Q(\lambda) e^{-N\lambda dt} d\lambda = \frac{1}{1+N dt \langle \lambda \rangle} = \frac{1}{11} = 0.091$. Durations are geometric with continuation probability $10/11$, mean 11 bins, exactly as observed. Sizes have an exponential tail.
- (3) The gamma rate is zero most of the time and occasionally huge. Silence probability $(1 + N dt \theta)^{-k} = 51^{-0.004} = 0.984$; avalanches almost always last one bin, but their sizes, $\text{Binomial}(N, \lambda dt)$ with λ heavy-tailed, are broadly distributed up to N , looking like a power law over a decade.

Neither case is critical. Correlations induced by a common input change the statistics completely. Analytically, with rates redrawn every bin the duration is always geometric, $P(T > n) = q^n$. Only rates that persist over an avalanche give power-law durations (next exercise).

Exercise 8.4

(Power-law durations from a quenched rate) Compute $P_{>}(n) = \gamma \int_0^\infty e^{-\gamma\lambda} (1 - e^{-\delta\lambda})^n d\lambda$ exactly and find its large- n behaviour. Check numerically for $\gamma = 1$, $\delta = 0.5$.

Solution

With $u = e^{-\delta\lambda}$, $P_{>}(n) = \frac{\gamma}{\delta} \int_0^1 u^{\gamma/\delta-1} (1-u)^n du = \frac{\gamma}{\delta} B(\frac{\gamma}{\delta}, n+1) = \frac{\Gamma(1+\gamma/\delta)\Gamma(n+1)}{\Gamma(n+1+\gamma/\delta)} \simeq \Gamma(1+\gamma/\delta) n^{-\gamma/\delta}$. For $\gamma/\delta = 2$: the ratio $P_{>}(n)/n^{-2}$ is 1.52, 1.94, 1.994, 1.9994 for $n = 10, 10^2, 10^3, 10^4$, converging to $\Gamma(3) = 2$. The duration density decays as n^{-3} here; with $\gamma = \delta$ it would decay as n^{-2} , the “critical” exponent, obtained without any criticality.

Exercise 8.5

(Spike-triggered average of a linear neuron) Suppose $r(t) = r_0 + \int_0^\infty D(\tau) s(t-\tau) d\tau$ with white-noise stimulus, $\langle s(t)s(t') \rangle = \sigma^2 \delta(t-t')$. Show that $C(\tau) = \frac{\sigma^2}{r_0} D(\tau)$.

Solution

$Q_{rs}(-\tau) = \frac{1}{T} \int r(t)s(t-\tau) dt = \int D(\tau') \langle s(t-\tau')s(t-\tau) \rangle d\tau' = \sigma^2 D(\tau)$, since the r_0 term averages to zero against a zero-mean stimulus. Dividing by $\langle r \rangle = r_0$ gives $C(\tau) = \sigma^2 D(\tau)/r_0$. For white-noise stimuli the spike-triggered average is the linear filter of the neuron, up to a constant: this is why reverse correlation identifies receptive fields.

8.7. Sources

Prof. S. Suweis, *PMLS Course Notes*, Chapter 5 and Sections 6.1–6.2; Lecture 10 slides (the brain as a complex system, measurement techniques, emergent patterns, model complexity, structure and function) and Lecture 11 slides (stimulus to response, tuning curves, psychophysical laws, Poisson processes, avalanches); Homework 4. P. Dayan and L. F. Abbott, *Theoretical Neuroscience*, MIT Press (2001), Chapter 1. J. M. Beggs and D. Plenz, *J. Neurosci.* **23**, 11167 (2003). M. A. Muñoz, *Criticality and dynamical scaling in living systems*, *Rev. Mod. Phys.* **90**, 031001 (2018). J. Touboul and A. Destexhe, *PRE* **95**, 012413 (2017). B. Mariani et al., *Sci. Rep.* **12**, 10770 (2022). C. R. Sims, *Science* **360**, 652 (2018).

9. Single-neuron dynamics

Chapter 8 treated neurons as black boxes that emit spikes. This chapter opens the box. The membrane potential is set by the balance between ion concentration gradients and electric forces. Treating the membrane as an electric circuit gives the leaky integrate-and-fire model. Voltage-dependent conductances give the Hodgkin–Huxley model of the action potential, which reduces to the two-variable FitzHugh–Nagumo model. At the other extreme of abstraction, the McCulloch–Pitts neuron and the perceptron capture computation with a single threshold unit.

9.1. The neuron

A neuron receives inputs on its **dendrites** through about 10^4 synapses, integrates them in the **soma**, and, when the membrane potential at the axon hillock crosses a threshold, emits an **action potential** that travels along the **axon** (sped up by the myelin sheath) to the synapses of other neurons. At rest the inside of the cell is about 70 mV more negative than the outside: the membrane is *polarised*. Ion pumps keep Na^+ and Cl^- concentrated outside and K^+ inside. Current flowing out of the cell (positive ions leaving) makes the potential more negative (**hyperpolarisation**); current flowing in (**depolarisation**) makes it less negative. A spike is a rapid depolarisation of about 100 mV lasting about 1 ms, followed by a refractory period.

9.2. Electrophysiology

9.2.1. The Nernst potential

Ions cross the membrane through selective **channels**. Two forces act on them: diffusion down the concentration gradient and the electric field across the membrane. The **sodium–potassium pump** uses the energy of ATP hydrolysis to move 3Na^+ out and 2K^+ in, maintaining both gradients and a net charge imbalance.

Consider a membrane permeable only to K^+ , initially electrically neutral but with more K^+ inside. Potassium diffuses out, leaving negative charge (impermeable organic anions) behind. The resulting field opposes further outflow, and equilibrium is reached when the two forces balance. Since only a tiny number of ions needs to move to build the field, the concentrations are essentially unchanged.

Proposition 9.1 (Nernst equation). *The equilibrium (reversal) potential of an ion species X of valence z_X is*

$$V_X = \frac{RT}{z_X F} \ln \frac{[X]_o}{[X]_i},$$

with R the gas constant, T the temperature and F Faraday’s constant.

Derivation. At constant T and pressure the relevant potential is the Gibbs free energy, $dG = -SdT + VdP + \mu dn$, and the chemical potential of an ideal solute at concentration x is $\mu = \mu_0 + RT \ln x$. A charged ion in a region at potential V has the **electrochemical potential** $\mu = \mu_0 + RT \ln x + z_X FV$. Equilibrium across the membrane requires equal electrochemical potentials inside and outside: $RT \ln x_i + z_X FV_i = RT \ln x_o + z_X FV_o$. Solving for $V_m = V_i - V_o$ gives the result. $\square$

9. Single-neuron dynamics

At room temperature ($RT/F \simeq 25$ mV), with $[K^+]_i = 140$ mM, $[K^+]_o = 5$ mM, $V_K \simeq -84$ mV; with $[Na^+]_i = 12$ mM, $[Na^+]_o = 140$ mM, $V_{Na} \simeq +62$ mV.

9.2.2. Several ions: the resting potential

With several permeable ions, each current follows Ohm's law, $I_X = g_X(V_m - V_X)$, with a conductance g_X . At rest the total current vanishes, so the membrane potential is a conductance-weighted average:

$$V_{\text{rest}} = \frac{\sum_X g_X V_X}{\sum_X g_X}.$$

For $g_{Na}/g_K \simeq 0.05$, $V_{\text{rest}} \simeq -77$ mV: close to V_K , since potassium dominates at rest. The rest state is a *dynamic* equilibrium: individual currents flow continuously and the pumps compensate them.

9.2.3. The membrane as an RC circuit

The lipid membrane is a thin insulator, i.e. a **capacitor** ($C \simeq 1 \mu\text{F}/\text{cm}^2$). Channels are (variable) **resistors** of conductance g_X in series with **batteries** V_X . Current conservation gives

$$C \frac{dV}{dt} = - \sum_X g_X (V - V_X) + I = -g_{\text{inp}} (V - V_{\text{rest}}) + I,$$

with $g_{\text{inp}} = \sum_X g_X$ the input conductance and I an injected current.

Voltage clamp. A feedback amplifier injects whatever current is needed to hold V at a command value V_s ; the injected current then measures the total ionic current at that voltage. With constant conductances the response to a current step is the charging of an RC circuit,

$$V(t) = V_s + (V_c - V_s)e^{-t/\tau}, \quad \tau = \frac{C}{g_{\text{inp}}},$$

and the steady current gives the linear **I–V relation**, whose slope is g_{inp} . Measured I–V curves are nonlinear because the conductances depend on V . Selectively blocking channels with drugs (tetrodotoxin for Na^+ , tetraethylammonium for K^+) isolates each contribution.

For a time-dependent current, with V measured from rest, the solution of $\tau \dot{V} = -V + RI(t)$ ($R = 1/g_{\text{inp}}$) is a convolution:

$$V(t) = \frac{R}{\tau} \int_{-\infty}^t e^{-(t-s)/\tau} I(s) ds,$$

i.e. the membrane is a **low-pass filter** with time constant $\tau \sim 10$ ms.

9.3. The integrate-and-fire model

Lapicque's (1907) **leaky integrate-and-fire** (LIF) neuron keeps only the subthreshold RC dynamics and replaces the spike by a rule:

$$\tau \frac{dV}{dt} = -(V - V_{\text{rest}}) + RI, \quad V(t^-) = V_{\text{th}} \Rightarrow \text{spike, } V(t^+) = V_{\text{reset}}.$$

For a constant current, starting from $V_{\text{reset}} = V_{\text{rest}}$, $V(t) = V_{\text{rest}} + RI(1 - e^{-t/\tau})$. The threshold is reached at t_{sp} such that $RI(1 - e^{-t_{\text{sp}}/\tau}) = V_{\text{th}} - V_{\text{rest}} \equiv \Delta$. The firing rate is

$$r = \frac{1}{t_{\text{sp}}} = \left[\tau \ln \frac{RI}{RI - \Delta} \right]^{-1} \Theta(RI - \Delta) \underset{RI \gg \Delta}{\simeq} \frac{RI - \Delta/2}{\tau \Delta}.$$

There is a current threshold $RI = \Delta$ (the rheobase) below which the neuron never fires. Above it the rate rises steeply, then linearly in I (the f–I curve). With $\tau = 10$ ms and $\Delta = 16$ mV, $RI = 18, 20, 25, 40$ mV gives 46, 62, 98, 196 Hz. An absolute refractory period τ_{ref} adds to t_{sp} and caps the rate at $1/\tau_{\text{ref}}$.

Real neurons are not perfect integrators. The threshold is not sharp: the same current pulse sometimes produces a spike and sometimes not. And some neurons are **resonators**: they fire preferentially to inputs at a particular frequency, and can even fire after an *inhibitory* burst (post-inhibitory rebound). These behaviours require the dynamics of the ion channels.

9.3.1. Spike-rate adaptation

Many neurons fire fast at the onset of a constant current and then slow down. A potassium conductance g_{sra} that increases by Δg_{sra} at each spike and decays with a time constant τ_{sra} of ~ 100 ms reproduces this:

$$\tau \frac{dV}{dt} = -(V - V_{\text{rest}}) - r_m g_{\text{sra}} (V - V_K) + RI, \quad \tau_{\text{sra}} \frac{dg_{\text{sra}}}{dt} = -g_{\text{sra}} + \tau_{\text{sra}} \Delta g_{\text{sra}} \sum_i \delta(t - t_i).$$

Each spike makes the next harder to trigger, so interspike intervals lengthen until a steady state is reached.

9.4. Voltage-dependent conductances: Hodgkin–Huxley

9.4.1. Gating variables

A channel contains q independent gates, each open with probability n . The channel conducts only if all gates are open, with probability n^q , so the conductance per unit area is $g = \bar{g} n^q$, with $\bar{g}$ the maximal conductance. Each gate opens at rate $\alpha_n(V)$ and closes at rate $\beta_n(V)$, a two-state Markov process whose mean obeys

$$\frac{dn}{dt} = \alpha_n(V)(1 - n) - \beta_n(V)n \iff \tau_n(V) \frac{dn}{dt} = n_\infty(V) - n,$$

with $\tau_n = 1/(\alpha_n + \beta_n)$ and $n_\infty = \alpha_n/\beta_n$. The rates follow from the Boltzmann factors of the gate's conformational change, $\alpha, \beta \propto \exp(\pm qBV/k_B T)$, which makes $n_\infty(V)$ a sigmoid. **Persistent** conductances (like the delayed-rectifier K^+ , $q = 4$) only activate. **Transient** conductances (like Na^+) have activation gates m that open with depolarisation and an inactivation gate h that closes with depolarisation: $g_{Na} = \bar{g}_{Na} m^3 h$.

9.4.2. The Hodgkin–Huxley model

From voltage-clamp experiments on the squid giant axon, Hodgkin and Huxley (1952) obtained

$$C_m \frac{dV}{dt} = -\bar{g}_K n^4 (V - V_K) - \bar{g}_{Na} m^3 h (V - V_{Na}) - \bar{g}_L (V - V_L) + I,$$

$$\tau_w(V) \frac{dw}{dt} = w_\infty(V) - w, \quad w = n, m, h.$$

The action potential unfolds as follows. A depolarisation opens the fast m gates. Sodium flows in and depolarises further (positive feedback, the upstroke toward V_{Na}). On a slower time scale h closes, inactivating sodium, and n opens, letting potassium out (repolarisation and an undershoot toward V_K , the after-hyperpolarisation). While h recovers and n closes, the neuron is refractory. The model reproduces the shape and threshold of the spike, refractoriness and propagation along the axon. It earned the 1963 Nobel Prize, but four nonlinear equations per neuron are hard to analyse and expensive to simulate in large networks.

9.4.3. The FitzHugh–Nagumo reduction

Two observations simplify HH. m is much faster than the other variables, so $m \simeq m_\infty(V)$. And h and n evolve on similar time scales and are roughly linearly related, so one variable can stand for both. The result is a fast **excitation** variable v (the membrane potential) and a slow **recovery** variable w :

$$\frac{dv}{dt} = v(v - a)(1 - v) - w + I, \quad \frac{dw}{dt} = \varepsilon(v - \gamma w), \quad \varepsilon \ll 1.$$

The cubic v -nullcline and the linear w -nullcline organise the phase plane. For small I they cross on the left branch of the cubic: a stable rest state, from which a small perturbation decays but a perturbation beyond the middle branch triggers a large excursion (a spike) before returning. The neuron is **excitable**. As I increases, the crossing moves to the middle branch, the rest state loses stability through a Hopf bifurcation, and the system settles on a **relaxation limit cycle**: periodic spiking. With $a = 0.1$, $\varepsilon = 0.01$, $\gamma = 0.5$, the rest state is stable up to $I \simeq 0.11$ and the model spikes periodically (with v swinging between about -0.28 and 1) beyond it.

Neurons are dynamical systems, and the bifurcation through which they start firing determines their computational class (Izhikevich). A saddle-node on an invariant circle gives integrators, which can fire at arbitrarily low rates (class 1). A Hopf bifurcation gives resonators with a non-zero minimum frequency (class 2), able to show post-inhibitory rebound.

9.5. Artificial neurons

9.5.1. McCulloch–Pitts neurons

McCulloch and Pitts (1943) abstracted the neuron into a threshold unit: inputs $\sigma_j \in \{0, 1\}$ are weighted and summed, and the output is

$$\sigma_{\text{out}} = \Theta\left(\sum_j w_j \sigma_j - \theta\right).$$

With suitable weights and thresholds a unit implements logic gates: NOT ($w = -1$, $\theta = -1/2$), AND ($w_1 = w_2 = 1$, $\theta = 3/2$), OR ($w_1 = w_2 = 1$, $\theta = 1/2$). Networks of such units can compute any Boolean function. Replacing Θ by smooth or piecewise-linear **activation functions** (sigmoid $1/(1 + e^{-x})$, tanh, ReLU $\max(0, x)$, leaky ReLU, ELU, maxout) gives the units of modern artificial neural networks.

9.5.2. The perceptron

Rosenblatt's **perceptron** (1958) is a threshold unit that *learns* its weights from labelled examples (σ^μ, t^μ) :

$$y = \Theta(\mathbf{w} \cdot \sigma - \theta), \quad \Delta \mathbf{w} = \eta(t^\mu - y^\mu) \sigma^\mu, \quad \Delta \theta = -\eta(t^\mu - y^\mu).$$

Weights change only when the unit errs, moving the decision hyperplane toward the misclassified example. The **perceptron convergence theorem** guarantees that the algorithm finds a separating hyperplane in a finite number of steps whenever one exists. The perceptron is a linear classifier, so it fails on problems that are not linearly separable, the simplest being XOR (Minsky and Papert, 1969). Solving XOR needs a hidden layer, and multi-layer networks trained by backpropagation descend from this observation.

9.6. Exercises

Exercise 9.1

(Nernst and resting potentials) (a) Compute V_K and V_{Na} at 37°C for the concentrations in the text. (b) Find V_{rest} for $g_{Na}/g_K = 0.05$. (c) What happens to V_{rest} if extracellular K^+ rises to 20 mM?

Solution

(a) $RT/F = 26.7$ mV at 310 K: $V_K = 26.7 \ln(5/140) = -89.0$ mV, $V_{Na} = 26.7 \ln(140/12) = +65.6$ mV. (b) $V_{\text{rest}} = (-89.0 + 0.05 \times 65.6)/1.05 = -81.6$ mV. (c) $V_K = 26.7 \ln(20/140) = -52.0$ mV and $V_{\text{rest}} = (-52.0 + 3.3)/1.05 = -46.4$ mV. The cell depolarises by about 35 mV, enough to inactivate sodium channels and block firing. This is why the potassium concentration in the blood is tightly regulated (hyperkalaemia is dangerous for heart and nerves).

Exercise 9.2

(f-I curve of the LIF neuron) Derive the firing rate of the LIF neuron for constant current, including an absolute refractory period τ_{ref} . Evaluate it for $\tau = 10$ ms, $\Delta = 16$ mV, $RI = 20$ mV, and $\tau_{\text{ref}} = 0$ or 2 ms. Derive the large-current approximation.

Solution

$r = [\tau_{\text{ref}} + \tau \ln \frac{RI}{RI - \Delta}]^{-1}$ for $RI > \Delta$, else 0. For $RI = 20$ mV: $\tau \ln 5 = 16.1$ ms, so $r = 62$ Hz without refractoriness and $1/18.1$ ms = 55 Hz with 2 ms. For $RI \gg \Delta$, $\ln \frac{1}{1-x} = x + x^2/2 + \dots$ with $x = \Delta/RI$ gives $t_{\text{sp}} \simeq \frac{\tau \Delta}{RI} (1 + \frac{\Delta}{2RI})$, so $r \simeq \frac{RI - \Delta/2}{\tau \Delta}$: linear in I with slope $1/(\tau \Delta)$. At $RI = 40$ mV this gives 200 Hz against the exact 196 Hz. Without the $-\Delta/2$ correction the approximation would give 150 Hz.

Exercise 9.3

(Gating kinetics) A gate with $\alpha_n = 0.5 \text{ ms}^{-1}$ and $\beta_n = 0.125 \text{ ms}^{-1}$ at a given voltage starts closed ($n = 0$). Find $n(t)$ and the potassium conductance $\bar{g}_K n^4$. Why does the conductance rise with a delay (sigmoidally) while n rises exponentially?

Solution

$\tau_n = 1/(0.625) = 1.6$ ms and $n_\infty = 0.5 \times 1.6 = 0.8$, so $n(t) = 0.8(1 - e^{-t/1.6})$. The conductance $\propto n^4 = 0.41(1 - e^{-t/1.6})^4$ starts as t^4 for small t : all four gates must open, so the conductance has an initial delay and an S-shaped rise. This is exactly the delayed rise Hodgkin and Huxley observed for potassium, which led them to the fourth power.

Exercise 9.4

(FitzHugh–Nagumo) For $\dot{v} = v(v-a)(1-v) - w + I$, $\dot{w} = \varepsilon(v - \gamma w)$ with $a = 0.1$, $\gamma = 0.5$, $\varepsilon = 0.01$: (a) find the fixed point and its stability for $I = 0$; (b) find numerically the current at which repetitive firing starts; (c) explain the shape of the limit cycle.

Solution

(a) With $w = v/\gamma = 2v$: $v(v-0.1)(1-v) = 2v$, whose only real root is $v = 0$, so the fixed point is

$(0,0)$. The Jacobian is $\begin{pmatrix} -a & -1 \\ \varepsilon & -\varepsilon\gamma \end{pmatrix} = \begin{pmatrix} -0.1 & -1 \\ 0.01 & -0.005 \end{pmatrix}$, with trace -0.105 and determinant 0.0105 : stable. (b) Simulations show a stable rest state up to $I \simeq 0.11$ and large periodic oscillations (v amplitude $\simeq 1.2$) for $I \gtrsim 0.12$. (c) Because $\varepsilon \ll 1$, the orbit alternates fast horizontal jumps between the outer branches of the cubic nullcline with slow drifts along them: a **relaxation oscillation**. The upstroke and downstroke are fast; the plateau and recovery are slow, like a real spike followed by refractoriness.

Exercise 9.5

(Logic gates and XOR) Build McCulloch–Pitts units for OR and NAND. Show that no single unit computes XOR, and build XOR with a hidden layer.

Solution

OR: $w = (1, 1)$, $\theta = 0.5$. NAND: $w = (-1, -1)$, $\theta = -1.5$. XOR would need $0 < \theta$ (input $00 \rightarrow 0$), $w_1 > \theta$ and $w_2 > \theta$ (inputs $10, 01 \rightarrow 1$), and $w_1 + w_2 < \theta$ (input $11 \rightarrow 0$). Adding the middle two gives $w_1 + w_2 > 2\theta > \theta$, a contradiction: XOR is not linearly separable. With a hidden layer, $\text{XOR}(x_1, x_2) = \text{AND}(\text{OR}(x_1, x_2), \text{NAND}(x_1, x_2))$: two hidden units (OR, NAND) and an output AND unit with $w = (1, 1)$, $\theta = 1.5$.

Exercise 9.6

(Perceptron learning) Train a perceptron on AND with $\eta = 1$, starting from $\mathbf{w} = 0$, $\theta = 0$, presenting $00, 01, 10, 11$ cyclically, with the convention $\Theta(0) = 0$. How many epochs does it take?

Solution

Epoch 1: 00 : $y = \Theta(0) = 0 = t$. 01 : correct. 10 : correct. 11 : $y = 0$, $t = 1$, so $\mathbf{w} = (1, 1)$, $\theta = -1$. Epoch 2: 00 : $\Theta(0 + 1) = 1 \neq 0$, so $\theta = 0$. 01 : $\Theta(1) = 1 \neq 0$, so $\mathbf{w} = (1, 0)$, $\theta = 1$. 10 : $\Theta(1 - 1) = 0$, correct. 11 : $\Theta(1 - 1) = 0 \neq 1$, so $\mathbf{w} = (2, 1)$, $\theta = 0$. Epoch 3: 00 correct; 01 : $\Theta(1) = 1$, error, so $\mathbf{w} = (2, 0)$, $\theta = 1$; 10 : $\Theta(1) = 1$, error, so $\mathbf{w} = (1, 0)$, $\theta = 2$; 11 : $\Theta(-1) = 0$, error, so $\mathbf{w} = (2, 1)$, $\theta = 1$. Epoch 4: 00 : 0 ; 01 : $\Theta(0) = 0$; 10 : $\Theta(1) = 1$, error, so $\mathbf{w} = (1, 1)$, $\theta = 2$; 11 : $\Theta(0) = 0$, error, so $\mathbf{w} = (2, 2)$, $\theta = 1$. Epoch 5: 01 : $\Theta(1) = 1$, error, so $\mathbf{w} = (2, 1)$, $\theta = 2$; 10 : $\Theta(0) = 0$; 11 : $\Theta(1) = 1$. Epoch 6: all four are correct ($00 \rightarrow 0$, $01 \rightarrow \Theta(-1)$, $10 \rightarrow \Theta(0)$, $11 \rightarrow \Theta(1)$). The weights stop changing after five epochs with errors (the sixth epoch is error-free), converging to $\mathbf{w} = (2, 1)$, $\theta = 2$, as guaranteed by the convergence theorem since AND is linearly separable.

9.7. Sources

Prof. S. Suweis, *PMLS Course Notes*, Sections 6.3–6.4; Lecture 12 slides (electrophysiology, RC circuit, integrate-and-fire and resonators, transient conductances, Hodgkin–Huxley, FitzHugh–Nagumo, McCulloch–Pitts, perceptron) and Lecture 13 review. P. Dayan and L. F. Abbott, *Theoretical Neuroscience*, Chapters 5–6. A. L. Hodgkin and A. F. Huxley, *J. Physiol.* **117**, 500 (1952). R. FitzHugh, *Biophys. J.* **1**, 445 (1961). E. M. Izhikevich, *Dynamical Systems in Neuroscience*, MIT Press (2007). W. S. McCulloch and W. Pitts, *Bull. Math. Biophys.* **5**, 115 (1943). F. Rosenblatt, *Psychol. Rev.* **65**, 386 (1958).

10. Firing-rate networks and Wilson–Cowan dynamics

Detailed spiking models do not scale to networks of many neurons. **Firing-rate models** describe each neuron (or population) by a continuous rate, obtained by averaging over spikes, and connect them through synaptic weights. This chapter derives the rate equations, studies two interacting populations, excitatory and inhibitory (the neural analogue of predator and prey), and their stochastic version, which shows the stochastic amplification of Chapter 5. It then analyses linear recurrent networks, which amplify selected input patterns and integrate signals, and stochastic networks of binary units, the bridge to the Hopfield and Boltzmann machines of Chapters 11–12.

10.1. From spikes to rates

Consider a neuron receiving inputs from N_u presynaptic neurons b with firing rates u_b through synaptic weights w_b (positive for excitatory, negative for inhibitory synapses). A presynaptic spike at time 0 produces in the postsynaptic soma a current $w_b K_s(t)$, where the **synaptic kernel** K_s (normalised, $\int_0^\infty K_s = 1$) describes the opening and closing of synaptic conductances. Summing over spikes and inputs, assuming linear superposition,

$$I_s(t) = \sum_b w_b \int_{-\infty}^t K_s(t-\tau) \rho_b(\tau) d\tau \simeq \sum_b w_b \int_{-\infty}^t K_s(t-\tau) u_b(\tau) d\tau,$$

where the spike trains are replaced by their rates, which is accurate with many inputs. For an exponential kernel $K_s(t) = e^{-t/\tau_s}/\tau_s$, differentiation gives

$$\tau_s \frac{dI_s}{dt} = -I_s + \mathbf{w} \cdot \mathbf{u}.$$

The output rate follows the input current through a static nonlinearity ψ (the f–I curve of Chapter 9), low-pass filtered by the membrane:

$$\tau_r \frac{dv}{dt} = -v + \psi(I_s(t)).$$

These two equations form the **firing-rate model**. If $\tau_s \ll \tau_r$, the current is always at equilibrium, $I_s = \mathbf{w} \cdot \mathbf{u}$, and

$$\tau_r \frac{dv}{dt} = -v + \psi(\mathbf{w} \cdot \mathbf{u});$$

if instead $\tau_r \ll \tau_s$, $v = \psi(I_s)$ and the current carries the dynamics. At stationarity $v_\infty = \psi(\mathbf{w} \cdot \mathbf{u})$: the perceptron’s integration rule, now derived from synaptic physiology.

10.2. Excitatory and inhibitory populations

Dale’s law: a neuron releases the same neurotransmitter at all its synapses, so it is either excitatory (its outgoing weights are all positive) or inhibitory (all negative). Neurons therefore come in two “species”, like predators and prey. For one excitatory and one inhibitory unit with rates v_E , v_I and external inputs h_E , h_I ,

$$\tau_E \frac{dv_E}{dt} = -v_E + \psi(h_E + M_{EE}v_E + M_{EI}v_I), \quad \tau_I \frac{dv_I}{dt} = -v_I + \psi(h_I + M_{IE}v_E + M_{II}v_I),$$

with $M_{EE}, M_{IE} > 0$ and $M_{EI}, M_{II} < 0$, and a rectifying activation $\psi(x) = [x]_+$.

Example 10.1 (An oscillating E–I pair (Dayan and Abbott))

Take $M_{EE} = 1.25$, $M_{IE} = 1$, $M_{EI} = -1$, $M_{II} = 0$, $h_E = 10$ Hz, $h_I = -10$ Hz, $\tau_E = 10$ ms, with τ_I as control parameter. With both rates positive, the fixed point solves a linear system: $v_E^* = 26.7$ Hz, $v_I^* = 16.7$ Hz, independent of τ_I . The Jacobian is

$$J = \begin{pmatrix} (M_{EE} - 1)/\tau_E & M_{EI}/\tau_E \\ M_{IE}/\tau_I & -1/\tau_I \end{pmatrix}, \quad \text{tr } J = \frac{0.25}{\tau_E} - \frac{1}{\tau_I}, \quad \det J = \frac{0.75}{\tau_E \tau_I} > 0.$$

Its eigenvalues are complex. For $\tau_I < 40$ ms the trace is negative and the fixed point is a stable focus: rates relax through **damped oscillations**. At $\tau_I = 40$ ms the eigenvalues cross the imaginary axis ($\pm 43i \text{ s}^{-1}$, about 7 Hz), a **Hopf bifurcation**. For $\tau_I > 40$ ms the system settles on a **limit cycle**: sustained oscillations, confined by the rectification. Slow inhibition destabilises the network, as delayed negative feedback does in any control system.

The **nullclines** $\dot{v}_E = 0$ and $\dot{v}_I = 0$ divide the phase plane into regions with opposite flow and give the qualitative picture at a glance.

10.3. Wilson–Cowan models

10.3.1. Deterministic populations

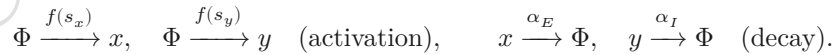
Wilson and Cowan (1972) described large homogeneous populations by the fractions x and y of active excitatory and inhibitory neurons:

$$\frac{dx}{dt} = -\alpha_E x + f(w_{EE}x - w_{EI}y + h_E), \quad \frac{dy}{dt} = -\alpha_I y + f(w_{IE}x - w_{II}y + h_I),$$

with a sigmoidal activation $f(s) = 1/(1 + e^{-s})$, decay rates $\alpha_{E,I}$, and input h . The model reproduces multistability, hysteresis and oscillations of cortical tissue.

10.3.2. A microscopic stochastic model

To include finite-size fluctuations, build a reaction scheme on two populations of V neurons each, where Φ denotes an inactive neuron (Benayoun et al., 2010; de Santo et al., 2018):



Choose the simplest symmetric case: $\alpha_E = \alpha_I = 1$, no self-coupling ($w_{EE} = w_{II} = 0$), $|w_{EI}| = |w_{IE}| = r$, and inputs that centre the activations:

$$s_x = -r\left(\frac{n_y}{V} - \frac{1}{2}\right), \quad s_y = r\left(\frac{n_x}{V} - \frac{1}{2}\right).$$

Inhibitory activity lowers excitatory activation, and excitatory activity raises inhibitory activation. With densities $x = n_x/V$, $y = n_y/V$, the transition rates are $T(n_x \rightarrow n_x \mp 1) = n_x$ and $T(n_x \rightarrow n_x + 1) = Vf(s_x)$, and similarly for y . The Kramers–Moyal expansion gives the Langevin equations

$$\dot{x} = -x + f(-r(y - \frac{1}{2})) + \frac{1}{\sqrt{V}} \sqrt{x + f(-r(y - \frac{1}{2}))} \eta_x,$$

$$\dot{y} = -y + f(r(x - \frac{1}{2})) + \frac{1}{\sqrt{V}} \sqrt{y + f(r(x - \frac{1}{2}))} \eta_y,$$

with independent white noises. For $V \rightarrow \infty$ the fixed point is $x^* = y^* = 1/2$, with Jacobian

$$J = \begin{pmatrix} -1 & -r/4 \\ r/4 & -1 \end{pmatrix}, \quad \lambda = -1 \pm i \frac{r}{4}.$$

The fixed point is a stable focus: the deterministic model predicts damped oscillations.

10.3.3. Stochastic amplification

Following Chapter 5, write $x = x^* + \xi_1/\sqrt{V}$, $y = y^* + \xi_2/\sqrt{V}$. The linear noise approximation gives $\dot{\xi} = J\xi + \mu$, with diffusion matrix $B = \text{diag}(x^* + f(0), y^* + f(0)) = \mathbb{1}$. Its power spectrum is

$$P_1(\omega) = [(i\omega - J)^{-1} B (-i\omega - J^\top)^{-1}]_{11} = \frac{1 + r^2/16 + \omega^2}{(\omega^2 - 1 - r^2/16)^2 + 4\omega^2}.$$

For $r > 4/\sqrt{3} \simeq 2.3$ it has a maximum at a non-zero frequency, approaching $r/4$ for large r : for $r = 8$ the peak is at $\omega \simeq 2.0$, about 2.6 times its zero-frequency value. Intrinsic noise sustains **quasi-cycles** around a stable state, a mechanism proposed for gamma oscillations in cortex. A related point: J is not symmetric (it is *non-normal* when $w_{EI} \neq w_{IE}$), and non-normal dynamics transiently amplify perturbations even when all eigenvalues are stable. This makes fluctuations much larger than the eigenvalues suggest.

10.3.4. Avalanches in the Wilson–Cowan model

In the variables $\Sigma = x + y$ (total activity) and $\Delta = x - y$ (imbalance), the stochastic model with excitatory–inhibitory balance produces bursts of activity separated by quiescence. Avalanche sizes and durations are power-law distributed, with exponents $\tau \simeq 3/2$, $\alpha \simeq 2$ and $\langle S \rangle(T) \sim T^\gamma$, $\gamma \simeq 2$, close to those of a critical branching process. They arise in a regime where the network is *not* at a phase transition but is dominated by inhibition with a weak, noisy excitatory drive (de Santo et al., 2018): another warning that power laws alone do not prove criticality.

10.3.5. Balanced networks

How do cortical neurons, each receiving thousands of inputs, fire irregularly at low rates? Let each neuron receive K excitatory and K inhibitory inputs, each a Poisson train at rate ν_0 , with synaptic strength W_0 . The mean excitatory and inhibitory currents are $\langle I_E \rangle = KW_0\nu_0$ and $\langle I_I \rangle = -KW_0\nu_0$, while their fluctuations have variance $\propto KW_0^2\nu_0$. If $W_0 = O(1)$, the means diverge with K and the neuron saturates. If $W_0 \sim 1/K$, fluctuations vanish and the neuron is deterministic. With the **balanced scaling** $W_0 = J_0/\sqrt{K}$, the means are each $O(\sqrt{K})$ but nearly cancel, $\langle I_E \rangle + \langle I_I \rangle = O(1)$, and the fluctuations are $O(1)$, $\sigma^2(I) = J_0^2\nu_0$. Neurons operate in a fluctuation-driven regime and fire irregularly. Balance is maintained dynamically by the network (van Vreeswijk and Sompolinsky, 1996), possibly helped by homeostatic plasticity.

10.4. Recurrent networks

10.4.1. Feedforward and recurrent architectures

In a **feedforward** network, output neurons $\mathbf{v}$ receive inputs $\mathbf{u}$ through a weight matrix W : $\tau_r \dot{\mathbf{v}} = -\mathbf{v} + \psi(W\mathbf{u})$. In a **recurrent** network the outputs are also coupled to each other by M :

$$\tau_r \frac{d\mathbf{v}}{dt} = -\mathbf{v} + \psi(\mathbf{h} + M\mathbf{v}), \quad \mathbf{h} = W\mathbf{u}.$$

With separate excitatory and inhibitory populations M has a block structure. Analytical work often assumes M symmetric, which violates Dale's law but can be justified when inhibition is fast. If $\tau_I \rightarrow 0$ and $M_{II} = 0$, then $\mathbf{v}_I = M_{IE}\mathbf{v}_E$, and the effective excitatory network has the (possibly symmetric) matrix $M_{EE} + M_{EI}M_{IE}$.

10.4.2. Linear recurrent networks

For a linear activation and symmetric M with orthonormal eigenvectors $\mathbf{e}_\mu$ and eigenvalues λ_μ , expand $\mathbf{v}(t) = \sum_\mu c_\mu(t)\mathbf{e}_\mu$:

$$\tau_r \frac{dc_\mu}{dt} = -(1 - \lambda_\mu)c_\mu + \mathbf{e}_\mu \cdot \mathbf{h}.$$

For constant input,

$$c_\mu(t) = \frac{\mathbf{e}_\mu \cdot \mathbf{h}}{1 - \lambda_\mu} \left(1 - e^{-t(1-\lambda_\mu)/\tau_r}\right) + c_\mu(0) e^{-t(1-\lambda_\mu)/\tau_r}.$$

- If some $\lambda_\mu > 1$ the network is unstable: activity grows exponentially.
- If all $\lambda_\mu < 1$, $\mathbf{v}_\infty = \sum_\mu \frac{\mathbf{e}_\mu \cdot \mathbf{h}}{1 - \lambda_\mu} \mathbf{e}_\mu$: each input component is multiplied by $1/(1 - \lambda_\mu)$. Eigenvalues close to 1 give **selective amplification** of the corresponding input patterns, and the network acts as a filter that picks out features.
- If $\lambda_1 = 1$ exactly (marginal), the network **integrates** its input: $\tau_r \dot{c}_1 = \mathbf{e}_1 \cdot \mathbf{h}$, so $c_1(t) = c_1(0) + \frac{1}{\tau_r} \int_0^t \mathbf{e}_1 \cdot \mathbf{h}(t') dt'$. After a brief pulse, activity persists indefinitely: a **memory**. The oculomotor integrator, which converts bursts of velocity commands into a sustained eye position, is thought to work this way. Its fine tuning (λ_1 must equal 1 very precisely) is a classic problem.

Example 10.2 (Orientation tuning by recurrence)

Neurons in visual cortex labelled by their preferred angle $\theta \in (-\pi, \pi]$, with density ρ_θ , coupled by a translation-invariant kernel $M(\theta - \theta')$:

$$\tau_r \frac{dv(\theta)}{dt} = -v(\theta) + h(\theta) + \rho_\theta \int_{-\pi}^{\pi} d\theta' M(\theta - \theta') v(\theta').$$

The eigenfunctions of a convolution on the circle are Fourier modes, $e_0 = (2\pi\rho_\theta)^{-1/2}$, $e_\mu \propto \cos \mu\theta, \sin \mu\theta$, with eigenvalues $\lambda_\mu = \rho_\theta \int M(\theta) \cos(\mu\theta) d\theta$. For $M(\theta) = \frac{\lambda_1}{\pi\rho_\theta} \cos \theta$ only $\mu = 1$ is amplified: a noisy, weakly tuned input becomes a smooth, sharply tuned output peaked at the input's preferred angle. With the rectification $[\cdot]_+$ the network stays stable even for $\lambda_1 > 1$ (e.g. 1.9), and the tuning curve becomes independent of the input contrast.

10.5. Stochastic recurrent networks

Let each unit be binary, $v_a \in \{0, 1\}$, with total input $I_a = h_a + \sum_{a'} M_{aa'} v_{a'}$, symmetric weights ($M_{aa'} = M_{a'a}$) and $M_{aa} = 0$. At each step a random unit is updated stochastically:

$$P[v_a = 1] = F(I_a) = \frac{1}{1 + e^{-I_a}}.$$

This Markov chain (Glauber dynamics) satisfies detailed balance with respect to the Boltzmann distribution

$$P[\mathbf{v}] = \frac{e^{-E(\mathbf{v})}}{Z}, \quad E(\mathbf{v}) = -\mathbf{h} \cdot \mathbf{v} - \frac{1}{2} \mathbf{v} \cdot M \mathbf{v},$$

which is therefore its unique stationary distribution. Flipping unit a changes the energy by $\Delta E = -I_a \Delta v_a$, and $F(I_a)/(1 - F(I_a)) = e^{I_a}$ is exactly the Boltzmann ratio. A **mean-field** version replaces the units by their average activity, $\tau_s \dot{\mathbf{I}} = -\mathbf{I} + \mathbf{h} + M\mathbf{F}(\mathbf{I})$, with factorised distribution $Q[\mathbf{v}] = \prod_a F(I_a)^{v_a} (1 - F(I_a))^{1-v_a}$. The Kullback–Leibler divergence $D_{KL}(Q\|P)$ is a Lyapunov function of these dynamics: they converge to the mean-field approximation of P . This is the starting point of Boltzmann machines (Chapter 12).

10.6. Exercises

Exercise 10.1

(E–I pair) For the example network, (a) verify the fixed point (26.7, 16.7) Hz; (b) show that the Hopf bifurcation occurs at $\tau_I = 40$ ms and find the oscillation frequency there; (c) what happens for $\tau_I = 50$ ms?

Solution

(a) $v_E = 1.25v_E - v_I + 10$ and $v_I = v_E - 10$ give $0.25v_E = v_I - 10 = v_E - 20$, so $v_E = 26.7$ and $v_I = 16.7$. (b) $\text{tr } J = 0.025 - 1/\tau_I$ (in ms^{-1}) vanishes at $\tau_I = 40$ ms, where $\det J = 0.75/(10 \times 40) = 1.875 \times 10^{-3} \text{ ms}^{-2}$, so $\omega = \sqrt{\det J} = 0.0433 \text{ ms}^{-1} = 43 \text{ s}^{-1}$, i.e. $f = 6.9$ Hz. (c) At $\tau_I = 50$ ms the eigenvalues are $(2.5 \pm 38.6i) \text{ s}^{-1}$: unstable focus. Trajectories spiral out until the rectification $[\cdot]_+$ clips v_I at zero, producing a stable limit cycle of about 6 Hz with large amplitude.

Exercise 10.2

(Stochastic Wilson–Cowan spectrum) For the symmetric stochastic Wilson–Cowan model at $x^* = y^* = 1/2$, derive the noise matrix B , the power spectrum $P_1(\omega)$, and the condition for a peak at $\omega > 0$. Evaluate for $r = 1, 4, 8$.

Solution

At the fixed point the activation and decay rates both equal 1/2 for each population, and $B_{xx} = x^* + f(0) = 1$, $B_{yy} = 1$, $B_{xy} = 0$ (the reactions change one population each). With $J = \begin{pmatrix} -1 & -a \\ a & -1 \end{pmatrix}$, $a = r/4$: $|\det(i\omega - J)|^2 = (1 + a^2 - \omega^2)^2 + 4\omega^2$ and the numerator is $B_{11}(\omega^2 + 1) + B_{22}a^2 = 1 + a^2 + \omega^2$. Writing $c = 1 + a^2$ and $u = \omega^2$, $P_1 = (c + u)/[(c - u)^2 + 4u]$ increases at $u = 0$ iff $3c^2 - 4c > 0$, i.e. $c > 4/3$: a peak at $\omega > 0$ exists iff $a > 1/\sqrt{3}$, i.e. $r > 4/\sqrt{3} \simeq 2.3$. Numerically: $r = 1$, maximum at $\omega = 0$; $r = 4$, a broad peak at $\omega \simeq 0.9$, 1.2 times $P_1(0)$; $r = 8$, a clear peak at $\omega \simeq 2.0$, 2.6 times

$P_1(0)$. Strong E–I coupling turns noise into quasi-periodic oscillations at the imaginary part $r/4$ of the eigenvalues.

Exercise 10.3

(Selective amplification) A symmetric linear network has eigenvalues 0.9, 0.5 and 0 with eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and receives $\mathbf{h} = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3$. Find $\mathbf{v}_\infty$ and the relaxation times of each mode with $\tau_r = 10$ ms.

Solution

$\mathbf{v}_\infty = \frac{1}{0.1}\mathbf{e}_1 + \frac{1}{0.5}\mathbf{e}_2 + \mathbf{e}_3 = 10\mathbf{e}_1 + 2\mathbf{e}_2 + \mathbf{e}_3$. The first pattern is amplified tenfold relative to the input. Relaxation times $\tau_r/(1 - \lambda_\mu)$: 100 ms, 20 ms, 10 ms. Amplification comes with slowing down: modes near $\lambda = 1$ are both large and slow, the hallmark of a system close to an instability (critical slowing down, Chapter 13).

Exercise 10.4

(Neural integrator) With $\lambda_1 = 1$ and $\tau_r = 10$ ms, an input pulse $\mathbf{e}_1 \cdot \mathbf{h} = 5$ Hz lasting 50 ms is applied from $c_1(0) = 0$. What is c_1 after the pulse? If instead $\lambda_1 = 0.99$, how long does the memory last?

Solution

$c_1 = \frac{1}{10} \times 5 \times 50 = 25$ Hz, and it stays there after the pulse: persistent activity. With $\lambda_1 = 0.99$ the mode decays with time constant $\tau_r/(1 - \lambda_1) = 1$ s, so the memory fades within a second or two. To hold a memory for ~ 10 s (as the oculomotor system does) requires $|1 - \lambda_1| \lesssim 10^{-3}$: fine tuning, which motivates models with plasticity or bistable units.

Exercise 10.5

(Glauber dynamics and detailed balance) For the stochastic binary network, show that the update rule $P[v_a = 1] = 1/(1 + e^{-I_a})$ satisfies detailed balance with respect to $P \propto e^{-E}$.

Solution

Let $\mathbf{v}$ and $\mathbf{v}'$ differ only in unit a , with $v_a = 0$ and $v'_a = 1$. Then $E(\mathbf{v}') - E(\mathbf{v}) = -h_a - \sum_{a'} M_{aa'} v_{a'} = -I_a$ (using symmetry and $M_{aa} = 0$). The ratio of transition probabilities is $\frac{W(\mathbf{v} \rightarrow \mathbf{v}')}{W(\mathbf{v}' \rightarrow \mathbf{v})} = \frac{F(I_a)}{1 - F(I_a)} = e^{I_a} = e^{-(E(\mathbf{v}') - E(\mathbf{v}))} = \frac{P(\mathbf{v}')}{P(\mathbf{v})}$, which is detailed balance. The Boltzmann distribution is stationary, and the chain is ergodic, so it is the unique stationary state.

10.7. Sources

Prof. S. Suweis, *PMLS Course Notes*, Chapter 7; Lecture 12 slides (firing-rate models), Lecture 13 slides (Wilson–Cowan model, stochastic version, power spectrum, avalanches, balance condition, olfactory bulb) and Lecture 14 slides (E–I recurrent networks, linear recurrent networks, stochastic networks). P. Dayan and L. F. Abbott, *Theoretical Neuroscience*, Chapter 7. H. R. Wilson and J. D. Cowan, *Biophys. J.* **12**, 1 (1972). M. Benayoun, J. D. Cowan, W. van Drongelen, E. Wallace, *PLoS Comput. Biol.* **6**, e1000846 (2010). S. di Santo, P. Villegas, R. Burioni, M. A. Muñoz, *J. Stat. Mech.* 073402 (2018). C. van Vreeswijk and H. Sompolinsky, *Science* **274**, 1724 (1996). Z. Li and J. J. Hopfield, *Biol. Cybern.* **61**, 379 (1989).

11. Plasticity, learning and associative memory

Long-term memories are thought to be stored in the strengths of synapses, which change with experience: **synaptic plasticity**. This chapter formalises Hebb's idea that "neurons that fire together wire together", shows why the naive rule is unstable, and develops stabilised rules (covariance, normalisation, BCM, Oja), which turn a neuron into a principal-component analyser. Hebbian learning in recurrent networks then produces **associative memories**: activity patterns stored as attractors of the dynamics. The Hopfield network makes this precise with an energy function, a capacity, and spurious states. A final section on reservoir computing shows a complementary, supervised use of recurrent dynamics.

11.1. Hebbian plasticity

11.1.1. Hebb's rule and its problems

Consider a single postsynaptic neuron with linear dynamics $\tau_r \dot{v} = -v + \mathbf{w} \cdot \mathbf{u}$ driven by inputs $\mathbf{u}$. Plasticity is much slower than neural dynamics, so $v = \mathbf{w} \cdot \mathbf{u}$ at all times. Hebb's (1949) conjecture becomes

$$\tau_w \frac{d\mathbf{w}}{dt} = \langle v \mathbf{u} \rangle = Q \mathbf{w}, \quad Q = \langle \mathbf{u} \mathbf{u}^\top \rangle,$$

averaged over the input patterns. The input **correlation matrix** Q is positive semi-definite, so $\frac{d}{dt} |\mathbf{w}|^2 = \frac{2}{\tau_w} \langle v^2 \rangle \geq 0$. Three problems follow:

1. **Unbounded growth**: weights grow exponentially along the leading eigenvector of Q , requiring saturation by hand ($0 \leq w \leq w_{\max}$).
2. **No depression**: with positive rates, weights can only increase, yet long-term depression (LTD) is observed alongside long-term potentiation (LTP).
3. **No competition**: all synapses grow together, while learning requires some synapses to win at the expense of others.

11.1.2. Covariance rules

Thresholds introduce depression. Plasticity is potentiating if the postsynaptic rate exceeds θ_v and depressing otherwise:

$$\tau_w \frac{d\mathbf{w}}{dt} = \langle (v - \theta_v) \mathbf{u} \rangle \quad \text{or} \quad \tau_w \frac{d\mathbf{w}}{dt} = \langle v(\mathbf{u} - \theta_u) \rangle.$$

Choosing $\theta_v = \langle v \rangle$ or $\theta_u = \langle \mathbf{u} \rangle$ gives $\tau_w \dot{\mathbf{w}} = C \mathbf{w}$, with $C = \langle (\mathbf{u} - \langle \mathbf{u} \rangle)(\mathbf{u} - \langle \mathbf{u} \rangle)^\top \rangle$ the input **covariance matrix**: these are the **covariance rules**. They allow LTD but still grow without bound.

11.1.3. Normalisation and competition

Subtractive normalisation keeps the total weight fixed:

$$\tau_w \frac{d\mathbf{w}}{dt} = v \mathbf{u} - \frac{v(\mathbf{1} \cdot \mathbf{u})}{N} \mathbf{1}, \quad \frac{d}{dt} (\mathbf{1} \cdot \mathbf{w}) = \frac{v \mathbf{1} \cdot \mathbf{u}}{\tau_w} \left(1 - \frac{\mathbf{1} \cdot \mathbf{1}}{N} \right) = 0.$$

11. Plasticity, learning and associative memory

It is highly competitive: weights are reduced by the same amount, so small weights shrink proportionally more and are driven to zero.

The **BCM rule** (Bienenstock, Cooper, Munro, 1982) uses a sliding threshold, $\tau_w \dot{\mathbf{w}} = v \mathbf{u}(v - \theta_v)$, with $\tau_\theta \dot{\theta}_v = v^2 - \theta_v$. If a synapse gets strong, the threshold rises and makes further potentiation harder: stability through homeostasis.

Oja's rule (1982) adds a multiplicative decay:

$$\tau_w \frac{d\mathbf{w}}{dt} = v \mathbf{u} - \alpha v^2 \mathbf{w}, \quad \alpha > 0.$$

Theorem 11.1 (Oja's rule extracts the first principal component). *For zero-mean inputs with covariance C , the averaged Oja dynamics $\tau_w \dot{\mathbf{w}} = C\mathbf{w} - \alpha(\mathbf{w}^\top C\mathbf{w})\mathbf{w}$ converges to $\mathbf{w}^* = \pm \mathbf{e}_1/\sqrt{\alpha}$, where $\mathbf{e}_1$ is the eigenvector of C with the largest eigenvalue.*

Proof. (i) Norm: $\tau_w \frac{d}{dt} |\mathbf{w}|^2 = 2(\mathbf{w}^\top C\mathbf{w})(1 - \alpha|\mathbf{w}|^2)$, so $|\mathbf{w}|^2 \rightarrow 1/\alpha$. (ii) Fixed points satisfy $C\mathbf{w} = \lambda\mathbf{w}$ with $\lambda = \alpha\mathbf{w}^\top C\mathbf{w}$: eigenvectors of C , normalised to $1/\sqrt{\alpha}$. (iii) Linearising around $\mathbf{e}_\mu/\sqrt{\alpha}$, a perturbation along $\mathbf{e}_\nu$ grows at rate $\lambda_\nu - \lambda_\mu$. Only $\mu = 1$, the largest eigenvalue, is stable. $\square$

The neuron's output $v = \mathbf{w} \cdot \mathbf{u}$ becomes the projection of the input onto its direction of largest variance. A single Hebbian neuron performs **principal component analysis**, an unsupervised compression of its input. Networks of such neurons with lateral inhibition (Sanger's rule) extract several components.

11.2. Associative memory

Recurrent networks can hold short-term memories (the integrator of Chapter 10). Long-term **associative memory** stores patterns so that a partial or noisy cue recalls the complete pattern, as Pavlov's dog associates the bell with food. The idea is to make the stored patterns **fixed points** of the network dynamics, each with a **basin of attraction**: all initial conditions sufficiently similar to a memory flow toward it. Learning is unsupervised. Given patterns $\mathbf{v}_m$, $m = 1, \dots, N_{\text{mem}}$, choose weights such that $\mathbf{v}_m = \psi(M\mathbf{v}_m)$.

For rate networks with sparse binary patterns, $v_m(i) \in \{0, c\}$ with a fraction α of active units, a weight matrix that works is the Hebbian **covariance** construction

$$M = \frac{\lambda}{c^2 \alpha N (1 - \alpha)} \sum_{n=1}^{N_{\text{mem}}} (\mathbf{v}_n - \alpha c \mathbf{1})(\mathbf{v}_n - \alpha c \mathbf{1})^\top - \frac{\mathbf{1}\mathbf{1}^\top}{\alpha N},$$

whose first term makes each pattern an approximate eigenvector, using that uncorrelated patterns overlap only by $\mathbf{v}_n \cdot \mathbf{v}_m \simeq N\alpha^2 c^2$. The second term is a global inhibition that keeps inactive units off. With a threshold-like ψ satisfying $\psi(-c(1 + \alpha\lambda)) = 0$ and $\psi(c(\lambda - 1 - \alpha\lambda)) = c$, each pattern is a fixed point. The construction is symmetric, which guarantees convergence to fixed points (a Lyapunov function exists). As more patterns are stored, the crosstalk between them grows and the memories degrade: capacity is finite.

11.3. The Hopfield network

11.3.1. Definition and energy

Hopfield (1982) considered N binary units $\sigma_i = \pm 1$ with symmetric couplings $J_{ij} = J_{ji}$, $J_{ii} = 0$, and thresholds h_i . Asynchronous dynamics picks a unit at random and aligns it with its local field:

$$\sigma_i \leftarrow \text{sign}\left(\sum_j J_{ij} \sigma_j - h_i\right).$$

The dynamics never increases the **energy**

$$\mathcal{E}(\sigma) = -\frac{1}{2} \sum_{i,j} J_{ij} \sigma_i \sigma_j + \sum_i h_i \sigma_i.$$

Flipping σ_i changes the energy by $\Delta\mathcal{E} = 2\sigma_i(\sum_j J_{ij}\sigma_j - h_i)$, and the update flips σ_i only when this is negative. Since $\mathcal{E}$ is bounded below and decreases at each effective flip, the dynamics reaches a local minimum in finite time: a Metropolis algorithm at zero temperature. Symmetry of J is essential. With asymmetric couplings no energy exists and the dynamics can cycle.

11.3.2. The Hebb (Hopfield) rule

To store p patterns $\xi^\mu \in \{\pm 1\}^N$, set $(h_i = 0)$

$$J_{ij} = \frac{1}{N} \sum_{\mu=1}^p \xi_i^\mu \xi_j^\mu, \quad i \neq j.$$

Units that agree in a pattern get an excitatory coupling, units that disagree an inhibitory one. The energy becomes

$$\mathcal{E} = -\frac{N}{2} \sum_{\mu} m_{\mu}^2 + \frac{p}{2}, \quad m_{\mu} = \frac{1}{N} \sum_i \xi_i^\mu \sigma_i,$$

where m_{μ} is the **overlap** with pattern μ . Each stored pattern ($m_{\mu} = 1$) is therefore a low-energy configuration. The local field on unit i in state ξ^ν is

$$h_i = \sum_j J_{ij} \xi_j^\nu = \xi_i^\nu + \frac{1}{N} \sum_{\mu \neq \nu} \sum_{j \neq i} \xi_i^\mu \xi_j^\mu \xi_j^\nu.$$

The first term is the **signal**, which stabilises the pattern. The second is **crosstalk** from the other patterns: a sum of about Np random ± 1 terms divided by N , Gaussian with variance p/N . A unit is unstable when the crosstalk exceeds 1 with the wrong sign, with probability $P_{\text{err}} \simeq \frac{1}{2} \text{erfc}(\sqrt{N/2p})$. For small p/N errors are rare. As p/N grows, errors cascade.

11.3.3. Capacity and spurious states

The statistical mechanics of the model (Amit, Gutfreund and Sompolinsky, 1985) treats it as a spin glass at temperature T . Random patterns can be retrieved, with overlaps close to 1, as long as the **load** satisfies

$$\alpha = \frac{p}{N} < \alpha_c \simeq 0.138.$$

At α_c the retrieval states disappear discontinuously and the network falls into a spin-glass phase with no relation to the stored memories (catastrophic forgetting). Besides the stored patterns and their inverses $-\xi^\mu$ (the energy is even), there are **spurious** minima. The best known are odd mixtures $\text{sign}(\pm \xi^{\mu_1} \pm \xi^{\mu_2} \pm \xi^{\mu_3})$, which have overlap $1/2$ with each of three patterns. At large loads there are also spin-glass states. A finite temperature (stochastic updates, Chapter 12) destabilises the shallow spurious minima before the stored ones, which helps retrieval.

11.3.4. Variants

- **Sparse patterns** with activity $a \ll 1$ and covariance couplings $J_{ij} \propto \sum_{\mu} (\xi_i^\mu - a)(\xi_j^\mu - a)$ have a much larger capacity, $\alpha_c \sim 1/(a|\ln a|)$.
- **Diluted** networks, where each unit connects to only a fraction of others (like cortex), keep a capacity proportional to the number of connections per unit.

- **Robustness:** deleting a random fraction of the couplings degrades retrieval gracefully, a property shared by biological memory.

The Hopfield model also appears in biology beyond neuroscience. Gene-expression patterns of cell types behave like stored attractors of regulatory networks (Chapters 16–17).

11.4. Supervised learning: reservoir computing

A different use of recurrent networks keeps the recurrent weights random and fixed and trains only a linear readout. The **reservoir** has state $\mathbf{x}$ obeying

$$\tau_x \frac{d\mathbf{x}}{dt} = -\mathbf{x} + W_{\text{in}} I(t) + W_{\text{rec}} \psi(\mathbf{x}), \quad y(t) = \mathbf{W}_{\text{out}} \cdot \psi(\mathbf{x}(t)).$$

With W_{rec} random and scaled near the edge of chaos (spectral radius $\lesssim 1$), the reservoir maps the input history into a rich, high-dimensional nonlinear representation. Training reduces to linear regression. With $\mathcal{U}$ the matrix whose columns are the reservoir states $\mathbf{u}_t = \psi(\mathbf{x}_t)$ and $\mathbf{v}$ the target outputs, minimising the mean squared error gives

$$\mathbf{W}_{\text{out}} = (\mathcal{U}\mathcal{U}^\top + \beta \mathbf{1}\mathbf{1})^{-1} \mathcal{U} \mathbf{v},$$

with a ridge parameter β against overfitting. Online alternatives are stochastic gradient descent and the FORCE algorithm, which also trains feedback weights (Sussillo and Abbott, 2009). Echo-state networks predict chaotic time series and generate complex motor patterns (Jaeger and Haas, 2004).

11.5. Exercises

Exercise 11.1

(Oja's rule, class exercise Q1) Train a single linear neuron with Oja's rule ($\eta = 0.001$, $\alpha = 0.1$, $w(0)$ of norm 5 in a random direction) on $p = 10^4$ points drawn from $\mathcal{N}(\mathbf{0}, \mathbf{1})$ in two dimensions. Repeat from several initial conditions. What do the weights converge to, and why?

Solution

The norm always converges to $1/\sqrt{\alpha} = 3.16$ (simulations: 3.1630 in three runs), but the direction is different in each run $((-2.61, 1.78), (-0.88, 3.04), (1.78, -2.61))$. For isotropic data $C = \mathbf{1}$, every direction is an eigenvector with the same eigenvalue: the first principal component is not defined. The rule fixes the norm, while the direction diffuses neutrally, driven only by sampling noise.

Exercise 11.2

(Oja's rule, Q2) Repeat with $p = 1000$ points of covariance $\text{diag}(10, 1)$, mean $(10, 10)$, rotated by 30° . Do the weights still find the first principal component?

Solution

No. Without centring the data the rule converges to the leading eigenvector of the *second-moment* matrix $\langle \mathbf{u}\mathbf{u}^\top \rangle = C + \mu\mu^\top$, which is dominated by the mean. The simulated weight direction is $(0.27, 0.96)$, essentially the mean direction $(0.26, 0.97)$, while the true first principal component is $(0.88, 0.48)$. Oja's rule performs PCA only on zero-mean inputs. Biologically, inputs must be centred first, e.g. by adaptation, which subtracts the mean rate.

Exercise 11.3

(Oja's rule, Q3) Compare Oja's rule with $\Delta \mathbf{w} = \eta [C\mathbf{w} - \alpha \mathbf{w}(\mathbf{w}^\top \mathbf{w})]$ on centred data.

Solution

At a fixed point of the new rule, $C\mathbf{w} = \alpha |\mathbf{w}|^2 \mathbf{w}$, so $\mathbf{w}$ is again an eigenvector, and the same stability argument selects $\mathbf{e}_1$. The norm is different: $|\mathbf{w}|^2 = \lambda_1/\alpha$ instead of $1/\alpha$. With covariance $\text{diag}(10, 1)$ rotated by 30° ($\lambda_1 \simeq 10.35$ in the sample), Oja's rule gives $|\mathbf{w}| = 3.16$ while the new rule gives $|\mathbf{w}| \simeq \sqrt{\lambda_1/\alpha} = 10.2$ (about 9.6 after five epochs of online learning, still converging). Both point along $(0.87, 0.50)$. The new rule encodes the *variance* of the leading component in the weight norm, whereas Oja's rule normalises it away.

Exercise 11.4

(Instability of Hebb's rule) For $\tau_w \dot{\mathbf{w}} = Q\mathbf{w}$ with $Q = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, solve for $\mathbf{w}(t)$ from $\mathbf{w}(0) = (1, 0)$.

Show that the direction converges while the norm diverges, and add subtractive normalisation to see what changes.

Solution

Eigenvalues 3 (along $(1, 1)/\sqrt{2}$) and 1 (along $(1, -1)/\sqrt{2}$). $\mathbf{w}(t) = \frac{1}{2}e^{3t/\tau_w}(1, 1) + \frac{1}{2}e^{t/\tau_w}(1, -1)$: the direction tends to $(1, 1)/\sqrt{2}$ but $|\mathbf{w}|$ grows as e^{3t/τ_w} . With subtractive normalisation the projection on $\mathbf{1}$ is frozen at $w_1 + w_2 = 1$, and only the component along $(1, -1)$ evolves, at rate $\lambda_2 = 1$: $\mathbf{w}(t) = \frac{1}{2}(1, 1) + \frac{1}{2}e^{t/\tau_w}(1, -1)$. This grows until the bounds are hit, ending at $(1, 0)$ or $(0, 1)$: one synapse wins and the other is eliminated. Normalisation turns Hebbian growth into competition.

Exercise 11.5

(Hopfield capacity, class exercise 1) Store p random patterns in a network of $N = 400$ units and test retrieval from a 5%-corrupted version of each pattern using asynchronous zero-temperature updates. Plot the fraction of successful retrievals (final overlap > 0.9) against $\alpha = p/N$.

Solution

Simulation (up to 20 patterns tested per load): $\alpha = 0.05$: 100%; 0.10: 100%; 0.12: 90%; 0.14: 95%; 0.16: 65%; 0.20: 30%. Retrieval is essentially perfect below $\alpha \approx 0.12$ and collapses between 0.14 and 0.2, consistent with the theoretical $\alpha_c \simeq 0.138$; finite-size effects smooth the transition. With the signal-to-noise estimate, the single-step error probability at $\alpha = 0.14$ is $\frac{1}{2} \text{erfc}(\sqrt{1/0.28}) \approx 0.4\%$. Errors then cascade, which is why the actual threshold is lower than a naive estimate would suggest.

Exercise 11.6

(HW 5) Playing with the Hopfield code. Use the course code `hopfield.py`: load images, convert them to ± 1 patterns, store them with the Hebb rule, and retrieve them from corrupted cues. Summarise your observations in half a page.

Solution

A typical report:

- **Storage and retrieval.** With a few (2–4) images of $N \sim 10^3$ – 10^4 pixels, corrupting 20–30% of a pattern and running the dynamics recovers it exactly within a few sweeps, while the energy decreases monotonically.
- **Inverse states.** Very corrupted cues ($> 50\%$ flipped) converge to the *inverted* image, the other minimum of the even energy.
- **Correlated patterns.** Natural images are correlated (large uniform regions, similar backgrounds), so their crosstalk is much larger than for random patterns. Storing several similar images produces mixtures: the capacity for correlated patterns is far below $0.138N$. Centring the patterns (subtracting the mean activity), as in the covariance rule, improves it.
- **Mixtures.** Starting from a superposition of three patterns often ends in the spurious mixture state.
- **Temperature.** Running at small T (stochastic updates) helps escape spurious states. Too much noise destroys the memories.
- **Damage.** Setting a fraction of weights to zero leaves retrieval nearly intact until a large fraction is removed.

Exercise 11.7

(Spurious mixture states) For three patterns, show that $\sigma = \text{sign}(\xi^1 + \xi^2 + \xi^3)$ has overlap $1/2$ with each, and compute its energy relative to a pure pattern for large N and $p = 3$.

Solution

At each site, the sign of the sum agrees with ξ_i^1 unless both ξ_i^2 and ξ_i^3 disagree with it, which has probability $1/4$. So $m_1 = \frac{3}{4} - \frac{1}{4} = \frac{1}{2}$, and likewise for m_2, m_3 . The energy is $\mathcal{E} = -\frac{N}{2} \sum_{\mu} m_{\mu}^2 = -\frac{N}{2} \cdot \frac{3}{4} = -0.375N$, against $-\frac{N}{2}(1 + O(1/N))$ for a pure pattern (whose overlaps with the other two are $O(N^{-1/2})$). The mixture is a local minimum, but shallower than the stored memories, so a small temperature removes it first.

11.6. Sources

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12. Boltzmann machines and maximum entropy

Hopfield networks store patterns as minima of an energy at zero temperature. **Boltzmann machines** add noise: units update stochastically, the network samples the Boltzmann distribution of its energy, and learning becomes the statistical problem of fitting a probability distribution to data. This chapter derives the Boltzmann machine learning rule, connects it to the maximum-entropy principle (used again for gene networks in Chapter 17), and introduces restricted Boltzmann machines and contrastive divergence, the basis of early deep learning.

12.1. Constraint satisfaction and stochastic networks

Many problems ask for variables $x_1, \dots, x_n$ that satisfy constraints. **Hard** constraints must hold exactly. **Soft** constraints may be violated at a cost: perception (consistent interpretations of an ambiguous image), scheduling, or reproducing the statistics of data such as $\langle x_i \rangle$ and $\langle x_i x_j \rangle$. A good solution satisfies the hard constraints and minimises the total cost of violated soft ones.

A **Boltzmann machine** (Hinton and Sejnowski, 1983) encodes such a problem in a network of binary units $s_i \in \{0, 1\}$ (or ± 1) with symmetric weights $W_{ij} = W_{ji}$ and thresholds θ_i , and the energy

$$\mathcal{E}(\mathbf{s}) = - \sum_{i < j} W_{ij} s_i s_j + \sum_i \theta_i s_i.$$

A positive weight rewards units i and j being on together; a negative weight penalises it. Low energy means well-satisfied constraints. Some units can be **clamped** to externally fixed values (inputs), and the network then looks for the best completion of the rest.

Example 12.1 (Three units)

Take $W_{12} = -1$, $W_{13} = W_{23} = 1$, $\theta = 0$, and clamp $s_1 = 1$. The four completions have energies $\mathcal{E}(s_2, s_3)$: $(0, 0)$: 0; $(0, 1)$: $-W_{13} = -1$; $(1, 0)$: $-W_{12} = +1$; $(1, 1)$: $-(W_{12} + W_{13} + W_{23}) = -1$. Two completions share the minimum: unit 3 should be on (it is rewarded by both others), while unit 2 is torn between its penalty with unit 1 and its reward with unit 3, which cancel.

12.1.1. Stochastic dynamics and annealing

Deterministic descent (the Hopfield rule) gets stuck in local minima. A Boltzmann machine updates a randomly chosen unit k with

$$P(s_k = 1) = \frac{1}{1 + e^{-\Delta \mathcal{E}_k / T}}, \quad \Delta \mathcal{E}_k = \mathcal{E}(s_k = 0) - \mathcal{E}(s_k = 1) = \sum_i W_{ki} s_i - \theta_k,$$

the energy gap of unit k . As $T \rightarrow 0$ this becomes the deterministic threshold rule. At $T > 0$ uphill moves occur, more often at higher temperature, which lets the network escape local minima. The Markov chain is ergodic and satisfies detailed balance (Chapter 10), so it converges to the **Boltzmann distribution**

$$P(\mathbf{s}) = \frac{e^{-\mathcal{E}(\mathbf{s})/T}}{Z}, \quad \frac{P(\mathbf{s}_\alpha)}{P(\mathbf{s}_\beta)} = e^{-[\mathcal{E}(\mathbf{s}_\alpha) - \mathcal{E}(\mathbf{s}_\beta)]/T}.$$

Simulated annealing (Kirkpatrick, Gelatt, Vecchi, 1983) lowers T slowly. At high T the system explores broadly; as T decreases it settles into deep minima. A logarithmic cooling schedule guarantees convergence to the global minimum; faster schedules work well in practice.

12.2. Learning

12.2.1. Visible and hidden units

Split the units into **visible** ones $\mathbf{v}$, which represent data, and **hidden** ones $\mathbf{h}$, which are free and can capture higher-order structure. Given M training patterns $\mathbf{v}^a$, the data define a target distribution

$$P_+(\mathbf{v}) = \frac{1}{M} \sum_{a=1}^M \delta_{\mathbf{v}, \mathbf{v}^a},$$

while the freely running network produces $P_-(\mathbf{v}) = \sum_{\mathbf{h}} P(\mathbf{v}, \mathbf{h})$. Learning means adjusting W so that P_- approximates P_+ .

12.2.2. The learning rule

Measure the mismatch with the Kullback–Leibler divergence

$$G = D_{KL}(P_+ \| P_-) = \sum_{\mathbf{v}} P_+(\mathbf{v}) \ln \frac{P_+(\mathbf{v})}{P_-(\mathbf{v})} \geq 0.$$

Minimising G is equivalent to maximising the log-likelihood of the data, since

$$\mathcal{L} = \sum_a \ln P_-(\mathbf{v}^a) = M \sum_{\mathbf{v}} P_+(\mathbf{v}) \ln P_-(\mathbf{v}) = -M[S(P_+) + D_{KL}(P_+ \| P_-)],$$

and the entropy $S(P_+)$ of the data does not depend on W .

Theorem 12.1 (Boltzmann machine learning rule).

$$\frac{\partial G}{\partial W_{ij}} = -\frac{1}{T} (\langle s_i s_j \rangle_+ - \langle s_i s_j \rangle_-),$$

where $\langle \cdot \rangle_+$ averages over the network with visible units clamped to the data (hidden units at equilibrium) and $\langle \cdot \rangle_-$ over the freely running network. Gradient descent gives

$$\Delta W_{ij} = \epsilon (\langle s_i s_j \rangle_{\text{clamped}} - \langle s_i s_j \rangle_{\text{free}}).$$

Proof. $P_-(\mathbf{v}) = Z^{-1} \sum_{\mathbf{h}} e^{-\mathcal{E}(\mathbf{v}, \mathbf{h})/T}$ with $\partial_{W_{ij}}(-\mathcal{E}) = s_i s_j$. Then

$$\frac{\partial \ln P_-(\mathbf{v})}{\partial W_{ij}} = \frac{1}{T} \left[\sum_{\mathbf{h}} P(\mathbf{h} | \mathbf{v}) s_i s_j - \sum_{\mathbf{v}', \mathbf{h}} P(\mathbf{v}', \mathbf{h}) s_i s_j \right].$$

Averaging over P_+ : $\partial G / \partial W_{ij} = -\sum_{\mathbf{v}} P_+(\mathbf{v}) \partial \ln P_-(\mathbf{v}) / \partial W_{ij} = -\frac{1}{T} [\langle s_i s_j \rangle_+ - \langle s_i s_j \rangle_-]$. $\square$

The rule is **local**: a synapse only needs the correlation of its two units in two phases. In the clamped (“wake”) phase it is Hebbian; in the free (“sleep”) phase it is anti-Hebbian, unlearning what the network produces spontaneously. Learning stops when the network reproduces the data correlations. Without hidden units G is convex in W and the optimum is unique. The practical difficulty is that both averages require equilibrium sampling, which is very slow for large, densely connected networks.

12.3. Maximum entropy

Without hidden units, the Boltzmann machine is a maximum-entropy model. Given data on binary variables σ_i , suppose we only trust their means $\langle \sigma_i \rangle_D$ and pairwise correlations $\langle \sigma_i \sigma_j \rangle_D$. The least biased distribution reproducing them maximises the entropy $S = -\sum_{\sigma} P \ln P$ subject to these constraints. With Lagrange multipliers h_i and J_{ij} ,

$$P(\sigma) = \frac{1}{Z} \exp\left(\sum_i h_i \sigma_i + \sum_{i < j} J_{ij} \sigma_i \sigma_j\right) :$$

an **Ising model** with inhomogeneous fields and couplings. The multipliers are found by minimising $D_{KL}(P_D \| P)$, whose gradient is exactly the Boltzmann rule:

$$\Delta J_{ij} \propto \langle \sigma_i \sigma_j \rangle_D - \langle \sigma_i \sigma_j \rangle_{\text{model}}, \quad \Delta h_i \propto \langle \sigma_i \rangle_D - \langle \sigma_i \rangle_{\text{model}}.$$

Pairwise maximum-entropy models describe the collective activity of retinal neurons remarkably well: weak pairwise correlations produce strongly correlated population states (Schneidman et al., 2006). The same inference is applied to gene expression (Chapter 17), protein sequences and flocks of birds. Note that couplings J_{ij} are *effective* interactions, not necessarily physical connections: correlation does not imply direct coupling, and the inverse Ising problem disentangles direct from indirect correlations.

12.4. Restricted Boltzmann machines

Learning is easier if visible and hidden units form a **bipartite** graph, with no visible-visible and no hidden-hidden connections (Smolensky, 1986). The energy of an RBM is

$$E(\mathbf{v}, \mathbf{h}) = -\sum_{i,\mu} W_{i\mu} v_i h_{\mu} - \sum_i f_i(v_i) - \sum_{\mu} f_{\mu}(h_{\mu}),$$

with single-unit potentials f (for binary units, $f_i(v_i) = a_i v_i$ and $f_{\mu}(h_{\mu}) = b_{\mu} h_{\mu}$). Because no hidden unit talks to another, given $\mathbf{v}$ the hidden units are **conditionally independent**:

$$P(\mathbf{h} | \mathbf{v}) = \prod_{\mu} \frac{e^{h_{\mu} I_{\mu} + f_{\mu}(h_{\mu})}}{Z_{\mu}(I_{\mu})}, \quad I_{\mu} = \sum_i W_{i\mu} v_i,$$

and symmetrically $P(\mathbf{v} | \mathbf{h}) = \prod_i P(v_i | \mathbf{h})$. For binary units,

$$P(h_{\mu} = 1 | \mathbf{v}) = \sigma(b_{\mu} + I_{\mu}), \quad P(v_i = 1 | \mathbf{h}) = \sigma\left(a_i + \sum_{\mu} W_{i\mu} h_{\mu}\right),$$

with $\sigma(x) = 1/(1 + e^{-x})$: each layer is a layer of logistic neurons driven by the other. The most probable hidden state given $\mathbf{v}$ minimises the energy, $h_{\mu}^* = (f'_{\mu})^{-1}(I_{\mu})$, which defines the effective **activation function** of the hidden unit. The choice of potential determines whether the unit behaves as a sigmoid, a ReLU or a linear unit.

Sampling is done by **block Gibbs sampling**: from $\mathbf{v}$ sample all of $\mathbf{h}$ at once, then from $\mathbf{h}$ sample all of $\mathbf{v}$, and so on. Summing out the hidden units gives the marginal of the visible layer, $P(\mathbf{v}) \propto e^{\sum_i a_i v_i} \prod_{\mu} (1 + e^{b_{\mu} + I_{\mu}})$: each hidden unit adds a “feature detector” term. An RBM with enough hidden units can approximate any distribution over $\mathbf{v}$.

12.4.1. Contrastive divergence

The learning rule is again $\Delta W_{i\mu} = \epsilon(\langle v_i h_{\mu} \rangle_{\text{data}} - \langle v_i h_{\mu} \rangle_{\text{model}})$. The data term is easy: clamp $\mathbf{v}$ to a data vector and use the exact $P(h_{\mu} = 1 | \mathbf{v})$. The model term would require running the Gibbs chain to

12. Boltzmann machines and maximum entropy

equilibrium. Hinton's **contrastive divergence** (CD- k) approximates it by starting the chain *at the data* and running only k steps (often $k = 1$):

$$\mathbf{v}^{(0)} = \text{data} \rightarrow \mathbf{h}^{(0)} \rightarrow \mathbf{v}^{(1)} \rightarrow \mathbf{h}^{(1)}, \quad \Delta W_{i\mu} = \epsilon(v_i^{(0)}h_\mu^{(0)} - v_i^{(1)}h_\mu^{(1)}).$$

It does not follow the exact likelihood gradient, but it works well in practice and made RBMs trainable. Stacking RBMs, each trained on the hidden activities of the previous one, gives deep belief networks (Hinton et al., 2006), a key step in the revival of deep learning.

12.5. Connection to statistical mechanics

Boltzmann machines are Ising spin systems with learned couplings, and their behaviour is governed by statistical mechanics.

- The **free energy** $F = -T \ln Z$ and its derivatives give the moments needed for learning; mean-field approximations (naive mean field, TAP, Bethe) replace sampling by self-consistent equations, $m_i = \tanh(\beta(h_i + \sum_j J_{ij}m_j))$ for ± 1 units.
- Learned models are often found near a **critical point** of their own parameter space: small changes of an effective temperature produce large changes in their statistics. Whether this reflects the data or the inference is an active debate, connected with brain criticality (Chapter 13).
- RBMs are equivalent to Hopfield networks: integrating out Gaussian hidden units yields $P(\mathbf{v}) \propto \exp(\frac{\beta}{2} \sum_\mu (\sum_i W_{i\mu} v_i)^2)$, a Hopfield energy with the hidden units' weights as stored patterns (Barra et al., 2012). Retrieval and learning are two faces of the same model.

12.6. Exercises

Exercise 12.1

(Boltzmann distribution of a small machine) For the three-unit example at temperature $T = 1$ with s_1 clamped to 1, compute the probabilities of the four states of (s_2, s_3) and the marginal $P(s_3 = 1)$.

Solution

Energies: $(0, 0): 0$; $(0, 1): -1$; $(1, 0): +1$; $(1, 1): -1$. Boltzmann weights $1, e, e^{-1}, e$; $Z = 1 + 2e + e^{-1} = 6.80$. Probabilities $0.147, 0.400, 0.054, 0.400$. $P(s_3 = 1) = 0.80$: the machine is confident about unit 3 and undecided about unit 2 ($P(s_2 = 1) = 0.45$), mirroring the energy analysis.

Exercise 12.2

(Learning rule without hidden units) A machine with two ± 1 units, $P(\sigma_1, \sigma_2) \propto e^{h(\sigma_1 + \sigma_2) + J\sigma_1\sigma_2}$, is trained on data with $\langle \sigma_i \rangle_D = 0$ and $\langle \sigma_1 \sigma_2 \rangle_D = 0.6$. Find the maximum-entropy parameters.

Solution

By symmetry $h = 0$. Then $\langle \sigma_1 \sigma_2 \rangle = \frac{2e^J - 2e^{-J}}{2e^J + 2e^{-J}} = \tanh J$. Matching, $\tanh J = 0.6$, so $J = \text{artanh } 0.6 = 0.693 = \ln 2$. The learning rule $\Delta J \propto 0.6 - \tanh J$ converges to this value from any start, since G is convex in J .

Exercise 12.3

(RBM conditionals) An RBM with two visible and one hidden binary unit has $W = (2, -1)^\top$, $a = (0, 0)$, $b = -0.5$. Compute $P(h = 1 \mid \mathbf{v})$ for all four $\mathbf{v}$, and $P(\mathbf{v})$ up to normalisation.

Solution

$I = 2v_1 - v_2$, so $P(h = 1 \mid \mathbf{v}) = \sigma(I - 0.5)$: $\mathbf{v} = (0, 0)$: $\sigma(-0.5) = 0.38$; $(0, 1)$: $\sigma(-1.5) = 0.18$; $(1, 0)$: $\sigma(1.5) = 0.82$; $(1, 1)$: $\sigma(0.5) = 0.62$. Marginal: $P(\mathbf{v}) \propto 1 + e^{b+I}$: $1 + e^{-0.5} = 1.61$, $1 + e^{-1.5} = 1.22$, $1 + e^{1.5} = 5.48$, $1 + e^{0.5} = 2.65$. Normalised: 0.15, 0.11, 0.50, 0.24. The hidden unit acts as a detector of “ v_1 on and v_2 off” and makes that pattern the most probable.

Exercise 12.4

(Simulated annealing) For an energy with two minima, $\mathcal{E}(x) = (x^2 - 1)^2 + 0.3x$ on a grid, explain why Metropolis sampling at fixed small T may stay in the wrong minimum, and how annealing helps.

Solution

The global minimum is near $x \simeq -1$ (the $0.3x$ term lowers it by 0.6 relative to $x \simeq 1$), separated by a barrier of height about 1 at $x \approx 0$. At $T = 0.05$ the escape rate $\propto e^{-\Delta\mathcal{E}/T} \simeq e^{-20}$ is negligible: a chain started near $x = 1$ stays there. At $T \sim 0.5$ barrier crossings are frequent and the chain visits both wells, spending more time in the deeper one. Lowering T slowly then traps it in the global minimum with high probability. Annealing trades speed for the ability to cross barriers early.

12.7. Sources

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13. Whole-brain models and brain criticality

Can the large-scale anatomy of the brain predict its spontaneous activity? And why would the brain operate close to a phase transition? This chapter combines the two questions. It starts from the theory of critical branching processes, which explains the exponents of neuronal avalanches. It then presents a minimal stochastic **whole-brain model**, the Haimovici–Tagliazucchi–Chialvo (HTC) model, which runs on the human connectome measured by DTI and reproduces resting-state functional connectivity best near its critical point. A mean-field analysis with the tools of Part I shows the transition. Applications to individual subjects and stroke patients suggest that criticality may be a biomarker of brain health.

13.1. The criticality hypothesis

Many systems with many interacting components show collective phases, and changes in parameters (temperature, coupling) can induce phase transitions between them. At a continuous transition, correlations extend over the whole system, fluctuations are maximal, responses to perturbations are largest, and observables follow power laws with universal exponents. The **criticality hypothesis** proposes that the brain, and perhaps other living systems, is poised near such a point, because criticality would confer:

- a large **dynamic range**: sensitivity to stimuli spanning many orders of magnitude (Kinouchi and Copelli, 2006);
- maximal **information transmission** and storage;
- a rich repertoire of metastable states, and fast, coordinated global responses.

The experimental support comes from two observations: **scale-free avalanches**, with sizes distributed as $s^{-3/2}$ and durations as T^{-2} , found in cortical slices, rats, monkeys and humans (MEG, EEG, fMRI); and **long-range temporal correlations** in the amplitude fluctuations of neural oscillations ($1/f$ spectra). Chapter 8 showed that power laws alone are not proof: what matters is the full set of critical signatures (exponent relations, data collapse, scaling with system size).

13.2. Critical branching processes

The simplest model of activity propagation is a **branching process**. Each active unit independently activates k units in the next time step, with k drawn from an offspring distribution p_k with mean σ , the **branching ratio**. An avalanche starts from one active unit and ends when activity dies out.

- $\sigma < 1$ (subcritical): activity dies quickly; avalanche sizes have an exponential cut-off.
- $\sigma > 1$ (supercritical): with finite probability activity explodes.
- $\sigma = 1$ (critical): activity always dies out, but avalanches of every size occur.

Theorem 13.1 (Avalanche exponents of the critical branching process). *At $\sigma = 1$, with finite offspring variance, the size and duration distributions decay as*

$$P(s) \sim s^{-3/2}, \quad P(T) \sim T^{-2},$$

and the mean size of avalanches of duration T scales as $\langle s \rangle_T \sim T^2$.

Sketch. Let $g(z) = \sum_k p_k z^k$ be the offspring generating function and $Q(z) = \sum_s P(s) z^s$ that of the total size. An avalanche is the root plus independent sub-avalanches from each child, so $Q(z) = z g(Q(z))$. Near $z = 1$, with $g(1) = 1$, $g'(1) = 1$ and $g''(1) = b > 0$, expand $Q = 1 - \epsilon$: $1 - \epsilon \simeq z(1 - \epsilon + \frac{b}{2}\epsilon^2)$, so $\epsilon \simeq \sqrt{2(1-z)/b}$. A square-root singularity of Q at $z = 1$ corresponds to $P(s) \sim s^{-3/2}$. For durations, the survival probability u_t obeys $1 - u_{t+1} = g(1 - u_t)$, so $u_{t+1} \simeq u_t - \frac{b}{2}u_t^2$, giving $u_t \simeq 2/(bt)$ and a duration density $\sim t^{-2}$. $\square$

These are the mean-field exponents observed by Beggs and Plenz. They satisfy the **crackling-noise relation** $\frac{\alpha-1}{\tau-1} = \gamma$, where $P(s) \sim s^{-\tau}$, $P(T) \sim T^{-\alpha}$ and $\langle s \rangle_T \sim T^\gamma$: $(2-1)/(3/2-1) = 2$. Experimental tests of criticality check this relation and the collapse of avalanche shapes, not only the exponents. The branching ratio can be estimated from data as the mean number of active units following each active unit; values close to 1 are found in cortical recordings, with refinements needed for subsampling.

13.3. Whole-brain models

13.3.1. From structure to function

Structural connectivity (SC): DTI tractography gives a weighted network W_{ij} between N cortical regions, its weights proportional to the number of fibres. **Functional connectivity** (FC): the Pearson correlation matrix of resting-state BOLD signals,

$$F_{ij} = \frac{\langle x_i x_j \rangle - \langle x_i \rangle \langle x_j \rangle}{\sigma_i \sigma_j}.$$

FC is organised into **resting-state networks** (default mode, visual, sensorimotor, attention...) that are reproducible across subjects. Whole-brain models put a dynamical system on each node, couple the nodes through the connectome, and compare simulated with empirical FC. “Make everything as simple as possible, but not simpler”: the HTC model uses the simplest possible node dynamics.

13.3.2. The HTC model

Each node i is in one of three states: quiescent Q , excited E (written $s_i = 1$), or refractory R ($s_i = 0$ for Q and R). Time is discrete and updates are synchronous, following a stochastic Greenberg–Hastings excitable automaton:

$$Q \rightarrow E \text{ if } \sum_j W_{ij} s_j > T, \text{ or spontaneously with probability } r_1; \quad E \rightarrow R \text{ with probability } 1; \quad R \rightarrow Q \text{ with probability } r_2.$$

A node fires if enough of its neighbours fire (weighted by the connectome) or spontaneously. It then becomes refractory and recovers after a geometric time with mean $1/r_2$. The threshold T is the control parameter. Typical values are $r_1 = 2/N$ (rare spontaneous activation) and $r_2 = r_1^{1/5}$.

Observables: the mean activity $\langle A \rangle$, with $A(t) = \frac{1}{N} \sum_i s_i(t)$, its standard deviation $\sigma(A)$, and the average sizes of the largest and second-largest **clusters** $\langle S_1 \rangle$, $\langle S_2 \rangle$. Clusters are sets of structurally connected nodes that are simultaneously active.

13.3.3. Two phases and a transition

- **Small T (supercritical):** even weak connections propagate activity. Activity is high and sustained, fluctuations are fast and uncorrelated, and one giant cluster dominates.
- **Large T (subcritical):** only the strongest connections transmit. Activity is low and sparse, made of short-lived local events.

- **Intermediate** T_c : a continuous transition. $\langle S_1 \rangle$ drops, $\sigma(A)$ peaks and $\langle S_2 \rangle$ peaks, the signature of a percolation-like transition, used to locate T_c . At T_c the cluster-size distribution is a truncated power law, $P(S) \sim S^{-\alpha}$ with $\alpha \simeq 3/2$.

Comparing with data: the Pearson correlation between simulated and empirical FC, $\rho(F_m, F_e)$, and the similarity of simulated resting-state networks to empirical templates are both **maximal near** T_c . A model of the healthy brain matches its spontaneous activity best at criticality.

13.3.4. Homeostatic plasticity

In the raw model, activity is dominated by hubs, nodes with large total input strength $W_i = \sum_j W_{ij}$. Rocha et al. (2018) proposed a homeostatic normalisation that equalises the input of every region,

$$\widetilde{W}_{ij} = \frac{W_{ij}}{\sum_j W_{ij}},$$

so every node has unit in-strength, as if synapses had adjusted to keep each region's excitability constant. With $\widetilde{W}$:

- the transition becomes sharper and the $\langle S_2 \rangle$ peak more pronounced;
- the cluster-size power law extends about three times further (cut-off ~ 15 instead of ~ 5) with $\alpha = 3/2$ (against $\alpha \simeq 1.75$ without normalisation);
- the match with empirical FC improves by a factor of about 1.5 at the group level and up to 2 for individuals.

13.3.5. Personalised models

Using the structural and functional matrices of 43 healthy subjects ($N = 264$ regions, Rudie et al., 2012), each individual connectome can be simulated. Without normalisation each subject has a different critical threshold, but the ratio $T_c/\langle W \rangle \simeq 0.16$ is nearly constant, as predicted by mean field (T_c scales with the mean weight). With normalisation all subjects collapse onto the same curves with $T_c \simeq 0.15$: homeostasis makes the critical point universal across individuals. This makes personalised modelling feasible.

13.3.6. Stroke and recovery

Applied to stroke patients (Corbetta et al., 2015), personalised models show that lesions push the brain away from criticality. Activity and cluster statistics of patients deviate from those of controls, with a weaker transition (lower $\sigma(A)$ and $\langle S_2 \rangle$ peaks). During recovery, from 3 months to 1 year after the stroke, patients move back toward critical behaviour. The changes in these criticality measures correlate with the recovery of behavioural scores across subjects (Rocha et al., 2022). Criticality is a candidate **biomarker** of healthy brain dynamics.

13.4. Mean-field theory of the HTC model

13.4.1. Master equation

Consider the fully connected, normalised case, $\widetilde{W}_{ij} = 1/N$. The input to a node is the fraction of active nodes, $\sum_j \widetilde{W}_{ij} s_j - T = x - T$. Let n_x, n_y be the numbers of excited and refractory nodes ($n_Q = N - n_x - n_y$). In continuous time the transitions are

$$(n_x, n_y) \rightarrow (n_x + 1, n_y) \text{ at rate } (N - n_x - n_y) r_{\text{act}}, \quad r_{\text{act}} = r_1 + (1 - r_1)\Theta(x - T),$$

13. Whole-brain models and brain criticality

$$(n_x, n_y) \rightarrow (n_x - 1, n_y + 1) \text{ at rate } n_x, \quad (n_x, n_y) \rightarrow (n_x, n_y - 1) \text{ at rate } r_2 n_y.$$

With densities $x = n_x/N$, $y = n_y/N$, the Kramers–Moyal expansion (Chapters 2 and 5) gives a Fokker–Planck equation with drift and diffusion

$$A_1 = (1 - x - y)r_{\text{act}} - x, \quad A_2 = x - r_2 y,$$

$$B_{11} = (1 - x - y)r_{\text{act}} + x, \quad B_{22} = r_2 y + x, \quad B_{12} = B_{21} = -x,$$

the off-diagonal term coming from the $E \rightarrow R$ reaction, which moves one unit from x to y .

13.4.2. Equilibria in the thermodynamic limit

For $N \rightarrow \infty$, $\dot{x} = A_1$, $\dot{y} = A_2$. Two regimes arise from the step function.

- **Supercritical branch** ($x > T$, $r_{\text{act}} = 1$): $y_+ = \frac{1}{2r_2+1}$, $x_+ = r_2 y_+$. It exists iff $x_+ > T$, i.e. $T < T_+ \equiv \frac{r_2}{2r_2+1}$. The activity is independent of r_1 : it is self-sustained by interactions.
- **Subcritical branch** ($x \leq T$, $r_{\text{act}} = r_1$): $y_- = \frac{r_1}{r_2+(r_2+1)r_1}$, $x_- = r_2 y_-$. It exists iff $T \geq T_- \equiv x_-$.

Since $r_1 < 1$, $T_- < T_+$. Three regions follow: for $T < T_-$ only the active phase exists, for $T > T_+$ only the quiescent phase, and for $T_- < T < T_+$ **both** are stable, each with its own basin of attraction. In mean field the transition is therefore discontinuous (first order), with **hysteresis**: increasing T the system jumps down at T_+ , decreasing it jumps up at T_- . On real, sparse and heterogeneous connectomes the transition is smoothed into the continuous-looking transition seen in simulations.

13.4.3. Stability

At the supercritical point,

$$J_+ = \begin{pmatrix} -2 & -1 \\ 1 & -r_2 \end{pmatrix}, \quad \lambda_+ = \frac{-(2+r_2) \pm \sqrt{r_2^2 - 4r_2}}{2} :$$

always stable, a focus for $r_2 < 4$ and a node for $r_2 > 4$. At the subcritical point,

$$J_- = \begin{pmatrix} -1-r_1 & -r_1 \\ 1 & -r_2 \end{pmatrix}, \quad \lambda_- = \frac{-(1+r_1+r_2) \pm \sqrt{(1+r_1+r_2)^2 - 4(r_1+r_2+r_1r_2)}}{2} :$$

always stable, and a focus iff $r_1 - 2\sqrt{r_1} + 1 < r_2 < r_1 + 2\sqrt{r_1} + 1$.

13.4.4. Fluctuations and oscillations

Around a stable equilibrium the linear noise approximation gives $\dot{\zeta} = J\zeta + \eta$, with $\langle \eta\eta^\top \rangle = B$ evaluated at the fixed point, and the power spectrum

$$P_x(\omega) = \frac{\alpha + \beta\omega^2}{(\omega^2 - \Omega_0^2)^2 + \Gamma^2\omega^2}, \quad \alpha = B_{11}J_{22}^2 - 2B_{12}J_{12}J_{22} + B_{22}J_{12}^2, \quad \beta = B_{11}, \quad \Omega_0^2 = \det J, \quad \Gamma = -\text{tr } J,$$

the formula of Chapter 5. In the fully connected model the foci are strongly damped ($\text{Re } \lambda \simeq -1$) and spectral peaks are weak. Network structure changes this: on connectomes and random graphs, near the critical threshold the model shows collective oscillations and a peak in the power spectrum. Activity propagates as waves on Erdős–Rényi graphs and as localised patterns on the connectome. Criticality and structure jointly drive emergent oscillations (Barzon et al., 2022).

13.5. Exercises

Exercise 13.1

(Branching ratio) In a branching process each active neuron activates each of $K = 10$ downstream neurons independently with probability p . (a) What is the branching ratio? (b) For which p is the process critical? (c) For $p = 0.08$, what is the mean avalanche size?

Solution

(a) $\sigma = Kp$. (b) $p_c = 1/K = 0.1$. (c) For $\sigma < 1$ the mean total size is $\sum_t \sigma^t = 1/(1 - \sigma) = 1/(1 - 0.8) = 5$ neurons. It diverges as $\sigma \rightarrow 1$ with exponent 1, the susceptibility of this mean-field transition.

Exercise 13.2

(Critical avalanche sizes) For binary branching (each active unit has 0 or 2 offspring with probability $1/2$), solve $Q(z) = z g(Q(z))$ exactly and verify $P(s) \sim s^{-3/2}$.

Solution

$g(z) = \frac{1}{2}(1 + z^2)$, so $Q = \frac{z}{2}(1 + Q^2)$, i.e. $zQ^2 - 2Q + z = 0$ and $Q(z) = \frac{1 - \sqrt{1 - z^2}}{z}$, taking the root regular at $z = 0$. Expanding $\sqrt{1 - z^2} = 1 - \sum_{k \geq 1} \frac{(2k-3)!!}{2^k k!} z^{2k}$ gives non-zero probabilities only for odd sizes, $P(s = 2k - 1) = \frac{(2k-3)!!}{2^k k!}$. By Stirling $(2k-3)!!/(2^k k!) \sim \frac{1}{2\sqrt{\pi}} k^{-3/2}$, so $P(s) \sim s^{-3/2}$. The square-root singularity of Q at $z = 1$ is the generating-function signature of the exponent $3/2$.

Exercise 13.3

(HTC mean field) For $r_1 = r_2 = 0.1$, compute $x_{\pm}$, $T_{\pm}$ and the nature of both equilibria. What happens to the activity if T is slowly increased from 0 to 0.1 and then decreased back?

Solution

$y_+ = 1/1.2 = 0.833$, $x_+ = 0.0833 = T_+$; $y_- = 0.1/(0.1 + 0.11) = 0.476$, $x_- = 0.0476 = T_-$. J_+ has eigenvalues $-1.05 \pm 0.31i$ (stable focus); J_- has -0.99 and -0.21 (stable node, since $r_2 = 0.1$ is outside $(1 - 2\sqrt{0.1} + 0.1, 1 + 2\sqrt{0.1} + 0.1) = (0.47, 1.73)$). Increasing T from 0, the system stays on the active branch ($x \simeq 0.083$) until $T_+ = 0.083$, then drops to $x_- \simeq 0.048$. Decreasing T , it stays low until $T_- = 0.048$ and then jumps back up. The hysteresis loop spans $[0.048, 0.083]$, the region of bistability.

Exercise 13.4

(Homeostatic normalisation) Why does the normalisation $\tilde{W}_{ij} = W_{ij}/\sum_j W_{ij}$ make the critical threshold the same for all subjects in mean field, and what does it change for hubs?

Solution

After normalisation every node's total input weight is 1, so a node's input is a weighted average of its neighbours' states, bounded by 1, whatever the subject's overall fibre density. In mean field the activation condition is $x > T$ for every node and every subject, hence a universal T_c . Without normalisation the input of node i is $\sim W_i x$, so $T_c \propto \langle W \rangle$, which differs between subjects (hence the constant ratio $T_c/\langle W \rangle$). For hubs, normalisation reduces their dominance: before it, high-strength

nodes are almost always active and enslave the dynamics; after it, every region contributes similarly, which enriches the repertoire of patterns and improves agreement with functional data.

Exercise 13.5

(Diffusion matrix of the HTC model) Derive B from the three reactions and evaluate it at the supercritical equilibrium for $r_2 = 0.1$.

Solution

Reactions with state changes and rates per N : activation $(+1, 0)$ at $(1 - x - y)r_{\text{act}}$; excited to refractory $(-1, +1)$ at x ; recovery $(0, -1)$ at $r_2 y$. Then $B = \sum_j \mathbf{v}_j \mathbf{v}_j^\top a_j$: $B_{11} = (1 - x - y)r_{\text{act}} + x$, $B_{22} = x + r_2 y$, $B_{12} = -x$. At $x_+ = 0.083$, $y_+ = 0.833$, $r_{\text{act}} = 1$: $B_{11} = 0.083 + 0.083 = 0.167$, $B_{22} = 0.083 + 0.083 = 0.167$, $B_{12} = -0.083$. The negative correlation reflects that each $E \rightarrow R$ event lowers x and raises y together.

13.6. Sources

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Part III.

Part III: Physics of the cell

14. The cell and stochastic gene expression

The third module moves inside the cell. Genes are the units, proteins and mRNA are the “populations”, and transcription factors play the role of interactions. Molecule numbers are small (a few to a few thousand copies per cell), so noise is not a correction but a defining feature: genetically identical cells in identical environments express genes at very different levels. This chapter introduces the molecular machinery of gene expression and solves the two classic stochastic models: the one-stage (birth–death) model, which gives Poisson statistics, and the two-stage (mRNA–protein) model, which gives protein **bursts** and a negative binomial distribution, directly comparable with single-cell data.

14.1. The cell and the central dogma

A cell is a crowded, compartmentalised reactor. Its genome (about 4.6 Mb in *E. coli*, 3 Gb in humans) encodes genes that are **transcribed** by RNA polymerase into messenger RNA (mRNA), which is **translated** by ribosomes into proteins. Each codon of three nucleotides ($4^3 = 64$ possibilities) codes for one of 20 amino acids, plus stop signals. A protein is a chain of 50–1000 amino acids that folds into a specific three-dimensional structure (Chapter 18). The folding is driven by the free energy $F = E - TS$: hydrophobic residues hide in the core, and the folded state has low energy but also low entropy.

Transcription factors (TFs) are proteins that bind specific DNA sequences (operators) near a gene’s promoter and activate or repress its transcription. Gene expression is therefore regulated by the products of other genes, forming **gene regulatory networks** (Chapters 16–17).

Typical numbers for *E. coli* show why noise matters:

quantity	value
cell volume	$\sim 1 \mu\text{m}^3$ (so 1 molecule $\simeq 1.7$ nM)
mRNA lifetime $1/d_0$	a few minutes
protein lifetime $1/d_1$	set by dilution, about one cell cycle (~ 20 –60 min)
proteins per mRNA	tens
copies of a typical TF	10–100

14.2. The one-stage model

Ignore mRNA and let a protein be produced at rate v and degraded (or diluted) at rate d per molecule:



The master equation for $P_n(t)$, the probability of n proteins, is the birth–death equation of Chapter 2 with constant birth rate:

$$\frac{dP_n}{dt} = v(P_{n-1} - P_n) + d[(n+1)P_{n+1} - nP_n].$$

With the generating function $G(z, t) = \sum_n P_n z^n$, using $\sum_n n P_n z^n = z \partial_z G$,

$$\partial_t G = (z - 1)(vG - d \partial_z G).$$

14. The cell and stochastic gene expression

At stationarity $\partial_z G = \frac{v}{d} G$, so $G = e^{(v/d)(z-1)}$: a **Poisson** distribution with mean $\langle n \rangle = v/d$ and **Fano factor** $\sigma^2/\langle n \rangle = 1$. The time-dependent solution from $n = 0$ is Poisson with mean $\frac{v}{d}(1 - e^{-dt})$, and relaxation happens on the time scale $1/d$.

A single molecule survives a time t with probability $P_>(t) = e^{-dt}$, so its **half-life** is $t_{1/2} = \ln 2/d$: minutes for mRNA, while stable proteins are mostly removed by dilution at cell division.

14.3. The two-stage model

Experiments find Fano factors much larger than one. The reason is that proteins are made from a small number of mRNAs, each producing many proteins. The two-stage model includes both species:



With m mRNAs and n proteins,

$$\frac{dP_{m,n}}{dt} = v_0(P_{m-1,n} - P_{m,n}) + v_1 m(P_{m,n-1} - P_{m,n}) + d_0[(m+1)P_{m+1,n} - mP_{m,n}] + d_1[(n+1)P_{m,n+1} - nP_{m,n}].$$

The generating function $F(z', z, t) = \sum_{m,n} P_{m,n} z'^m z^n$ obeys, with $u = z' - 1$ and $v = z - 1$,

$$\partial_t F = v_0 u F + v_1 (1+u) v \partial_u F - d_0 u \partial_u F - d_1 v \partial_v F.$$

14.3.1. Moments

From the equations for the means and second moments (or by differentiating F at $u = v = 0$),

$$\langle m \rangle = \frac{v_0}{d_0}, \quad \langle n \rangle = \frac{v_0 v_1}{d_0 d_1}, \quad \frac{\sigma_n^2}{\langle n \rangle} = 1 + \frac{v_1}{d_0 + d_1}.$$

Define the **burst frequency** $a = v_0/d_1$ (mRNAs made per protein lifetime) and the **burst size** $b = v_1/d_0$ (proteins made per mRNA). Then $\langle n \rangle = ab$, and when $\gamma = d_0/d_1 \gg 1$ (mRNA much shorter-lived than protein), the Fano factor is $1 + b$. Protein noise is set by the burst size, not by the mean.

14.3.2. Bursts and the negative binomial

When $\gamma \gg 1$, each mRNA lives briefly and produces a geometrically distributed burst of proteins with mean b (each step, translation with probability $\frac{v_1}{v_1+d_0}$ or decay). Bursts arrive as a Poisson process at rate v_0 , and proteins decay slowly. Solving the PDE by characteristics in this limit (Shahrezaei and Swain, 2008), the mRNA variable relaxes adiabatically to $u \simeq \frac{bv}{1-bv}$, and the stationary protein generating function becomes

$$F(z) = \left(\frac{1}{1+b-bz} \right)^a \implies P_n = \frac{\Gamma(a+n)}{\Gamma(n+1)\Gamma(a)} \left(\frac{b}{1+b} \right)^n \left(\frac{1}{1+b} \right)^a :$$

a **negative binomial** with mean ab and variance $ab(1+b)$. The same distribution appeared for species abundances (Chapter 3): there, too, it reflected multiplicative “bursty” growth with immigration. The continuum limit is a Gamma distribution with shape a and scale b .

Example 14.1 (Same mean, different noise)

Two genes with the same mean of 50 proteins:

- $a = 20$, $b = 2.5$: frequent small bursts. The distribution is nearly Gaussian, with Fano factor

3.5 (Gillespie simulation: 3.4).

- $a = 0.5$, $b = 100$: rare large bursts. The distribution is strongly skewed, with a peak at zero and a long tail, and the Fano factor is 101 (simulation: 102).

Time traces of the second gene show long silent periods interrupted by sudden jumps, exactly what is seen in single-molecule experiments (Cai, Friedman and Xie, 2006).

14.3.3. Intrinsic and extrinsic noise

Elowitz et al. (2002) put two copies of the same promoter, driving different fluorescent proteins, into the same *E. coli* cell. Fluctuations that affect both copies equally (cell size, ribosome and polymerase numbers, cell-cycle phase) are **extrinsic**. Fluctuations that make the two copies differ (the randomness of individual reaction events) are **intrinsic**. With normalised expressions x_1, x_2 ,

$$\eta_{\text{int}}^2 = \frac{\langle (x_1 - x_2)^2 \rangle}{2\langle x_1 \rangle \langle x_2 \rangle}, \quad \eta_{\text{ext}}^2 = \frac{\langle x_1 x_2 \rangle - \langle x_1 \rangle \langle x_2 \rangle}{\langle x_1 \rangle \langle x_2 \rangle}.$$

Both contributions are comparable in bacteria. The models of this chapter describe intrinsic noise; extrinsic noise enters as slow fluctuations of the rates.

14.4. Why noise matters

Noise is not only a nuisance. **Bet hedging**: in fluctuating environments, populations in which some cells randomly express a stress-response program survive sudden stresses, as in bacterial persistence to antibiotics. **Cell-fate decisions**: noise can push cells across thresholds of bistable circuits (Chapter 16), as in the decision of *B. subtilis* to become competent or of phage λ to choose lysis or lysogeny. Conversely, cells *suppress* noise where precision matters, with negative feedback (Chapter 16) and with multiple gene copies.

14.5. Exercises

Exercise 14.1

(One-stage model) Show that $G(z, t) = \exp[\frac{v}{d}(1 - e^{-dt})(z - 1)]$ solves the one-stage generating-function equation with $P_n(0) = \delta_{n0}$. What are the mean, variance and half-life?

Solution

With $\mu(t) = \frac{v}{d}(1 - e^{-dt})$, $G = e^{\mu(z-1)}$, so $\partial_t G = \dot{\mu}(z-1)G = ve^{-dt}(z-1)G$ and $\partial_z G = \mu G$. The right-hand side is $(z-1)(v-d\mu)G = (z-1)ve^{-dt}G$, equal to the left. At $t = 0$, $G = 1$. The distribution is Poisson at all times, with mean = variance = $\mu(t) \rightarrow v/d$. One molecule survives with e^{-dt} , so $t_{1/2} = \ln 2/d$.

Exercise 14.2

(Two-stage moments) Derive $\langle m \rangle$, $\langle n \rangle$ and the protein Fano factor from the moment equations of the two-stage model.

Solution

Means: $\langle \dot{m} \rangle = v_0 - d_0 \langle m \rangle$ and $\langle \dot{n} \rangle = v_1 \langle m \rangle - d_1 \langle n \rangle$, so $\langle m \rangle = v_0/d_0$ and $\langle n \rangle = v_0 v_1 / (d_0 d_1)$. Second moments at stationarity (writing covariances): $\sigma_m^2 = \langle m \rangle$ (Poisson); $\frac{d}{dt} \text{Cov}(m, n) = v_1 \sigma_m^2 - (d_0 + d_1) \text{Cov} = 0$, so $\text{Cov} = \frac{v_1 \langle m \rangle}{d_0 + d_1}$; $\frac{d}{dt} \sigma_n^2 = v_1 \langle m \rangle + d_1 \langle n \rangle + 2v_1 \text{Cov} - 2d_1 \sigma_n^2 = 0$. Using $v_1 \langle m \rangle = d_1 \langle n \rangle$: $\sigma_n^2 = \langle n \rangle + \frac{v_1}{d_1} \text{Cov} = \langle n \rangle (1 + \frac{v_1}{d_0 + d_1})$. Fano factor $1 + \frac{v_1}{d_0 + d_1} \rightarrow 1 + b$ for $d_0 \gg d_1$.

Exercise 14.3

(Bursts by simulation) Simulate the two-stage model with the Gillespie algorithm for $d_0 = 1$, $d_1 = 0.01$ (in min^{-1}), and $(a, b) = (20, 2.5)$ and $(0.5, 100)$. Compare the time-averaged histograms with the negative binomial.

Solution

Rates: $v_0 = ad_1$, $v_1 = bd_0$. Time-weighted statistics over 3×10^5 min give mean 50.6 and Fano factor 3.40 for $(20, 2.5)$, and mean 49.7 and Fano factor 101.7 for $(0.5, 100)$, matching $ab = 50$ and $1 + b = 3.5$, 101. Histograms agree with $\text{NB}(a, b/(1 + b))$: the first is unimodal around 50, the second has its mode at 0 and a long tail up to several hundred. With $a < 1$ the gene is “off” most of the time.

Exercise 14.4

(Negative binomial from bursts) Assume bursts arrive at rate v_0 , each adding a geometric number of proteins with mean b , and proteins decay at rate d_1 . Show that the stationary generating function is $(1 + b - bz)^{-a}$ with $a = v_0/d_1$.

Solution

A geometric burst has generating function $g(z) = \frac{1}{1+b-bz}$. The master equation in generating-function form is $\partial_t F = v_0[g(z) - 1]F - d_1(z - 1)\partial_z F$. At stationarity $\partial_z \ln F = \frac{a[g(z)-1]}{z-1} = \frac{ab}{1+b-bz}$, which integrates to $\ln F = -a \ln(1 + b - bz)$, using $F(1) = 1$. Hence $F = (1 + b - bz)^{-a}$, the negative binomial. Its mean is $F'(1) = ab$ and its variance $ab(1 + b)$.

Exercise 14.5

(Intrinsic noise) For the one-stage model, compute $\eta^2 = \sigma^2/\langle n \rangle^2$. How many proteins are needed for 10% noise? What if the protein is made in bursts of size $b = 20$?

Solution

Poisson: $\eta^2 = 1/\langle n \rangle$, so 10% noise ($\eta^2 = 0.01$) needs $\langle n \rangle = 100$. With bursts, $\eta^2 = (1 + b)/\langle n \rangle$, so $\langle n \rangle = 2100$ proteins are needed for the same precision: bursting is costly for precision, which is one reason why highly expressed, noise-sensitive genes tend to use high transcription and low translation rates (small b).

14.6. Sources

Lecture 17 slides (A. Maritan: the cell, transcription factors, mRNA translation, stochastic gene expression, one-stage and two-stage models) and course materials *one-stage-model.pdf* and *2-stage model.pdf*. V. Shahrezaei and P. S. Swain, *Analytical distributions for stochastic gene expression*, PNAS **105**, 17256 (2008). M. B. Elowitz, A. J. Levine, E. D. Siggia, P. S. Swain, Science **297**, 1183 (2002). L. Cai, N. Friedman, X. S.

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15. Life at the micro-scale: information, ratchets and molecular motors

Inside a cell, molecular machines transport cargo, copy DNA and contract muscles, all while being kicked around by thermal fluctuations with energies comparable to their own working energy. Can such machines extract work from noise, apparently violating the second law? This chapter revisits the paradoxes of Maxwell's demon and Feynman's ratchet, introduces Landauer's principle (information is physical), and builds the stochastic models of molecular and Brownian motors. Their consistency with thermodynamics rests on the principle of **local detailed balance**.

15.1. Energy scales of the cell

At $T = 300$ K, the thermal energy is $k_B T = RT/N_A = 4.1 \times 10^{-21}$ J $\simeq 4.1$ pN $\cdot$ nm $\simeq 0.6$ kcal/mol $\simeq 25$ meV. Molecular processes involve energies of a few to a few tens of $k_B T$:

- **ATP hydrolysis**, $\text{ATP} \rightleftharpoons \text{ADP} + \text{P}_i$, releases $\Delta G^0 \simeq -30$ kJ/mol $\simeq 5 \times 10^{-20}$ J $\simeq 12 k_B T$ under standard conditions, and about 20 – $25 k_B T$ under cellular conditions, where ATP is far from equilibrium with its products.
- A water molecule at 300 K moves at $\sqrt{\langle v^2 \rangle} = \sqrt{3RT/M} \simeq 650$ m/s and undergoes a collision every $\sim 10^{-13}$ s. A protein in water experiences $\sim 10^{13}$ collisions per second, so its motion is overdamped Brownian motion: inertia is irrelevant (low Reynolds number), and any directed motion must be produced against incessant random forces.

15.2. Maxwell's demon and the cost of information

Maxwell (1867) imagined a demon controlling a trapdoor between two gas compartments at the same temperature. Letting fast molecules pass only from left to right and slow ones from right to left, it would create a temperature difference, $T_R > T_L$, without doing work, decreasing the entropy of an isolated system. This contradicts the second law.

Szilard's engine (1929) makes it quantitative with one molecule in a box at temperature T . Insert a partition in the middle, *measure* on which side the molecule is (one bit of information), and let it push the partition isothermally to the far wall. The extracted work is

$$W = \int_{V/2}^V p dV = k_B T \ln 2,$$

seemingly from a single heat bath.

The resolution came from the physics of computation. Bennett (1982) showed that measurement can be done without dissipation, but the demon's memory must eventually be **reset** to be reused, and:

Theorem 15.1 (Landauer's principle (1961)). *Erasing one bit of information in an environment at*

temperature T dissipates at least

$$Q \geq k_B T \ln 2,$$

i.e. it increases the entropy of the environment by at least $\Delta S \geq k_B \ln 2$.

Including the memory, the total entropy balance $\Delta S_{\text{gas}} + \Delta S_{\text{memory}} + \Delta S_{\text{environment}} \geq 0$ is respected: the work extracted by the Szilard engine is paid for by erasing the measurement record. Information is physical. Landauer's bound was verified experimentally with a colloidal particle in a double-well optical trap (Bérut et al., 2012): erasing the bit (forcing the particle into one well, whatever the initial well) dissipates on average $k_B T \ln 2$ in the slow limit. Conversely, an **information engine** can convert information into work at most $k_B T \ln 2$ per bit.

15.3. Feynman's ratchet and pawl

Feynman (1962) analysed a microscopic ratchet: a paddle wheel immersed in a gas at temperature T_2 is connected by an axle to a ratchet with a spring-loaded pawl at temperature T_1 . The pawl lets the ratchet rotate only one way, so it seems that random collisions with the paddles would rotate the axle and lift a weight.

They do not when $T_1 = T_2$. The pawl itself is thermally agitated: with the same probability $\propto e^{-\epsilon/k_B T}$ with which a collision supplies the energy ϵ to advance one tooth, the pawl is thermally lifted and the ratchet slips back. Forward and backward rates balance, and no net motion occurs. If $T_2 > T_1$, the device works as a heat engine, with efficiency bounded by Carnot's, $\eta \leq 1 - T_1/T_2$. (Feynman's original claim that it reaches the Carnot efficiency is incorrect: heat conduction through the axle makes it irreversible; Parrondo and Español, 1996.) Rectifying equilibrium fluctuations is impossible; directed motion needs a source of free energy that drives the system out of equilibrium.

15.4. Brownian motors

15.4.1. Equilibrium in a periodic potential

Consider an overdamped particle in a periodic, asymmetric ("sawtooth") potential $V(x + L) = V(x)$. At equilibrium its density is the Boltzmann distribution $P_{\text{eq}}(x) \propto e^{-\beta V(x)}$. On an infinite line the normalisation diverges, but the reduced density

$$\hat{P}(x) = \sum_{n \in \mathbb{Z}} P(x + nL), \quad \hat{P}_{\text{eq}}(x) = \frac{e^{-\beta V(x)}}{\hat{Z}}, \quad \hat{Z} = \int_0^L e^{-\beta V(x)} dx,$$

is well defined on one period. Being an equilibrium state, it carries **zero current**, $J = 0$: asymmetry of the potential alone does not produce motion. With an external force F , the tilted potential $V(x) - Fx$ produces a drift in the direction of the force, whatever the asymmetry.

15.4.2. Rectifying noise out of equilibrium

Directed motion appears when the potential is switched or tilted in time, which requires an energy input. In a **flashing ratchet** the sawtooth is periodically turned on and off. While off, the particle diffuses symmetrically. When the potential is switched on again, the particle is captured in the well toward which it has a shorter distance to the nearest maximum. Because of the asymmetry, more particles cross the short side than the long side, and a net current flows, even *against* a small load. A **rocking ratchet** (a periodic, zero-mean tilt) or temperature oscillations work the same way. Brownian motors are theoretical and artificial

devices (optical ratchets, colloids, electrons in asymmetric structures) that turn unbiased fluctuations into directed motion with the help of a non-equilibrium driving.

15.5. Molecular motors

15.5.1. Biological motors

Cells contain three families of **molecular motors**, proteins that convert the chemical free energy of ATP into mechanical work:

- **kinesins** walk along microtubules toward their plus end, carrying vesicles in axons, in 8 nm steps, one ATP per step, at about $1 \mu\text{m/s}$;
- **dyneins** walk toward the minus end of microtubules (retrograde transport, cilia and flagella);
- **myosins** move on actin filaments (muscle contraction, transport).

Compared with Brownian motors, they are tightly coupled to chemistry: each step is associated with a conformational change triggered by binding and hydrolysis of ATP. Directionality comes from the asymmetric structure of the track and of the motor, and their motion is nearly deterministic at high ATP concentration.

15.5.2. A two-state model

The simplest model has the motor on a periodic track with sites $x = na$ and two internal states, 1 and 2 (e.g. ATP-free and ATP-bound). The transitions are:

- **chemical**: $1 \rightarrow 2$ by ATP binding/hydrolysis, at rate K_+ ; backward at rate K_- , both at the same site;
- **mechanical**: from state 2 at site n to state 1 at site $n + 1$ at rate W_+ , and backward at rate W_- .

For the site-summed probabilities $P_i(t) = \sum_n P_i(n, t)$,

$$\dot{P}_1 = (K_- + W_+)P_2 - (K_+ + W_-)P_1, \quad P_1 + P_2 = 1,$$

so at stationarity

$$P_2^* = \frac{K_+ + W_-}{K_+ + K_- + W_+ + W_-}.$$

The mean velocity is a times the net rate of forward mechanical steps:

$$v = a(W_+P_2^* - W_-P_1^*) = a \frac{W_+K_+ - W_-K_-}{K_+ + K_- + W_+ + W_-}.$$

The motor moves forward iff $W_+K_+ > W_-K_-$: the product of the forward rates around a cycle must exceed that of the backward rates. In equilibrium the two products are equal (Kolmogorov's criterion for detailed balance), and $v = 0$.

15.5.3. Local detailed balance and thermodynamic consistency

How are the rates constrained by thermodynamics? For a transition $x' \rightarrow x$ that exchanges heat with a bath, **local detailed balance** requires

$$\frac{W_{x' \rightarrow x}}{W_{x \rightarrow x'}} = e^{Q_{\text{bath}}/k_B T} = e^{\Delta S_{\text{bath}}/k_B},$$

where Q_{bath} is the heat released to the environment. By the first law, $\Delta E = Q + W + \sum_i \mu_i \Delta N_i$, where W is mechanical work done on the system and the last term is the chemical work. Over a full cycle the motor

returns to the same internal state, advances by a against a load F (work $-Fa$), and converts one ATP into ADP and P_i , with $\Delta\mu = \mu_{\text{ATP}} - \mu_{\text{ADP}} - \mu_{P_i}$. Hence

$$\frac{W_+K_+}{W_-K_-} = e^{\beta(\Delta\mu - Fa)}.$$

- With $\Delta\mu = 0$ and $F = 0$ (equilibrium), forward and backward cycles are equally likely, and $v = 0$.
- The motor moves forward iff $\Delta\mu > Fa$. The **stall force** is $F_s = \Delta\mu/a$; beyond it the motor walks backward, and at high enough force it can even synthesise ATP. For kinesin ($a = 8$ nm, $\Delta\mu \simeq 20$ – $25 k_B T$) this gives $F_s \simeq 10$ – 13 pN, while measured stall forces are 6–7 pN: kinesin works at an efficiency of about 50%.
- The entropy production per cycle is $\beta(\Delta\mu - Fa) \geq 0$, and the thermodynamic efficiency $\eta = Fa/\Delta\mu \leq 1$, with equality only at stall, where the speed vanishes.

These are instances of **stochastic thermodynamics**, which extends the first and second laws to single trajectories of small systems. One of its results is that the ratio of the probabilities of a trajectory and its time reverse equals $e^{\Delta S_{\text{tot}}/k_B}$ (fluctuation theorems).

15.6. Exercises

Exercise 15.1

(Szilard engine) Compute the work extracted by isothermal expansion of a one-molecule gas from $V/2$ to V , and the minimal heat dissipated by resetting the memory. Express both at 300 K in joules and in units of the ATP free energy.

Solution

$W = \int_{V/2}^V \frac{k_B T}{V'} dV' = k_B T \ln 2 = 2.9 \times 10^{-21}$ J at 300 K. Resetting costs $Q \geq k_B T \ln 2$, the same amount, so the cycle produces no net work, as the second law requires. ATP releases about 5×10^{-20} J ($12 k_B T$, standard conditions), so one ATP could pay for erasing about 17 bits at the Landauer limit. Real cellular information processing (e.g. sensing by receptors) dissipates many times more per bit.

Exercise 15.2

(No current at equilibrium) For an overdamped particle in a periodic potential, the Fokker–Planck current is $J = -D(\partial_x P + \beta V'(x)P)$. Show that the stationary periodic solution has $J = 0$ when $V(x+L) = V(x)$, and compute J for a tilted potential $V(x) - Fx$ to first order in F .

Solution

At stationarity J is constant. Write $J = -De^{-\beta V} \partial_x (e^{\beta V} P)$, so $\partial_x (e^{\beta V} P) = -\frac{J}{D} e^{\beta V}$. Integrating over a period with P and V periodic, the left side integrates to zero, while the right side integrates to $-\frac{J}{D} \int_0^L e^{\beta V} dx \neq 0$. Hence $J = 0$ and $P \propto e^{-\beta V}$. With a tilt, $e^{\beta(V-Fx)} P$ increases by the factor $e^{-\beta FL}$ over a period, and to first order in F the particle drifts with velocity $v = \beta D_{\text{eff}} F$, where $D_{\text{eff}} = \frac{DL^2}{\int_0^L e^{\beta V} dx \int_0^L e^{-\beta V} dx} \leq D$ (the Lifson–Jackson formula). The drift is along the force, with a mobility reduced by the potential barriers, whatever the asymmetry of V .

Exercise 15.3

(Two-state motor) For $K_+ = 100$, $K_- = 1$, $W_+ = 200$, $W_- = 2$ (in s^{-1}) and $a = 8$ nm, compute P_2^* , the velocity, and the free energy per cycle $k_B T \ln \frac{W_+ K_+}{W_- K_-}$. What load stalls the motor?

Solution

Sum of rates = 303 s^{-1} . $P_2^* = (100 + 2)/303 = 0.337$. $v = 8 \text{ nm} \times \frac{200 \times 100 - 2 \times 1}{303} = 8 \times 66.0 = 528 \text{ nm/s}$. Cycle affinity $\ln(20000/2) = \ln 10^4 = 9.2$, so $\Delta\mu - Fa = 9.2 k_B T$ at zero load. The stall force is $F_s = 9.2 k_B T / a = 9.2 \times 4.1/8 = 4.7 \text{ pN}$. Under a load F the backward rates increase (or forward rates decrease) so that the ratio becomes $e^{9.2 - Fa/k_B T}$, and $v \rightarrow 0$ at F_s .

Exercise 15.4

(Efficiency of kinesin) Kinesin steps 8 nm per ATP and stalls at 6 pN under conditions where $\Delta\mu = 25 k_B T$. Compute the maximum mechanical work per step, the thermodynamic efficiency at stall, and the entropy produced per step at half the stall force.

Solution

Work at stall: $6 \text{ pN} \times 8 \text{ nm} = 48 \text{ pN} \cdot \text{nm} = 11.7 k_B T$. Efficiency $\eta = 11.7/25 = 47\%$. At $F = 3 \text{ pN}$, the work per step is $24 \text{ pN} \cdot \text{nm} = 5.9 k_B T$, and the entropy produced per step is $\Delta S/k_B \geq 25 - 5.9 = 19 k_B T$: most of the free energy is dissipated as heat when the motor moves at a finite speed.

15.7. Sources

Lecture 18 slides (A. Maritan: life at the micro/nano-scale, violating the second law, Maxwell's demon, Szilard engine, Landauer's principle, Feynman ratchet, molecular and Brownian motors, a simple model of a molecular motor, local detailed balance). R. Landauer, IBM J. Res. Dev. **5**, 183 (1961). A. Bérut et al., *Experimental verification of Landauer's principle linking information and thermodynamics*, Nature **483**, 187 (2012). R. P. Feynman, *The Feynman Lectures on Physics*, Vol. I, Chapter 46. J. M. R. Parrondo and P. Español, Am. J. Phys. **64**, 1125 (1996). R. D. Astumian and P. Hänggi, *Brownian motors*, Phys. Today **55**(11), 33 (2002). F. Jülicher, A. Ajdari, J. Prost, Rev. Mod. Phys. **69**, 1269 (1997). U. Seifert, Rep. Prog. Phys. **75**, 126001 (2012). C. Maes, *Local detailed balance*, SciPost Phys. Lect. Notes 32 (2021).

16. Gene regulation and genetic circuits

Genes do not act alone: transcription factors produced by some genes regulate the expression of others, forming circuits with recognisable functions. This chapter builds the standard models of transcriptional regulation from binding equilibria (Hill functions), and analyses the basic **network motifs**: negative autoregulation, which gives homeostasis, faster responses and noise reduction; delayed negative feedback, which gives oscillations; and positive feedback and mutual repression, which give bistable switches and cellular memory. It ends with a combined transport–metabolism circuit (Homework 6) and the generalisation to whole regulatory networks.

16.1. Regulator dynamics

16.1.1. Binding of a regulator

A transcription factor R binds an operator site O near a promoter: $R + O \xrightleftharpoons[k_-]{k_+} RO$. At equilibrium

$$\frac{[R][O]}{[RO]} = \frac{k_-}{k_+} = K,$$

the **dissociation constant** (the concentration of R at which the site is occupied half the time). Since the total number of sites is fixed, $[O] + [RO] = [O_T]$, the occupied fraction is

$$f_B([R]) = \frac{[RO]}{[O_T]} = \frac{[R]}{K + [R]}, \quad f_U = 1 - f_B = \frac{K}{K + [R]}.$$

Often h regulator molecules bind cooperatively (e.g. as a multimer or at several adjacent sites). For $hR + O \rightleftharpoons R_h O$ the occupied fraction becomes the **Hill function**

$$f_B = \frac{(R/K)^h}{1 + (R/K)^h}, \quad f_U = \frac{1}{1 + (R/K)^h},$$

with Hill coefficient h (typically 1–4). The larger h , the more switch-like the response: for $h \rightarrow \infty$ it becomes a step at $R = K$.

16.1.2. Activators and repressors

If R is an **activator**, the gene is transcribed when the operator is occupied, and its production rate is γf_B . If R is a **repressor**, transcription happens when the operator is free, at rate γf_U . Including a basal “leaky” production l and removal (degradation and dilution) with lifetime τ , a protein X regulated by R obeys

$$\frac{dX}{dt} = l + \frac{\gamma}{1 + (R/K)^h} - \frac{X}{\tau} \quad (\text{repression}), \quad \frac{dX}{dt} = l + \gamma \frac{(R/K)^h}{1 + (R/K)^h} - \frac{X}{\tau} \quad (\text{activation}).$$

This is a deterministic, mean-field description. Chapter 14 showed how noise is added.

16.2. Negative autoregulation

16.2.1. Homeostasis

The most common motif in *E. coli* (about 40% of transcription factors) is **negative autoregulation** (NAR): a repressor represses its own gene,

$$\frac{dR}{dt} = l + \frac{\gamma}{1 + (R/K)^h} - \frac{R}{\tau}.$$

For $h = 1$ the steady state solves a quadratic,

$$R^* = \frac{1}{2} \left[l\tau - K + \sqrt{K^2 + \tau(l^2\tau + 2Kl + 4K\gamma)} \right],$$

and for $l = 0$, $R^* \simeq \sqrt{K\gamma\tau}$ when $\gamma\tau \gg K$. Without regulation $R^* = (l + \gamma)\tau$, proportional to the production rate γ , which depends on the environment (growth conditions, polymerase availability). With NAR it grows only as $\sqrt{\gamma}$ for $h = 1$, and more slowly for larger h : as $h \rightarrow \infty$, $R^* \rightarrow K$ whatever γ . The protein level becomes **robust** to changes in the environment.

Example 16.1 (*Homeostasis in numbers*)

With $l = 0.1$, $K = \tau = 1$, increasing γ from 1 to 100 changes the unregulated level from 1.1 to 100, the $h = 1$ level from 0.69 to 9.6, and the $h = 3$ level only from 0.78 to 3.2.

16.2.2. Faster responses

NAR also speeds up the response. A gene switched on at $t = 0$ without regulation reaches half its final level in $t_{1/2} = \tau \ln 2$, independently of the production rate. With NAR the production rate is initially at its maximum (no repressor yet) and is then throttled as R approaches K , so the steady state is reached much faster. With $l = 0.5$, $\gamma = 5$, $K = \tau = 1$, the time to reach half of R^* is 0.34, 0.22, 0.14, 0.12 for $h = 1, 2, 5, 8$, against 0.69 without regulation. NAR also reduces noise, because fluctuations above the set point lower production and fluctuations below it raise it.

16.2.3. Stress response

Negative feedback through a metabolite gives adaptive stress responses. Let a repressor R (constant total R^*) be inactivated by a signal molecule S , $R_{\text{free}} = R^*/(1 + S)$, and repress an enzyme E that removes S :

$$\frac{dE}{dt} = \frac{1}{1 + R_{\text{free}}^2} - \frac{E}{\tau_E}, \quad \frac{dS}{dt} = \gamma_r - \alpha ES.$$

A sudden increase of the stress input γ_r raises S , frees the gene from repression, and produces the enzyme that brings S back down. The response is a transient peak followed by adaptation.

16.2.4. Delayed negative feedback: oscillations

Transcription, translation, maturation and transport take time. If the repression acts with a delay t_d ,

$$\frac{dR}{dt} = l + \frac{\gamma}{1 + (R(t - t_d)/K)^h} - \frac{R}{\tau},$$

the system overshoots: by the time the repressor acts, too much has been made, then too little. For a sufficiently long delay and steep repression the fixed point loses stability through a Hopf bifurcation and sustained oscillations appear. With $l = 0.1$, $\gamma = 10$, $K = \tau = 1$, $h = 4$: for $t_d \leq 0.5$ the level settles at

1.56; for $t_d = 1$ it oscillates between 0.9 and 3.5; for $t_d = 2$ between 0.5 and 7.5. Delayed negative feedback underlies circadian clocks, the p53–Mdm2 oscillations after DNA damage, and the synthetic **repressilator** (three genes repressing each other in a ring; Elowitz and Leibler, 2000).

16.3. Positive feedback and bistability

16.3.1. The toggle switch

Two repressors R and L that repress each other form a **double-negative**, i.e. effectively positive, feedback loop:

$$\frac{dR}{dt} = l_R + \frac{\gamma}{1 + L^{h_L}} - \frac{R}{\tau_R}, \quad \frac{dL}{dt} = l_L + \frac{\gamma}{1 + R^{h_R}} - \frac{L}{\tau_L},$$

in units where $K = 1$. For $h = 1$ there is a single symmetric steady state. For $h > 1$ and strong enough repression there are three: two stable states, one with R high and L low and one with the opposite, separated by an unstable symmetric state. For example, with $l = 0$, $\gamma = 4$, $\tau = 1$, $h = 2$ the steady states are (3.73, 0.27), (0.27, 3.73) and (1.38, 1.38).

The circuit is a **genetic switch**: a transient signal that pushes the system across the separatrix flips it to the other state, which then persists after the signal is gone. The cell has a **memory** of past signals, inherited by its daughters, an *epigenetic* state not encoded in the DNA sequence. The phage lysis–lysogeny decision is the classic natural example (the cI and Cro repressors), and Gardner, Cantor and Collins (2000) built a synthetic toggle switch in *E. coli*. Cooperativity ($h > 1$) is essential: without it the nullclines cross only once.

16.3.2. Self-activation

A gene that activates itself, $\dot{X} = l + \gamma \frac{X^h}{1 + X^h} - X/\tau$, is bistable for $h > 1$ in a range of parameters: an “off” state at $X \simeq l\tau$ and an “on” state near $(l + \gamma)\tau$. Positive feedback converts graded inputs into all-or-none, irreversible decisions (differentiation, sporulation, competence), and noise can trigger spontaneous switching between the states (Chapter 14).

16.4. Combining loops: transport and metabolism

Real circuits combine both kinds of feedback. Consider a cell taking up a nutrient S (e.g. lactose) through a transport protein T and consuming it with an enzyme E , both repressed by a regulator R that is inactivated by S , $R_{\text{free}} = R^*/(1 + S)$:

$$\frac{dS}{dt} = \sigma T - \gamma ES - S, \quad \frac{dT}{dt} = l + \frac{1}{1 + (R_{\text{free}}/K_T)^h} - T, \quad \frac{dE}{dt} = l + \frac{1}{1 + (R_{\text{free}}/K_E)^h} - E.$$

The **transport loop** is positive: more S frees more repressor, which makes more transporter, which imports more S . The **metabolic loop** is negative: more S makes more enzyme, which consumes S . The positive loop makes the response to the external supply σ steep and possibly bistable (the “all-or-none” induction of the *lac* operon observed by Novick and Weiner, 1957). The negative loop keeps the internal concentration of S in check. With $R^* = 10$, $K_T = K_E = 1$, $h = 2$, $l = 0.1$, $\gamma = 1$: for $\sigma = 1$ and 10 the circuit stays almost uninduced ($T = E \simeq 0.11$ – 0.15), while for $\sigma = 100$ it is fully induced ($T = E \simeq 1.07$, $S \simeq 52$).

16.5. From motifs to networks

For many genes the same approach gives a generalised dynamics

$$\frac{dx_i}{dt} = -x_i + \sum_{j=1}^N A_{ij} \frac{x_j^h}{1 + x_j^h},$$

where A_{ij} encodes the **gene regulatory network** (activation $A_{ij} > 0$, repression $A_{ij} < 0$). This is formally a recurrent neural network (Chapter 10) or a generalised Lotka–Volterra system (Chapter 6). Its many stable states can be identified with **cell types**, an idea going back to Kauffman and formalised as a Hopfield network whose attractors are cell identities (Waddington’s epigenetic landscape). Chapter 17 asks the inverse question: how to infer A from measured expression patterns.

16.6. Exercises

Exercise 16.1

(HW 6.1) Feedback loops. Choose one of the feedback loops discussed in class (negative or positive) and simulate it.

Solution

Negative autoregulation ($l = 0.5$, $\gamma = 5$, $K = \tau = 1$). Integrating from $R(0) = 0$, the trajectories converge to $R^* = 2.11, 1.74, 1.37, 1.24$ for $h = 1, 2, 5, 8$, matching the root of $l + \gamma/(1 + R^h) = R$ and the closed form for $h = 1$. As h grows, R^* approaches the large- h limit, where $R^* \simeq K = 1$ if $\tau(l + \gamma) > K$. Half-times are 0.34, 0.22, 0.14, 0.12, all shorter than the unregulated $\ln 2 = 0.69$: NAR speeds up the response. Homeostasis: as γ varies from 1 to 100 (with $l = 0.1$), R^* grows tenfold for $h = 1$ and fourfold for $h = 3$, against a hundredfold without regulation.

Positive feedback (toggle switch, $\gamma = 4$, $h = 2$). Trajectories from different initial conditions end in one of two stable states, (3.73, 0.27) or (0.27, 3.73), according to which side of the diagonal they start. A transient pulse of inducer that lowers one repressor flips the switch permanently.

Exercise 16.2

(HW 6.2) Combining loops. Combine two feedback loops (for instance the transport–metabolism network shown in class), simulate it, and describe what happens.

Solution

With the model of the text ($R^* = 10$, $K_T = K_E = 1$, $h = 2$, $l = 0.1$, $\gamma = 1$), integrating to steady state for increasing supply: $\sigma = 1$, $(S, T, E) = (0.10, 0.11, 0.11)$; $\sigma = 10$, $(1.31, 0.15, 0.15)$; $\sigma = 100$, $(51.6, 1.07, 1.07)$. At low supply the repressor is mostly free and the genes are off. When enough nutrient enters, it titrates the repressor, transporter and enzyme are induced, and uptake jumps (positive loop), while the enzyme limits the accumulation of S (negative loop). The induction is switch-like in σ . Making the transporter more sensitive than the enzyme ($K_T < K_E$) shifts the induction to lower supplies and can produce bistability and hysteresis: cells grown with the nutrient remain induced at supplies that cannot induce naive cells, a memory of past exposure.

Exercise 16.3

(Hill functions) Show that the steepness of the Hill function at $R = K$ is $\frac{df_B}{d \ln R} = h/4$, and compute the fold change in R needed to go from 10% to 90% occupancy.

Solution

$f_B = \frac{x^h}{1+x^h}$ with $x = R/K$, so $\frac{df_B}{d \ln R} = hf_B(1-f_B) = h/4$ at $x = 1$. $f_B = 0.1$ at $x^h = 1/9$ and 0.9 at $x^h = 9$, so $x_{90}/x_{10} = 81^{1/h}$: 81-fold for $h = 1$, 9-fold for $h = 2$, 3-fold for $h = 4$. Cooperativity turns a gradual response into a switch.

Exercise 16.4

(Bistability of self-activation) For $\dot{X} = \frac{X^2}{1+X^2} - \frac{X}{\tau}$ (no leak, $\gamma = K = 1$), find the steady states and the range of τ for bistability.

Solution

$X = 0$ is always a steady state. Non-zero ones satisfy $\frac{X}{1+X^2} = \frac{1}{\tau}$, i.e. $X^2 - \tau X + 1 = 0$, so $X_{\pm} = \frac{\tau \pm \sqrt{\tau^2 - 4}}{2}$, real for $\tau > 2$. For $\tau > 2$ there are three states: 0 (stable), X_- (unstable) and X_+ (stable). The cell is bistable once the protein lives long enough to sustain its own production. With $h = 1$ instead, $\frac{X}{1+X} = \frac{X}{\tau}$ has only $X = \tau - 1$ besides 0, so there is no bistability.

16.7. Sources

Lecture 19 slides (S. Suweis: regulator dynamics, transcription factors, negative feedback loops for homeostasis, stress response and oscillations, positive feedback loops, genetic switches in phages, combined transport-metabolism network, *S*-regulator dynamics) and the course material *genetic-circuits.pdf*; Homework 6. U. Alon, *An Introduction to Systems Biology*, CRC Press (2019), Chapters 2–6. M. Ptashne, *A Genetic Switch: Phage Lambda Revisited*, CSHL Press (2004). T. S. Gardner, C. R. Cantor, J. J. Collins, *Nature* **403**, 339 (2000). M. B. Elowitz and S. Leibler, *Nature* **403**, 335 (2000). A. Novick and M. Weiner, *PNAS* **43**, 553 (1957).

17. Inferring gene interaction networks by maximum entropy

Chapter 16 started from known regulatory interactions and predicted the dynamics. In practice the problem is reversed: we measure the expression of thousands of genes in many samples or single cells, and want to infer who regulates whom. This chapter treats network inference as a statistical-physics inverse problem. The **maximum-entropy principle** gives the least biased model that reproduces the measured means and correlations. For Gaussian data the inferred interaction matrix is the inverse of the covariance matrix, which separates direct interactions from indirect correlations. For binarised single-cell data, a kinetic (asymmetric) Ising model learned from time-ordered cells yields directed, signed networks able to predict the effect of perturbations.

17.1. Gene expression data

A typical dataset is a matrix of counts n_{it} : the number of mRNA reads of gene $i = 1, \dots, N$ in sample (or cell) $t = 1, \dots, T$. Before analysis, raw counts are normalised for sequencing depth ($\sum_i n_{it}$) and gene length L_i , and log-transformed:

$$x_{it} = \log \frac{n_{it}}{L_i \sum_j n_{jt}} \quad (\text{up to constants}).$$

Bulk RNA-seq averages millions of cells per sample. **Single-cell** RNA-seq (scRNA-seq) measures individual cells, revealing heterogeneity and transitions between states, but each cell is measured only once (it is destroyed). Dynamics must be reconstructed from snapshots, e.g. by ordering cells along a **pseudotime** of differentiation.

From the data we can estimate the means $\langle x_i \rangle$ and the pairwise correlations $\langle x_i x_j \rangle$. Ideally we would like the full joint distribution $p(x_1, \dots, x_N)$, but with $N \sim 10^4$ genes and $T \ll$ the number of possible states, it cannot be estimated directly. A principled proxy is needed.

17.2. The maximum-entropy principle

Among all distributions consistent with what we know, choose the one that assumes nothing else: the one maximising the entropy $S[\rho] = -\sum_{\mathbf{x}} \rho(\mathbf{x}) \ln \rho(\mathbf{x})$ (Jaynes, 1957). Imposing normalisation, the means and the second moments, with Lagrange multipliers ν , ψ_i and λ_{ij} ,

$$\mathcal{S} = S[\rho] - \nu \left(\sum \rho - 1 \right) - \sum_i \psi_i \left(\sum \rho x_i - \langle x_i \rangle \right) - \sum_{i \leq j} \lambda_{ij} \left(\sum \rho x_i x_j - \langle x_i x_j \rangle \right).$$

Setting $\delta \mathcal{S} / \delta \rho = 0$:

$$\rho(\mathbf{x}) = \frac{1}{Z} \exp \left(-\psi \cdot \mathbf{x} - \frac{1}{2} \mathbf{x}^\top M \mathbf{x} \right), \quad M_{ij} = \lambda_{ij} + \lambda_{ji}.$$

17.2.1. Continuous expression levels

For continuous x the maximum-entropy distribution is a **multivariate Gaussian**. Completing the square, $\mathbf{y} = \mathbf{x} + M^{-1}\psi$, it is centred at $\langle \mathbf{x} \rangle = -M^{-1}\psi$ with covariance

$$C_{ij} = \langle x_i x_j \rangle - \langle x_i \rangle \langle x_j \rangle = (M^{-1})_{ij}.$$

So the interaction matrix is the **inverse of the covariance matrix** (the precision matrix):

$$M = C^{-1}.$$

The model has $N + N(N+1)/2$ parameters, exactly the number of constraints, and the “Hamiltonian” $H = \frac{1}{2}\mathbf{x}^\top M \mathbf{x} + \psi \cdot \mathbf{x}$ reads M_{ij} as the effective **interaction** between genes i and j .

! Important

Correlation is not interaction. If gene A regulates B and B regulates C , then A and C are correlated although they do not interact directly. Biologists often build “co-expression networks” by thresholding correlations, which mixes direct and indirect effects. The inverse covariance removes the indirect contributions: M_{ij} is proportional to the **partial correlation** of i and j given all other genes, and is zero when i and j are conditionally independent.

17.2.2. The undersampled regime

With $T < N$ samples the empirical covariance matrix has rank at most T and cannot be inverted. Lezon et al. (2006) used the spectral decomposition

$$C = \sum_{k=1}^N \omega_k \mathbf{v}^{(k)} \mathbf{v}^{(k)\top},$$

keeping only the $T-1$ non-zero eigenvalues and defining the **pseudo-inverse** $M = \sum_{k:\omega_k>0} \omega_k^{-1} \mathbf{v}^{(k)} \mathbf{v}^{(k)\top}$. Regularised alternatives (ridge, the graphical lasso with an ℓ_1 penalty that favours sparse M) are used in practice.

17.2.3. Are pairwise interactions enough?

The maximum-entropy model can be extended with three-body terms, $H \supset \sum T_{ijk} x_i x_j x_k$. Applied to time series of yeast gene expression during the metabolic cycle, the distribution of inferred pairwise couplings is broad, while that of triplet couplings is narrowly centred on zero (Lezon et al., 2006): **pairwise interactions dominate**. Significance is assessed with a **null model**: shuffling the data table independently for each gene destroys correlations while preserving single-gene statistics, and couplings within the null distribution are discarded. The inferred networks show a hub-dominated structure, with modules matching functional categories (translation, transcription, cell division, mitochondria, ribosome biogenesis).

17.3. Binary expression and the Ising model

For many purposes it is enough to know whether a gene is “on” or “off”. Binarising each gene’s expression relative to a threshold (e.g. half its maximum),

$$s_i(t) = \begin{cases} +1 & x_i(t) \geq \frac{1}{2} \max_t x_i \\ -1 & \text{otherwise,} \end{cases}$$

the maximum-entropy distribution with fixed means and pairwise correlations is the Ising model of Chapter 12,

$$P(\mathbf{s}) = \frac{1}{Z} \exp\left(\sum_i h_i s_i + \sum_{i < j} J_{ij} s_i s_j\right),$$

with symmetric couplings. Symmetric couplings cannot represent the **direction** of regulation (A activates B but not vice versa) and are learned from equal-time statistics only.

17.4. A kinetic Ising model for single-cell data

Ordering single cells along pseudotime turns snapshots into an effective time series. A **kinetic (asymmetric) Ising model** then describes how each gene's state depends on the previous state of the others (Glauber dynamics, as in Chapter 10):

$$p(s_i(t+dt) | \mathbf{s}(t)) = \frac{\gamma dt}{2} \left[1 + s_i(t+dt) \tanh \theta_i(t) \right] + (1 - \gamma dt) \delta_{s_i(t+dt), s_i(t)}, \quad \theta_i(t) = h_i + \sum_j J_{ij} s_j(t),$$

where γ is the update rate and $J_{ij} \neq J_{ji}$ in general. The log-likelihood of the observed sequence over the update times τ_i of each gene,

$$\mathcal{L}(\{h_i\}, \{J_{ij}\}) = \sum_i \sum_{\tau_i} [s_i(\tau_i + dt) \theta_i(\tau_i) - \ln 2 \cosh \theta_i(\tau_i)],$$

is maximised by gradient ascent:

$$\delta J_{ij} \propto \frac{\partial \mathcal{L}}{\partial J_{ij}} = \frac{\dot{C}_{ij}(0)}{\gamma} + C_{ij}(0) - \langle \tanh \theta_i(t) s_j(t) \rangle,$$

with $C_{ij}(0) = \langle s_i(t) s_j(t) \rangle$ the equal-time correlation and $\dot{C}_{ij}(0)$ its time derivative. The derivative term carries the directional information that equal-time statistics lack.

The resulting pipeline (IGNITE, Corridori et al., 2024): quality control and normalisation, pseudotime ordering, binarisation, inference of (h, J) with hyperparameters selected by how well networks simulated with the learned parameters reproduce the data's correlation matrix. Applied to mouse embryonic stem cells exiting pluripotency (from a naive to a formative state, sampled at 0–48 h), it gives:

- a directed, signed, weighted gene regulatory network among key pluripotency and differentiation genes;
- couplings well outside a null model built from ~ 2500 networks inferred from shuffled data, i.e. statistically significant;
- a fraction of correctly inferred known interactions comparable to or better than other methods (about 0.67, similar to MaxEnt, SCODE and CellOracle);
- a **generative** model. Simulating the learned dynamics reproduces the clustering of the data and its trajectory in principal-component space, and switching genes off in silico predicts the effect of knock-outs, which can be compared with perturbation experiments.

17.5. Exercises

Exercise 17.1

(Maximum entropy with fixed mean and variance) Show that the maximum-entropy density on $\mathbb{R}$ with fixed mean μ and variance σ^2 is the Gaussian, and that with only the mean fixed on $[0, \infty)$ it is the exponential.

Solution

Stationarity of $-\int \rho \ln \rho - \nu(\int \rho - 1) - \psi(\int x \rho - \mu) - \lambda(\int x^2 \rho - \sigma^2 - \mu^2)$ gives $-\ln \rho - 1 - \nu - \psi x - \lambda x^2 = 0$, so $\rho \propto e^{-\psi x - \lambda x^2}$, a Gaussian whose parameters are fixed by the constraints ($\lambda = 1/2\sigma^2$, $\psi = -\mu/\sigma^2$). With only the mean on $[0, \infty)$, $\rho \propto e^{-\psi x}$: the exponential with $\psi = 1/\mu$. Maximum entropy reproduces the familiar distributions as the least informative ones given the constraints, and the Lagrange multipliers play the role of couplings.

Exercise 17.2

(Correlation versus interaction) Three genes form a chain $A - B - C$ with Gaussian interaction matrix $M = \begin{pmatrix} 1 & -0.4 & 0 \\ -0.4 & 1 & -0.4 \\ 0 & -0.4 & 1 \end{pmatrix}$. Compute the correlation matrix. Would thresholding correlations at 0.15 recover the true network?

Solution

$C = M^{-1} = \begin{pmatrix} 1.235 & 0.588 & 0.235 \\ 0.588 & 1.471 & 0.588 \\ 0.235 & 0.588 & 1.235 \end{pmatrix}$, giving correlation coefficients $\rho_{AB} = \rho_{BC} = 0.44$ and $\rho_{AC} = 0.19$.

Thresholding at 0.15 would add a spurious $A-C$ link. The partial correlations $-M_{ij}/\sqrt{M_{ii}M_{jj}}$ are 0.4 for $A-B$ and $B-C$ and exactly 0 for $A-C$: inverting the covariance removes the indirect correlation mediated by B .

Exercise 17.3

(Pseudo-inverse) With $N = 1000$ genes and $T = 50$ samples, how many eigenvalues of the empirical covariance matrix are non-zero? What does the pseudo-inverse do, and what is its main limitation?

Solution

After subtracting the sample mean, the centred data matrix has rank at most $T - 1 = 49$, so at most 49 eigenvalues are non-zero and 951 vanish. The pseudo-inverse inverts C on the 49-dimensional subspace spanned by the data and sets M to zero on its complement. Interactions are thus inferred only along the directions the data explore, and the smallest non-zero eigenvalues, dominated by sampling noise, are amplified by $1/\omega_k$. Regularisation (truncating small eigenvalues, ridge or lasso penalties) controls this noise at the cost of bias.

Exercise 17.4

(Kinetic Ising learning) For a single gene i driven by a single gene j , $\theta_i = h + Js_j$, show that the maximum-likelihood conditions are $\langle s_i^{\text{next}} \rangle = \langle \tanh \theta_i \rangle$ and $\langle s_i^{\text{next}} s_j \rangle = \langle \tanh \theta_i s_j \rangle$. If $s_j = \pm 1$ with equal probability and $s_i^{\text{next}} = s_j$ in 90% of the updates, what are h and J ?

Solution

$\partial \mathcal{L} / \partial h = \sum [s_i^{\text{next}} - \tanh \theta_i]$ and $\partial \mathcal{L} / \partial J = \sum [s_i^{\text{next}} - \tanh \theta_i] s_j$; setting both to zero gives the conditions. By symmetry $h = 0$. Then $\tanh \theta_i s_j = \tanh(J)$ for both values of s_j , and $\langle s_i^{\text{next}} s_j \rangle = 0.9 - 0.1 = 0.8$. So $\tanh J = 0.8$ and $J = \text{artanh} 0.8 = 1.10$. The coupling from j to i is positive (activation). The reverse coupling J_{ji} is learned separately from the updates of j , and may be zero.

17.6. Sources

Lecture 20 slides (S. Suweis: gene regulation networks and expression patterns, maximum entropy inference, pairwise versus triplet interactions, null models, the IGNITE pipeline, asymmetric kinetic Ising model, benchmarks and simulated gene activity) and course materials *Max_ent_calculations.pdf*, Lezon et al. (2006) and Corridori et al. (2024). T. R. Lezon, J. R. Banavar, M. Cieplak, A. Maritan, N. V. Fedoroff, *Using the principle of entropy maximization to infer genetic interaction networks from gene expression patterns*, PNAS **103**, 19033 (2006). E. T. Jaynes, Phys. Rev. **106**, 620 (1957). H. C. Nguyen, R. Zecchina, J. Berg, *Inverse statistical problems: from the inverse Ising problem to data science*, Adv. Phys. **66**, 197 (2017).

18. Proteins as physical objects: polymers, folding and the tube model

Proteins are the working molecules of the cell: enzymes, motors, channels, transcription factors. Each is a chain of amino acids that folds into a specific three-dimensional shape that determines its function. Can polymer physics explain why protein structures are what they are? This chapter recalls the scaling theory of homopolymers (ideal chains, self-avoiding chains, compact globules), lists the experimental facts that make proteins special, and presents the **tube model** of Banavar and Maritan. Treating the chain as a tube of finite thickness, rather than a string of beads, generates the menu of secondary structures (helices and sheets) from geometry and symmetry alone. The chapter ends with an outlook on topics treated in the course projects.

18.1. Homopolymers: scaling laws

A polymer is a chain of L similar monomers at positions $\mathbf{r}_1, \dots, \mathbf{r}_L$. Its size is measured by the end-to-end distance $R_L = |\mathbf{r}_L - \mathbf{r}_1|$ or the radius of gyration $R_G^2 = \frac{1}{L} \sum_i (\mathbf{r}_i - \mathbf{R}_c)^2$. Both scale with length as

$$R \sim L^\nu,$$

with a **universal** exponent ν that depends only on the dimension and on the phase of the polymer, not on chemical details, beyond the persistence length.

18.1.1. Ideal chains

If the chain can cross itself (no excluded volume), it is a random walk of L/ℓ_p independent segments of length $\sim \ell_p$ (the persistence length). By the central limit theorem the end-to-end vector is Gaussian,

$$P_{\text{ideal}}(\mathbf{R} | L) = \left(\frac{3}{2\pi L\ell_p} \right)^{3/2} \exp\left(-\frac{3R^2}{2L\ell_p}\right), \quad \langle R^2 \rangle = L\ell_p, \quad \nu = \frac{1}{2}.$$

The Gaussian form has an entropic interpretation: the free energy $F(R) = -k_B T \ln P = \frac{3k_B T R^2}{2L\ell_p}$ is that of an **entropic spring**. Stretching a chain reduces the number of its conformations, which is why rubber is elastic.

18.1.2. Self-avoiding chains: Flory's argument

Real chains cannot overlap. Excluded volume swells them, with $\nu > 1/2$. Flory's estimate balances the elastic entropy against the repulsion between monomers in a volume R^d :

$$\frac{F(R)}{k_B T} \simeq \frac{R^2}{L\ell_p} + v \frac{L^2}{R^d} \implies R \sim L^{\nu_F}, \quad \nu_F = \frac{3}{d+2}.$$

This gives $\nu = 3/5$ in $d = 3$ (the exact value is 0.588), $3/4$ in $d = 2$ (exact), and $1/2$ at the upper critical dimension $d = 4$. The end-to-end distribution of a self-avoiding chain is non-Gaussian, with the scaling form

$$P_{\text{real}}(\mathbf{R} | L) \propto x^\theta e^{-x^\delta}, \quad x = R/L^\nu, \quad \delta = 1/(1 - \nu).$$

18.1.3. Good and poor solvents

Monomers also attract each other with an energy that competes with solvent affinity. In a **good solvent** (high temperature) the chain is swollen ($\nu = 0.588$). In a **poor solvent** (low temperature) attraction wins and the chain collapses into a **compact globule** that fills space, so $R \sim L^{1/3}$. At the **Θ -point** between them, the second virial coefficient vanishes and the chain is nearly ideal ($\nu = 1/2$). Flory's theorem states that in a dense, compact polymer a sub-chain of length ℓ behaves as ideal, $R_G(\ell) \sim \ell^{1/2}$, because excluded volume is screened by the other chains.

The **compact phase** of standard models (beads on a string with pair potentials, or the Edwards continuum model $H = \int ds [\dot{\mathbf{r}}^2 + \kappa \ddot{\mathbf{r}}^2] + v \iint ds ds' \delta(\mathbf{r}(s) - \mathbf{r}(s'))$) is either a crystal of Hamiltonian walks or a disordered liquid-like globule. Its number of conformations grows exponentially, $\mathcal{N}(L) \sim L^{\gamma-1} \mu^L$, and none of them is special. This is very far from proteins.

18.2. What makes proteins special

A protein is a heteropolymer of 20 amino acids, with a backbone (N-C $_{\alpha}$ -C') and side chains of different sizes and chemistry (hydrophobic, polar, charged). Experiments show:

1. Proteins fold rapidly (microseconds to seconds) and reproducibly into a **native state**.
2. The native geometry determines function.
3. The number of distinct folds is small, a few thousand, much less than the number of sequences.
4. Many different sequences fold into the same structure: folds are robust to mutations.
5. Structures are built from a few modular **secondary motifs**: α -helices and almost-planar β -sheets, as predicted by Pauling and Corey (1951) from hydrogen-bond geometry.
6. Proteins with the same fold have similar folding rates and transition states.
7. Native structures are flexible and interact with ligands and with each other in versatile ways. Misfolded proteins can aggregate into **amyloid fibrils** (sheet-rich structures common to Alzheimer's and Parkinson's diseases) that any sequence can form.

Levinthal's paradox (1969): if each residue of a 100-residue protein had 3 conformations, a random search among $3^{100} \simeq 5 \times 10^{47}$ states at 10^{13} per second would take $\sim 10^{27}$ years. Folding must follow a biased **funnel-shaped energy landscape**, not a random search. These facts suggest that the menu of folds is determined by general physical principles shared by all sequences, with the sequence choosing among a few pre-sculpted possibilities.

18.3. The tube model

18.3.1. Thickness as a missing ingredient

A string of beads with pair potentials has a symmetry that a real chain lacks. Each backbone unit is not a sphere but closer to a coin, a flat object with a preferred orientation, because of the planar peptide bond and the direction of the side chain. In the continuum limit, a chain of spheres becomes a line of zero thickness, while a chain of coins becomes a **tube of finite thickness**. A pair potential between points of a curve is singular: nearby points along the chain are always close, so the potential needs a cut-off, i.e. a fundamental length the model lacks.

The thickness of a curve is captured by **three-body** interactions (Gonzalez and Maddocks, 1999). For any three points s, s', s'' of the curve, let $R(s, s', s'')$ be the radius of the circle through them. As the points

merge it tends to the local radius of curvature. The **thickness** of the curve is

$$\Delta = \min_{s,s',s''} R(s, s', s''),$$

the radius of the largest tube that can be inflated around the curve without self-intersection. A tube of thickness Δ is described by forbidding conformations with any $R(s, s', s'') < \Delta$: a non-singular, three-body constraint that respects the symmetry of the chain.

18.3.2. Optimal packing: the magic helix

What shapes does a compact tube adopt? Maximise compactness at fixed thickness, the polymer analogue of Kepler's sphere packing. The answer is a **helix** with a specific geometry: the densest packed single helix has pitch-to-radius ratio $p/r \simeq 2.512$, within 5% of real α -helices (Maritan, Micheletti, Trovato, Banavar, Nature 2000). The densest double helix has $p/r = 2\pi$, within 4% of B-DNA (6.03). Other optimal configurations are **planar zig-zag sheets**, which pack side by side like β -sheets. Secondary structures emerge from geometry, not from the chemical details of hydrogen bonds, which then fit into them.

18.3.3. The marginally compact phase

Add a solvophobic attraction proportional to the buried surface (energy $\simeq -$ area of the tube not exposed to solvent), with a range comparable to the thickness. As temperature and the ratio of interaction range to thickness vary, the tube has a swollen phase, a compact phase, and between them a **marginally compact phase**. There the tube is just compact enough to bury its surface but still constrained by its thickness. In this phase:

- the number of low-energy conformations is **small**: the degeneracy that plagues standard polymers is lifted;
- the ground states are rich in **helices and sheets**, assembled into a few thousand distinct topologies resembling real folds;
- the free-energy landscape is **pre-sculpted** by geometry and symmetry, and the sequence only has to choose one of a few competing minima. This explains robustness to mutations, fast folding, and the fact that unrelated sequences share folds.

Proteins appear to sit near a phase transition, in the marginally compact phase. The same structures make them prone to aggregation (amyloids, a generic sheet-rich state) and give intrinsically disordered proteins a natural place: sequences that do not select any minimum.

18.4. Outlook: from cells to communities

The course projects connect the three modules:

- **Clonal stem-cell dynamics.** Stem cells divide symmetrically or asymmetrically and differentiate stochastically. Master equations for clone sizes, like those of Chapters 2–3, predict clone-size distributions to compare with lineage-tracing data (Greulich, 2019).
- **The Waddington landscape.** Cell types as attractors of gene regulatory networks, modelled as Hopfield networks whose stored patterns are expression profiles (Chapters 11 and 16). Differentiation is a descent in a landscape, and reprogramming a forced jump between basins.
- **Microbiomes.** Human gut microbial communities show macroecological patterns (abundance distributions, Taylor's law, abundance fluctuations) captured by stochastic logistic models with environmental noise (Chapter 2) and by disordered Lotka–Volterra models (Chapter 6). Changes of their parameters can describe the transition to dysbiosis.

- **Eco-evolutionary dynamics.** Strains within a species interact and evolve on ecological time scales; consumer–resource models with adaptive metabolism (Chapter 7) connect metabolic networks to community assembly.

A guest lecture (A. Forli) illustrated the neuroscience frontier. In freely flying Egyptian fruit bats, hippocampal neurons encode the bat’s own position and also social information about group members. Ensembles of place cells “replay” flight trajectories in time-compressed form during rest. Event-based cameras enable microsecond imaging of neural activity. Comparative studies across species test whether mechanisms such as theta sequences, believed to be universal, really are.

18.5. Exercises

Exercise 18.1

(Ideal chain) Show that for an ideal chain of $n = L/\ell_p$ independent segments of length ℓ_p with isotropic directions, $\langle R^2 \rangle = n\ell_p^2$, and derive the entropic force needed to hold the ends at distance R .

Solution

$\mathbf{R} = \sum_k \mathbf{b}_k$ with $|\mathbf{b}_k| = \ell_p$ and $\langle \mathbf{b}_k \cdot \mathbf{b}_l \rangle = \ell_p^2 \delta_{kl}$, so $\langle R^2 \rangle = n\ell_p^2 = L\ell_p$. By the central limit theorem $P(\mathbf{R}) \propto e^{-3R^2/(2n\ell_p^2)}$, and $F = -k_B T \ln P = \frac{3k_B T}{2n\ell_p^2} R^2 + \text{const.}$ The force is $f = \partial F / \partial R = \frac{3k_B T}{n\ell_p^2} R$: a Hookean spring with stiffness proportional to temperature. Because it is entropic, the chain gets *stiffer* when heated.

Exercise 18.2

(Flory exponent) Minimise $F/k_B T = R^2/(L\ell_p) + \nu L^2/R^d$ with respect to R and derive ν_F . Evaluate it in $d = 1, 2, 3, 4$ and compare with exact values.

Solution

$\partial_R F = 2R/(L\ell_p) - d\nu L^2/R^{d+1} = 0$ gives $R^{d+2} \propto L^3$, so $\nu_F = 3/(d+2)$: 1 in $d = 1$ (exact, a stretched chain), 3/4 in $d = 2$ (exact), 3/5 in $d = 3$ (exact 0.588), 1/2 in $d = 4$ (exact, mean field from here on). Flory’s argument works so well because errors in the two terms nearly cancel.

Exercise 18.3

(Levinthal’s paradox) Estimate the time for an exhaustive search of the conformations of a 100-residue protein with 3 states per residue, sampling 10^{13} conformations per second. How does a funnel resolve the paradox?

Solution

$3^{100} = 5.2 \times 10^{47}$ conformations, so the search time is $5.2 \times 10^{34} \text{ s} \approx 1.6 \times 10^{27}$ years, against a real folding time of milliseconds. In a funnel each residue’s conformation is stabilised as soon as it is locally correct: errors are not frozen but corrected, so the search is biased toward the native state. If each residue finds its local state independently in time τ , folding takes $\sim \tau \ln L$ instead of $\sim \tau 3^L$. A modest energetic bias per residue ($\sim k_B T$) suffices.

Exercise 18.4

(Compact-state entropy) A compact polymer of L monomers has $\mathcal{N} \sim \mu^L$ conformations with $\mu \approx 1.4$ (cubic lattice). Compare the number of compact conformations of a 100-residue chain with the number of distinct protein folds. What does this say about standard polymer models?

Solution

$1.4^{100} \simeq 4 \times 10^{14}$ compact conformations, all of comparable energy in a homopolymer model, against about 10^3 observed folds. A standard compact homopolymer has an enormous ground-state degeneracy and no reason to prefer helices or sheets. The tube model reduces the number of competing low-energy structures to a few, rich in secondary motifs, matching the observed “menu” of folds.

18.6. Sources

Lecture 21 slides (A. Maritan, J. R. Banavar, T. X. Hoang, F. Seno, A. Trovato: a unified perspective on proteins, homopolymers, protein structure, compact phases, the tube model, the magic helix, the marginally compact phase), the course materials in *system-bio-part/proteins* (*Physics of protein folding*, *Optimal shapes of compact strings*, *Central limit theorem for the Edwards model*), Lecture 23 (guest lecture by A. Forli) and Lecture 24 (project topics). P.-G. de Gennes, *Scaling Concepts in Polymer Physics*, Cornell (1979). A. Maritan, C. Micheletti, A. Trovato, J. R. Banavar, *Optimal shapes of compact strings*, *Nature* **406**, 287 (2000). J. R. Banavar and A. Maritan, *Physics of proteins*, *Annu. Rev. Biophys. Biomol. Struct.* **36**, 261 (2007). O. Gonzalez and J. H. Maddocks, *PNAS* **96**, 4769 (1999). P. Greulich, *Mathematical modelling of clonal stem cell dynamics*, *Methods Mol. Biol.* **1975**, 107 (2019). A. Forli and M. M. Yartsev, *Nature* **621**, 796 (2023).

